# MATHEMATICS

# *VECTOR 3D*

# *THEORY SHEET*

# *ENGLISH*

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## 1.1 INTRODUCTION

Physical quantities are broadly divided in two categories

**(a) Vector quantities :**

Any quantity, such as velocity, momentum, or force, that has both magnitude and direction and for which vector addition is defined and meaningful; is treated as vector quantities.

**(b) Scalar quantities :**

A quantity, such as mass, length, time, density or energy, that has size or magnitude but does not involve the concept of direction is called scalar quantity.

**Directed line segment :**

Any given portion of a given straight line where the two end points are distinguished as **Initial** and **Terminal** is called a **Directed Line Segment**.

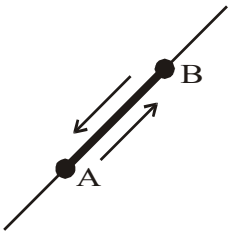
The directed line segment with initial point A and terminal point B is denoted by the symbol $\overrightarrow{AB}$.

The two end points of a directed line segment are not interchangeable and the directed line segments.

$\overrightarrow{AB}$ and $\overrightarrow{BA}$ must be thought of as different.

**Vector :**

A directed line segment is called vector. Every directed line segment have three essential characteristics.




**(i) Length :** The length of $\overrightarrow{AB}$ will be denoted by the symbol $|\overrightarrow{AB}|$

Clearly, we have $|\overrightarrow{AB}| = |\overrightarrow{BA}|$

**(ii) Line of support :** The line of unlimited length of which a directed line segment is a part is called its line of support or simply the Support.

## 1.2 IMPORTANT DEFINITIONS

**1. Zero or null vector :** A vector whose magnitude is zero is called **zero or null vector** and it is denoted by **O** or $\overrightarrow{O}$.

The initial and terminal points of the directed line segment representing zero vector are coincident and its direction is arbitrary.

**2. Unit vector :** A vector of unit magnitude is called a **unit vector**. A unit vector in the direction of a is denotedby $\hat{a}$.

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\vec{a}}{\vec{a}} = \frac{\text{vector a}}{\text{magnitude of a}}$$

**Note :**

(i) Unit vectors parallel to x- axis, y- axis and z- axis are denoted by $\hat{i}, \hat{j} \& \hat{k}$ respectively.

(ii) If $\vec{a} = x\hat{i} + y\hat{j}$ then $|\vec{a}| = \sqrt{x^2 + y^2}$

(iii) If $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$ then $|\vec{a}| = \sqrt{x^2 + y^2 + z^2}$

**3. Equal Vector :** Two vectors $\vec{a}$ and $\vec{b}$ are said to be equal , if

(a) $|\vec{a}| = |\vec{b}|$ (b) they have the same direction

**Note :** Components of two equal vectors taken in any arbitrary direction are equal.

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ , where $\hat{i}$, $\hat{j}$ & $\hat{k}$ are the unit vectors taken along co-ordinate axes, then

$\vec{a} = \vec{b} \Leftrightarrow a_1 = b_1,\ a_2 = b_2,\ a_3 = b_3$

**Ex.1** Find unit vector of $\hat{i} - 2\hat{j} + 3\hat{k}$

Ans $\frac{1}{\sqrt{14}}\ \hat{i} - \frac{2}{\sqrt{14}}\ \hat{j} + \frac{3}{\sqrt{14}}\ \hat{k}$

**Sol. :** $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$

if $\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$

then $|\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$

$\therefore\ |\vec{a}| = \sqrt{14}$

$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{\sqrt{14}}\ \hat{i} - \frac{2}{\sqrt{14}}\ \hat{j} + \frac{3}{\sqrt{14}}\ \hat{k}$

**Ex.2** Let $\vec{r} = 3\hat{i} + 2\hat{j} - 5\hat{k}$, $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{c} = -2\hat{i} + \hat{j} - 3\hat{k}$ . $\vec{r} = \lambda\vec{a} + \mu\vec{b} + \nu\vec{c}$ , then find $\lambda + \mu + \nu$ .

Ans 6

**Sol. :** $3\hat{i} + 2\hat{j} - 5\hat{k} = \lambda(2\hat{i} - \hat{j} + \hat{k}) + \mu(\hat{i} + 3\hat{j} - 2\hat{k}) + \nu(-2\hat{i} + \hat{j} - 3\hat{k})$

$= (2\lambda + \mu - 2\nu)\hat{i} + (-\lambda + 3\mu + \nu)\hat{j} + (\lambda - 2\mu - 3\nu)\hat{k}$

Equating components of equal vectors

$2\lambda + \mu - 2\nu = 3$ .....(i)

$-\lambda + 3\mu + \nu = 2$ .....(ii)

$\lambda - 2\mu - 3\nu = -5$ .....(iii)

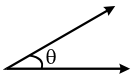
on solving (i), (ii) & (iii)

we get $\lambda = 3,\ \mu = 1,\ \nu = 2$

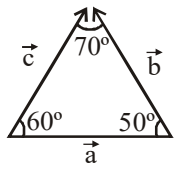
So $\lambda + \mu + \nu = 6$

**4. Angle between two vectors :**

It is the smaller angle formed when the initial points or the terminal points of the two vectors are brought together. Note that $0^\circ \le \theta \le 180^\circ$ .






Ex.3 


(i) Angle between $\vec{a}$ and $\vec{c}$

Ans. 60°

(ii) Angle between $\vec{a}$ and $\vec{b}$

Ans. 130°

(iii) Angle between $\vec{b}$ and $\vec{c}$

Ans. 70°

**5. Collinear vectors**:

Two vectors are said to be collinear if their directed line segments are parallel irrespective of their directions. Collinear vectors are also called **parallel vectors**. If they have the same direction they are named as **like vectors** and if they are in oppositedirections then **unlike vectors**.

**Note:**

1. Symbolically, two non-zero vectors $\vec{a}$ and $\vec{b}$ are collinear if and only if, $\vec{a} = \lambda\vec{b}$, where $\lambda \in R$

$\vec{a} = \lambda\vec{b} \Leftrightarrow (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) = \lambda\,(b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) \Leftrightarrow \quad a_1 = \lambda b_1,\ a_2 = \lambda b_2,\ a_3 = \lambda b_3$

$\Leftrightarrow \dfrac{a_1}{b_1} = \dfrac{a_2}{b_2} = \dfrac{a_3}{b_3}\ (= \lambda)$

2. Vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are collinear if $\dfrac{a_1}{b_1} = \dfrac{a_2}{b_2} = \dfrac{a_3}{b_3}$

3. Vectors parallel to $\vec{p}$ will be taken as $\lambda\vec{p}$

**Ex.4** Find values of x & y for which the vectors $\vec{a} = (x+2)\hat{i} - (x-y)\hat{j} + \hat{k}$ & $\vec{b} = (x-1)\hat{i} + (2x+y)\hat{j} + 2\hat{k}$ are parallel.

**Ans.** $x = -5,\ y = -20$

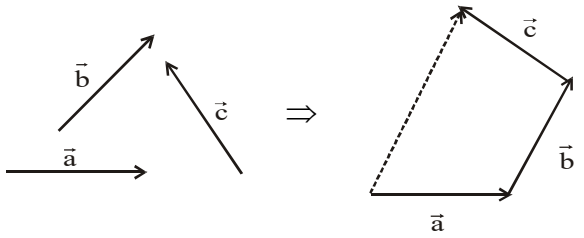
**Sol. :** $\vec{a}$ and $\vec{b}$ are parallel if $\dfrac{x+2}{x-1} = \dfrac{y-x}{2x+y} = \dfrac{1}{2}$

$x = -5,\ y = -20$

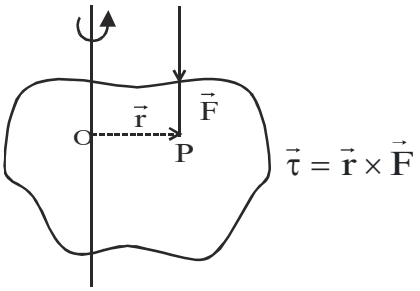
**6. Coplanar Vectors :** If the directed line segment of some given vectors lie in a plane then they are called **coplanar vectors**. It should be noted that two vectors having the same initial point are always coplanar but such three or more vectors may not be coplanar.

There is a very important classification of vector in which vectors are divided into two categories.

**(a) Free vectors :** If a vector can be translated anywhere in space without changing its magnitude & direction, then such a vector is called **free vector**. In other words, the initial point of free vector can be taken anywhere in space keeping its magnitude & direction same.




**(b)** **Localised vectors :** For a vector of given magnitude and direction, if its initial point is fixed in space, then such a vector is called **localised vector**.



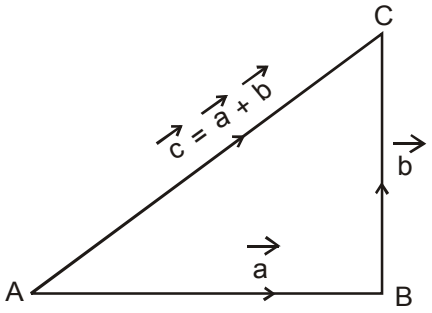

**Unless & until stated, vectors are treated as free vectors.**

## 1.3 ADDITION OF VECTORS

The vectors have magnitude as well as direction, therefore their addition is different than addition of real numbers.

Let $\vec{a}$ and $\vec{b}$ be two vectors in a plane, which are represented by $\overrightarrow{AB}$ and $\overrightarrow{CD}$. Their addition can be performed in the following two ways :

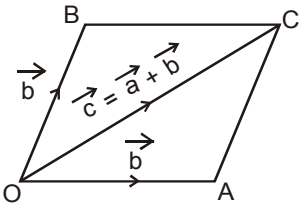
**Triangle law of addition :** If two vectors are represented by two consecutive sides of a triangle then their sum is represented by the third side of the triangle but in opposite direction. This is known as the **triangle law of addition of vectors**.




Thus, if $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{BC} = \vec{b}$, and $\overrightarrow{AC} = \vec{c}$

then $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ i.e. $\vec{a} + \vec{b} = \vec{c}$

**Parallelogram Law of Addition :** If two vectors are represented by two adjacent sides of a parallelogram, then their sum is represented by the diagonal of the parallelogram whose initial point is the same as the initial point of the given vectors. This is known as **parallelogram law of addition of vectors.**




Thus if $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \vec{c}$ then $\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ i.e. $\vec{a} + \vec{b} = \vec{c}$

Where $\overrightarrow{OC}$ is a diagonal of the parallelogram OABC.

**Addition in component form :** If the vectors are defined in terms of $\hat{i}$, $\hat{j}$ and $\hat{k}$. i.e. if $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then their sum is defined as $\vec{a} + \vec{b} = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$

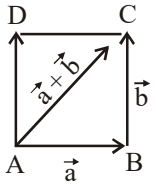
Properties

(i) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (ii) $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$ (iii) $\vec{0} + \vec{a} = \vec{a} = \vec{a} + \vec{0}$

(iv) **Additive inverse :** For every vector $\vec{a}$ its negative vector $-\vec{a}$ exists such that $\vec{a} + (-\vec{a}) = (-\vec{a}) + \vec{a} = \vec{0}$ i.e. (–a) is the additive inverse of the vector $\vec{a}$.

(v) For any three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ $\left.\begin{matrix}\vec{a}+\vec{b}=\vec{a}+\vec{c}\\ \vec{b}+\vec{a}=\vec{c}+\vec{a}\end{matrix}\right\} \Rightarrow \vec{b} = \vec{c}$

(vi) **Triangle inequality**




In $\Delta ABC$

$|AB - BC| < AC < AB + BC$

$\left||\vec{a}| - |\vec{b}|\right| \le |\vec{a} + \vec{b}| \le |\vec{a}| + |\vec{b}|$

## 1.4 MULTIPLICATION OF A VECTOR BY A SCALAR

On multiplying any vector by $\lambda$

(1) It's magnitude is multiplied by $|\lambda|$

(2) If $\lambda > 0$ then direction will remain same.

(3) If $\lambda < 0$ then direction will be opposite.

$\therefore$ magnitude of $m\vec{a} = |m\vec{a}| \Rightarrow$ m (magnitude of a) = $m|\vec{a}|$

Again if $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ then $m\vec{a} = (ma_1)\hat{i} + (ma_2)\hat{j} + (ma_3)\hat{k}$

**Properties**

If $\vec{a}$, $\vec{b}$ are any two vectors and m, n are any scalar then

(i) $m(\vec{a}) = (\vec{a})\, m = m\vec{a}$ (commutativity)

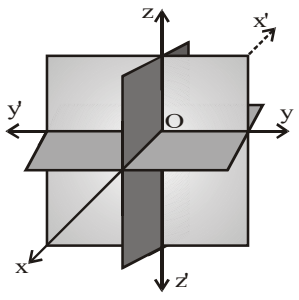
(ii) $m(n\vec{a}) = n(m\vec{a}) = (mn)\,\vec{a}$ (Associativity)

(iii) $(m + n)\,\vec{a} = m\vec{a} + n\vec{a}$

(iv) $m(\vec{a} + \vec{b}) = m\vec{a} + m\vec{b}$ } Distributivity

## 1.5 COORDINATE SYSTEM IN 3D

In cartesian system of the three lines are mutually perpendicular, such a system is called rectangular cartesian co-ordinate system.



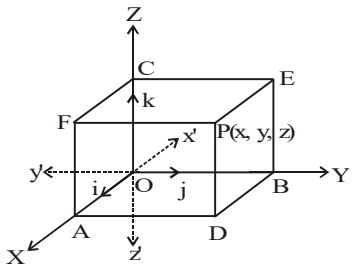

**Co-ordinate axes and co-ordinate planes :**

When three mutually perpendicular planes intersect at a point, then mutually perpendicular lines are obtained and these lines also pass through that point.

If we assume the point of intersection as origin, then the three planes are known as co-ordinate planes and the three lines are known as co-ordinate axes.

**Octants :**

Every plane bisects the space. Hence three co-ordinate plane divide the space in eight parts. These parts are known as octants.




**Coordinates of a Point in Space**

Let O be a fixed point, known as origin and let OX, OY and OZ be three mutually perpendicular lines, taken as x-axis, y-axis and z-axis respectively, in such a way that they form a right handed system.

The planes XOY, YOZ and ZOX are known as xy-plane, yz-plane and zx-plane respectively.

Let P be a point in space and distances of P from yz, zx and xy planes be x, y, z respectively (with proper signs) then we say that coordinates of P are (x, y, z). Also OA = |x|, OB = |y|, OC = |z|

If we assume $\hat{i}$, $\hat{j}$, $\hat{k}$ unit vectors along OX, OY, OZ respectively, then position vector of point P isthe vector $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$ or simply (x, y, z).

**Note : Any point on -**

- x-axis = $\{(x, y, z) \mid y = z = 0\}$
- y-axis = $\{(x, y, z) \mid x = z = 0\}$
- z-axis = $\{(x, y, z) \mid x = y = 0\}$
- xy plane = $\{(x, y, z) \mid z = 0\}$
- yz plane = $\{(x, y, z) \mid x = 0\}$
- zx plane = $\{(x, y, z) \mid y = 0\}$

Let P(a, b, c)

1. Foot of perpendicular from P on x-y plane = $P_{xy}$ (a, b, 0)
   Foot of perpendicular from P on y-z plane= $P_{yz}$ (0, b, c)
   Foot of perpendicular from P on zx plane = $P_{zx}$ (a, 0, c)
2. Distance of P from x-axis = $\sqrt{b^2 + c^2}$

   Distance of P from y-axis = $\sqrt{a^2 + c^2}$

   Distance of P from z-axis = $\sqrt{a^2 + b^2}$
3. Distance of P from xy plane = |c|

   Distance of P from yz plane = |a|

   Distance of P from zx plane = |b|

**Q.1** If the origin is shifted (1, 2, -3) without changing the directions of the axes then find the new coordinates of the point (0, 4, 5) with respect to new frame.

Ans. (-1, 2, 8)

**Sol. :** $x' = x - x_1$, where $(x_1, y_1, z_1)$ is shifted origin

$y' = y - y_1$

$z' = z - z_1$

$x' = 0 - 1 = -1$

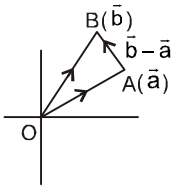
$y' = 4 - 2 = 2$

$z' = 5 + 3 = 8$ [ The coordinate of the point w.r.t. to new coordinate frame is (-1, 2, 8).

## 1.6 DISTANCE FORMULA

If $\vec{a}$ and $\vec{b}$ are position vectors of two points A and B, then

$\overrightarrow{AB} = \vec{b} - \vec{a}$ = position vector **(p.v.)** of B – position vector **(p.v.)** of A.




**Distance formula**

Distance between the two points A $(\vec{a})$ and B $(\vec{b})$ is $AB = |\vec{a} - \vec{b}|$

The distance between two points A $(x_1, y_1, z_1)$ and B $(x_2, y_2, z_2)$ is given by

$$AB = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$$

**(a) Distance from Origin :**

Let O be the origin and P (x, y, z) be any point, then $OP = \sqrt{(x^2 + y^2 + z^2)}$

**Q.1** If position vectors of A and B are $3\hat{i} + 4\hat{j} - 5\hat{k}$ and $2\hat{i} - 3\hat{j} + 4\hat{k}$ then $\overrightarrow{AB}$ equals-

Ans. $-\hat{i} - 7\hat{j} + 9\hat{k}$

**Sol. :** $\overrightarrow{AB}$ = p.v. of B – p.v. of A

$= (2\hat{i} - 3\hat{j} + 4\hat{k}) - (3\hat{i} + 4\hat{j} - 5\hat{k})$

$= (2 - 3)\hat{i} + (-3 - 4)\hat{j} + (4 + 5)\hat{k}$

$= -\hat{i} - 7\hat{j} + 9\hat{k}$

**Q.2** If $A = \hat{i} - \hat{j} + 2\hat{k}$ and $B = 2\hat{i} + 3\hat{j} - 4\hat{k}$ then $|\overrightarrow{AB}|$ equals-

Ans. $\sqrt{53}$

**Sol. :** $\overrightarrow{AB} = (2\hat{i} + 3\hat{j} - 4\hat{k}) - (\hat{i} - \hat{j} + 2\hat{k})$

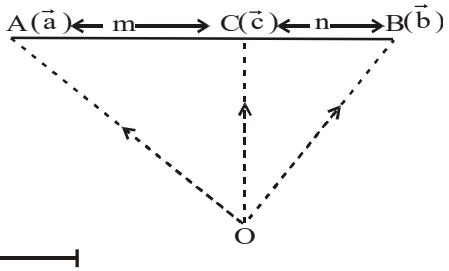
$= \hat{i} + 4\hat{j} - 6\hat{k}$ $\therefore$ $= \sqrt{(1)^2 + (4)^2 + (6)^2}$

$= \sqrt{1 + 16 + 36} = \sqrt{53}$

## 1.7 SECTION FORMULAE

If $\vec{a}$ & $\vec{b}$ are the position vectors of two points A & B then the p.v. of a point C $(\vec{r})$ which divides AB in the ratio m : n is given by :

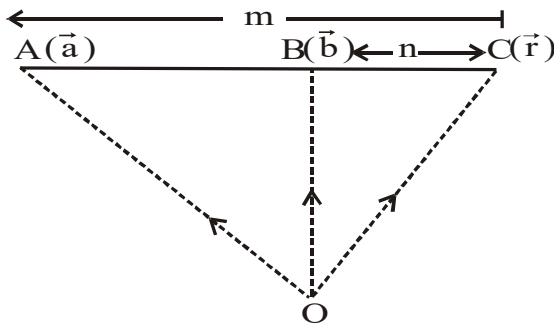
**(a) Internal Division :**




$$\overrightarrow{OC} = \vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

**Note :** Position vector of mid point of **AB** $= \frac{\vec{a} + \vec{b}}{2}$

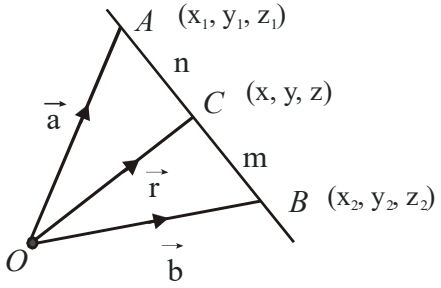
**(b) External division :**




$$\overrightarrow{OC} = \vec{r} = \frac{m\vec{b} - n\vec{a}}{m-n}$$

Proof:

Let A, B and C be three collinear points in space having position vectors $\vec{a}$, $\vec{b}$ and $\vec{r}$.




Let $\frac{AC}{CB} = \frac{n}{m}$

or, $\quad m\,AC = n\,CB$

or, $\quad m\overrightarrow{AC} = n\overrightarrow{CB}$ $\quad \ldots$ (i)

(As vectors are in same direction)

Now, $\overrightarrow{OA} + \overrightarrow{AC} = \overrightarrow{OC} \Rightarrow \overrightarrow{AC} = \vec{r} - \vec{a}$ $\quad \ldots$ (ii)

$\vec{r} + \overrightarrow{CB} = \vec{b} \Rightarrow \overrightarrow{CB} = \vec{b} - \vec{r}$ $\quad \ldots$ (iii)

Using (i), we get $\vec{r} = \frac{m\vec{a} + n\vec{b}}{m+n}$

Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be two points and

(i) let R (x, y, z) divide AB in the ratio $m_1 : m_2$ internally.

Then co-ordinates of $R(x, y, z) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}, \frac{m_1z_2 + m_2z_1}{m_1 + m_2}\right)$

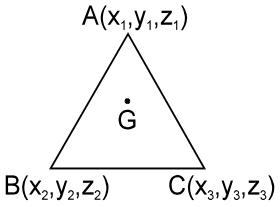
(ii) let R (x, y, z) divide AB in the ratio $m_1 : m_2$ externally.

Then co-ordinates of R(x, y, z) = $\left(\frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 - m_2y_1}{m_1 - m_2}, \frac{m_1z_2 - m_2z_1}{m_1 - m_2}\right)$

(iii) If R is mid point of AB

$R\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$

**Results for a triangle**




1. Centroid = $\left(\frac{\vec{a} + \vec{b} + \vec{c}}{3}\right)$

2. $I\left(\frac{\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c}}{\alpha + \beta + \gamma}\right)$; $I_1\left(\frac{\beta\vec{b} + \gamma\vec{c} - \alpha\vec{a}}{-\alpha + \beta + \gamma}\right)$; $I_2\left(\frac{\alpha\vec{a} + \gamma\vec{c} - \beta\vec{b}}{\alpha - \beta + \gamma}\right)$; $I_3\left(\frac{\alpha\vec{a} + \beta\vec{b} - \gamma\vec{c}}{\alpha + \beta - \gamma}\right)$

3. Orthocentre $\left(\frac{\vec{a}\tan A + \vec{b}\tan B + \vec{c}\tan C}{\tan A + \tan B + \tan C}\right)$

4. Circumentre $\left(\frac{\vec{a}\sin 2A + \vec{b}\sin 2B + \vec{c}\sin 2C}{\sin 2A + \sin 2B + \sin 2C}\right)$

## 1.8 COLLINEARITY OF 3 POINTS

Three points A, B and C are collinear, if any two vectors $\overrightarrow{AB}, \overrightarrow{BC}$ and $\overrightarrow{CA}$ are parallel i.e. one of them is scalar multiple of any one of the remaining vectors.

Note:

If $\vec{a}, \vec{b}, \vec{c}$ be position vectors of three points A,B and C respectively and x, y,z be three scalars so that all are not zero, then the necessary and sufficient conditions for three points to be collinear is that $x\vec{a} + y\vec{b} + z\vec{c} = 0$ and $x + y + z = 0$

**Q.1** If $A \equiv (2\hat{i} + 3\hat{j})$, $B \equiv (p\hat{i} + 9\hat{j})$ and $C \equiv (\hat{i} - \hat{j})$ are collinear, then the value of p is-

Ans. 7/2

**Sol. :** $\overrightarrow{AB} = (p - 2)\hat{i} + 6\hat{j}$, $\overrightarrow{AC} = -\hat{i} - 4\hat{j}$

Now A,B,C are collinear $\Leftrightarrow$ $\overrightarrow{AB} \parallel \overrightarrow{AC}$

$\Leftrightarrow \frac{p-2}{-1} = \frac{6}{-4}$ $\Leftrightarrow$ p = 7/2

Q.2 If point $A(\vec{a})$, $B(\vec{b})$, $C(\vec{c})$ are collinear points. Find x and $(x^2-2)\vec{a}+3x\vec{b}+4\vec{c}=0$, find x.

Ans. $x=-1,-2$

Sol. For A, B, C to be collinear

$x^2-2+3x+4=0$

$\Rightarrow x^2+3x+2=0$

$\Rightarrow (x+1)(x+2)=0$

$\Rightarrow x=-1,-2$

Q.3 If ABCD is a parallalogram and P is point of intersection of diagonals and O is origin such that $\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}+\overrightarrow{OD}=\lambda\overrightarrow{OP}$. Find $\lambda$.

Ans. $\lambda=4$

Sol. In parallalogram diagonals bisect call other.

So $\frac{\overline{OA}+\overline{OC}}{2}=\overline{OP}$ & $\frac{\overline{OB}+\overline{OD}}{2}=\overline{OP}$

$\Rightarrow \overline{OA}+\overline{OB}+\overline{OC}+\overline{OD}=4\left(\overline{OP}\right)$

Q.4 If $\vec{a}$ and $\vec{b}$ are adjacent side vectors of a parallalogram such that $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|$. Then find angle between $\vec{a}$ and $\vec{b}$.

Ans. 90º

Sol. It means length of diagonals are same so the parallelogram becomes rectangle.

Q.5 If $\overrightarrow{PO}+\overrightarrow{OQ}=\overrightarrow{QO}+\overrightarrow{OR}$. Prove that P, Q, R are collinear.

Ans.

Sol. $\Rightarrow -\overrightarrow{OP}+2\overrightarrow{OQ}-\overrightarrow{OR}=0$

sum of Coefficient $=-1+2-1=0$

so P, Q, R are collinear points.

Q.6 For $\Delta ABC$ $\overrightarrow{AB}=3\hat{i}+4\hat{k}$, $\overrightarrow{BC}=2\hat{j}-\hat{k}$ find length of side AC

Ans. $\sqrt{22}$

Sol. $\Rightarrow \overrightarrow{AC}=\overrightarrow{AB}+\overrightarrow{BC}$

$=3\hat{i}+4\hat{k}+2\hat{j}-\hat{k}$

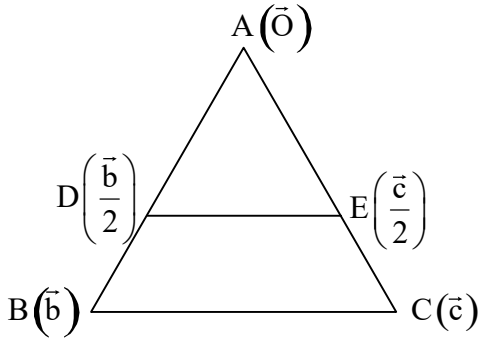
$=3\hat{i}+2\hat{j}+3\hat{k}$

$\Rightarrow \left|\overrightarrow{AC}\right|=\sqrt{22}$

Q.7 Prove that line joining mid-points of two sides of a triangle is parallel to the third side and is half of its length.

Ans.

Sol.




D, E are mid-points of AB and AC respectively

$\overrightarrow{DE} = \frac{\vec{c} - \vec{b}}{2}$ and $\overrightarrow{BC} = \vec{c} - \vec{b}$

Q.8 A(0,1,2), B(3,1,2) and C(0,5,2) are vertices of triangle ABC

(i) Find length of sides.

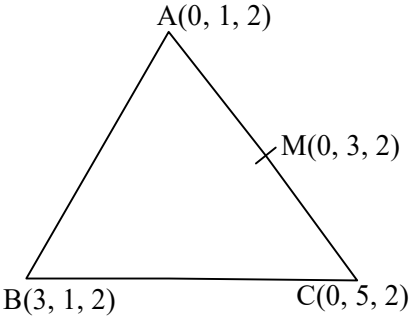
(ii) Find length of median from vertex B.

(iii) Find length of internal angle bisector from vertex A.

(iv) Find centroid, incentre, circumcentre and orthocentre.

Ans. (i) 3, 4, 5, (ii) $\sqrt{13}$ , (iii) $\frac{12}{7}\sqrt{2}$ , (iv) $G\left(1, \frac{7}{3}, 2\right), I(1,2,2), C\left(\frac{3}{2}, 3, 2\right), O(0,1,2)$

Sol.




As the z-coordinate of all the three vertices is same so lengths can be calculate neglecting it.

($\Delta$ ABC is in plane z = 2)

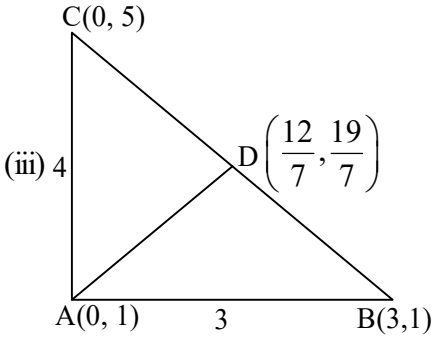
(i) $\left|\overrightarrow{AB}\right| = 3$

$\left|\overrightarrow{BC}\right| = 5$

$\left|\overrightarrow{CA}\right| = 4$

(ii) mid point ofAc is M (0, 3, 2)

$\left|\overrightarrow{BM}\right| = \sqrt{13}$




Angle bisector theores stares that

$$\frac{BD}{CD}=\frac{AB}{AC}=\frac{3}{4}$$

so co-ordinate of D $\left(\frac{12}{7},\frac{19}{7}\right)$

$$\left|\overrightarrow{AD}\right|=\sqrt{\left(\frac{12}{7}\right)^2+\left(\frac{19}{7}-1\right)^2}$$

$$=\frac{12}{7}\sqrt{2}$$

(iv) Centroid $=\left(\frac{\sum x_1}{3},\frac{\sum y_1}{3},\frac{\sum z_1}{3}\right)$

$$=\left(1,\frac{7}{3},2\right)$$

In right angle triangle or-thocentre is right angled vertex and circuncentre is mid point of hypotenues.
So Ortho-centre = A(0, 1, 2)
and circum-centre = mid point of BC

$$=\left(\frac{3}{2},3,2\right)$$

In-centre $=\left(\frac{ax_1+bx_2+cx_3}{a+b+c},\frac{ay_1+by_2+cy_3}{a+b+c},2\right)$

$$=\left(\frac{3\times0+4\times3+5\times0}{3+4+5},\frac{3\times5+4\times1+5\times1}{3+4+5},2\right)$$

= (1, 1, 2)

Q.9 Find slope of a line in x-y plane which is parallel to $3\hat{i}+4\hat{j}$

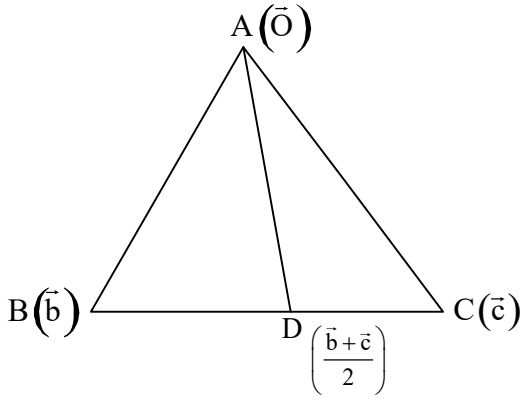
Ans. $\frac{4}{3}$

Sol. $\tan\theta=\frac{4}{3}$

Q.10 For $\Delta ABC$ if AD is median then prove that $\overrightarrow{AD} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$.

Ans.

Sol.




$$\overrightarrow{AD} = \frac{\vec{b}+\vec{c}}{2} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

## 1.9 LINEARLY DEPENDENT AND LINEARLY INDEPENDENT VECTORS

$x\vec{a} + y\vec{b}$ is linear combination of $\vec{a}$ and $\vec{b}$. $x\vec{a} + y\vec{b} + 3\vec{c}$ is linear combination of $\vec{a}$, $\vec{b}$, $\vec{c}$

1. A set of vectors are said to linearly independent if none of them can be written as linear combination of remaining vectors.

If $\vec{x}_1, \vec{x}_2, .....\vec{x}_n$ are n non zero vectors, & $k_1, k_2, .....k_n$ are n scalars & if the linear combination

$\mathbf{k_1\vec{x}_1 + k_2\vec{x}_2 + ....k_n\vec{x}_n = 0 \Rightarrow k_1 = 0, k_2 = 0 ....k_n = 0}$ then we say that vectors $\vec{x}_1, \vec{x}_2, .....\vec{x}_n$ are **linearly independent vectors.**

2. If $\vec{a}_1, \vec{a}_2, ....., \vec{a}_n$ are Linearly independent and

$$x_1\vec{a}_1 + x_2\vec{a}_2 + ..... + x_n\vec{a}_n = y_1\vec{a}_2 + y_2\vec{a}_2 + ..... + y_n\vec{a}_n$$

$$(x_1 - y_1)\vec{a}_2 + (x_2 - y_2)\vec{a}_2 + ..... + (x_n - y_n)\vec{a}_n = 0$$

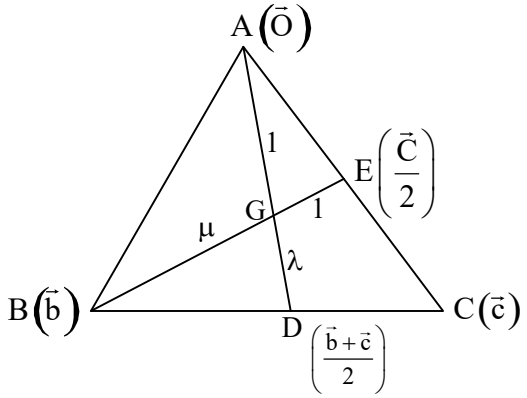
$x_1 = y_1$; $x_2 = y_2$..... $x_n = y_n$

3. We can compare coefficent on both side if vectors on both side are linearly independent.

Q.1 Prove that centroid divide the median is 2 : 1 internally.

Ans. $\frac{1}{2}$

Sol.




D and E are mid points of BC and AC

Equation PV of G

$$\frac{\vec{b}+\vec{c}}{2(\lambda+1)}=\frac{\mu\frac{\vec{c}}{2}+\vec{b}}{(\mu+1)}$$

$$\Rightarrow \frac{\vec{b}+\vec{c}}{2(\lambda+1)}=\frac{\mu\vec{c}+2\vec{b}}{2(\mu+1)}$$

as $\vec{b}$ and $\vec{c}$ are linearly independent so equating coefficients

$$\frac{1}{2(\lambda+1)}=\frac{1}{\mu+1} \qquad .....(1)$$

and $$\frac{1}{2(\lambda+1)}=\frac{\mu}{2(\mu+1)} \qquad .....(2)$$

From (1)/(2)

$$1=\frac{\mu}{2}\Rightarrow \mu=2$$

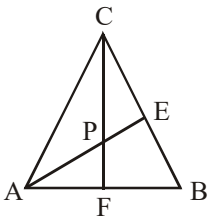
and from (1)

$$\frac{1}{2(\lambda+1)}=\frac{1}{3}$$

$$\Rightarrow \lambda+1=\frac{3}{2}$$

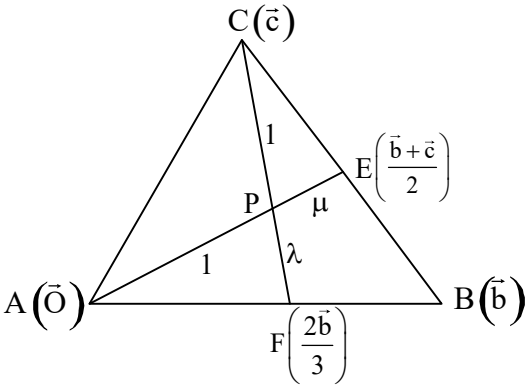
$$\Rightarrow \lambda=\frac{1}{2}$$

Q.2 In $\Delta ABC$, E is mid point of BCand F lies on AB such that $\frac{FB}{FA}=\frac{1}{2}$ Find $\frac{PC}{PF}$




Ans. $\frac{3}{2}$

Sol.




Let $\frac{PE}{PA}=\frac{\mu}{1}$ and $\frac{PF}{PC}=\frac{\lambda}{1}$

Equation PV of P

$$\frac{\lambda\vec{c}+\frac{2\vec{b}}{3}}{(\lambda+1)}=\frac{\vec{b}+\vec{c}}{2(\mu+1)}$$

as $\vec{b}$ and $\vec{c}$ are linearly independent so equating coefficient of $\vec{b}$ and $\vec{c}$

$$\frac{2}{3(\lambda+1)}=\frac{1}{2(\mu+1)} \quad .....(1)$$

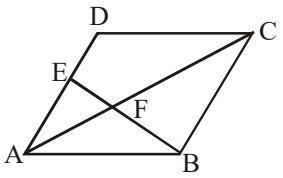
and $$\frac{\lambda}{\lambda+1}=\frac{1}{2(\mu+1)} \quad .....(2)$$

solving (1)/(2)

$$\frac{2}{3\lambda}=1\Rightarrow\lambda=\frac{3}{2}$$

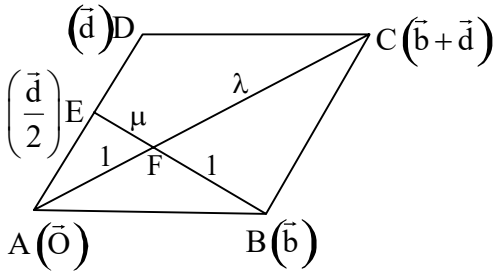
$$\Rightarrow\frac{PC}{PF}=\frac{3}{2}$$

Q.3 ABCD is a parallelogram E is mid point of AD. Find FB/FE




Ans. $\frac{2}{1}$

Sol.




Let $\frac{FC}{FA}=\frac{\lambda}{1}$ and $\frac{FE}{FB}=\frac{\mu}{1}$

Equating PV of F

$$\frac{\vec{b}+\vec{d}}{\lambda+1}=\frac{\mu\vec{b}+\frac{\vec{d}}{2}}{\mu+1}$$

as $\vec{b}$ and $\vec{d}$ are linearly independent so equating coefficient

$$\frac{1}{\lambda+1}=\frac{\mu}{\mu+1} \quad .....(1)$$

and $\frac{1}{\lambda+1}=\frac{1}{2(\mu+1)}$ .....(2)

solving (1)/(2)

$$1=2\mu$$

$$\Rightarrow \mu=\frac{1}{2}$$

So $\frac{FB}{FE}=\frac{1}{\mu}=\frac{2}{1}$

Q4. In a triangle ABC, D divides BC in the ratio 3 : 2 and E divides CA in the ratio 1 : 3. The lines AD and BE meet at H and CH meets AB in F. Find the ratio in which F divides AB.

Ans. $\frac{2}{1}$

**Sol. :** Take C as the origin of reference. Let a, b be the position vectors of the points A and B respectively. Now the points D is b and the point E is a. The equation of AD and BE are

$r = a + k\left(\frac{2}{5}b - a\right)$ and $r = b + t\left(\frac{1}{4}a - b\right)$ respectively.

For the point of intersection H, we have

$$a + k\left(\frac{2}{5}b - a\right) = b + t\left(\frac{1}{4}a - b\right) \quad \text{.....(i)}$$

$$\Longrightarrow 1 - k = \frac{t}{4},\ \frac{2}{5}k = 1 - t \Rightarrow k = \frac{5}{6};\ t = \frac{2}{3}$$

Putting these values of k and t in the equations of the lines AD and BE, we see that the point H is $\frac{1}{6}a + \frac{1}{3}b$ .

The equations of CH and AB being $r = t\left(\frac{a}{6} + \frac{b}{3}\right)$ and $r = a + k(b - a)$ respectively.

For the point of intersection F, $t\left(\frac{a}{6} + \frac{b}{3}\right) = a + k(b - a)$

$$\Longrightarrow \left(\frac{t}{6} - 1 + k\right) = 0, \left(\frac{t}{3} - k\right) = 0$$

These give $t = 2;\ k = \frac{2}{3}$

Thus, the point F is , $\frac{a + 2b}{3}$

so that F divides AB in the ratio 2 : 1.

**Q.5** Let 'O' be the point of intersection of diagonals of a parallelogram ABCD. The points M, N, K and P are the mid points of OA, MB, NC and KD respectively. Show that N, O and P are collinear.

**Sol. :** Let $O = (\vec{o}),\ A(\vec{a}),\ B(\vec{b})$

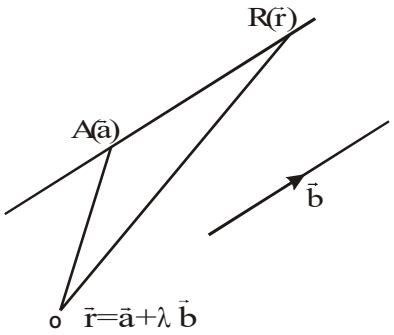
Now $M \equiv \frac{\vec{a}}{2},\ N \equiv \frac{\frac{\vec{a}}{2} + \vec{b}}{2} = \frac{\vec{a} + 2\vec{b}}{4}$

$$K \equiv \frac{\frac{\vec{a} + \vec{b}}{4} - \vec{a}}{2} = \frac{2\vec{b} - 3\vec{a}}{8} \qquad P \equiv \frac{-\vec{b} + \frac{2\vec{b} - 3\vec{a}}{8}}{2} = \frac{-6\vec{b} - 3\vec{a}}{16}$$

$\Rightarrow \quad \overrightarrow{OP} = -\frac{3}{16}(2\vec{b}+\vec{a})$ Also, $\overrightarrow{ON} = \frac{1}{4}(\vec{a}+2\vec{b}) = -\frac{1}{6}\left(\overrightarrow{OP}\right)$

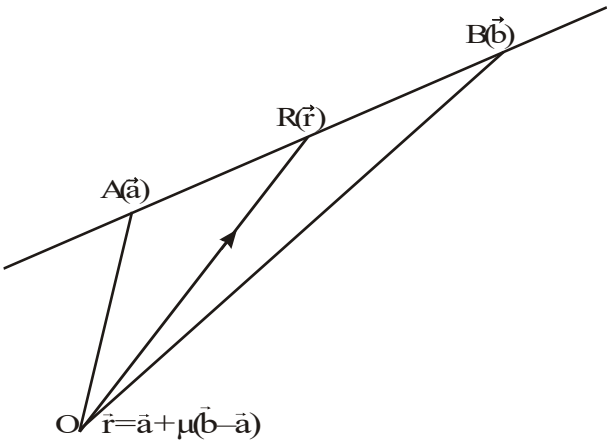
Hence points N, O and P are collinear.

## 2.1 VECTOR EQUATION OF LINE



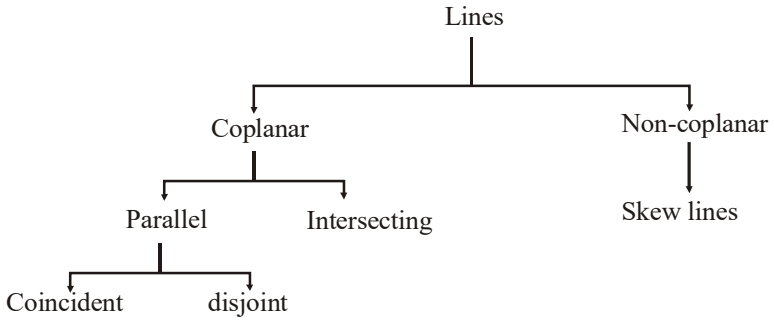

If the line passes through the point $A(\vec{a})$ & is parallel to the vector $\vec{b}$ then its equation is $\vec{r} = \vec{a} + t\vec{b}$.

$\vec{r}$ = (position vector of fixed point on line) + (Parameter) [direction vector]




Vector equation of a line passing through two points $A(\vec{a})$ & $B(\vec{b})$ is given by, $\vec{r} = \vec{a} + t(\vec{b} - \vec{a})$ where t is a parameter

## 2.2 CLASSIFICATION OF LINES




**Q.1** Determine nature of lines $\vec{r} = \hat{i} - \hat{j} - 10\hat{k} + \lambda(2\hat{i} - 3\hat{j} + 8\hat{k})$; $\vec{r} = 4\hat{i} - 3\hat{j} - \hat{k} + \mu(\hat{i} - 4\hat{j} + 7\hat{k})$

Ans. coplanar

**Sol. :** $L_1 : \vec{r} = (2\lambda + 1)\hat{i} - (1 + 3\lambda)\hat{j} + (8\lambda - 10)\hat{k}$

$L_2 : \vec{r} = (4+\mu)\hat{i} - (4\mu+3)\hat{j} + (7\mu-1)\hat{k}$

The given lines are not parallel. For coplanarity, the lines must intersect.

$\therefore \quad (2\lambda+1)\hat{i} - (1+3\lambda)\hat{j} + (8\lambda-10)\hat{k} = (4+\mu)\hat{i} - (4\mu+3)\hat{j} + (7\mu-1)\hat{k}$

$2\lambda + 1 = 4 + \mu$ .....(i)

$1 + 3\lambda = 4\mu + 3$ .....(ii)

$8\lambda - 10 = 7\mu - 1$ .....(iii)

Solving (i) & (ii), $\lambda = 2, \mu = 1$

and $\lambda = 2, \mu = 1$ satisfies equation (iii)

Point of Intersection(5, -7, 6)

Given lines are intersecting & hence coplanar.

**Q.2** Determine nature of lines $\vec{r} = \hat{i} - \hat{j} + 10\hat{k} + \lambda(2\hat{i} - 3\hat{j} + 8\hat{k})$ ; $\vec{r} = 4\hat{i} - 3\hat{j} - \hat{k} + \mu(\hat{i} - 4\hat{j} + 7\hat{k})$

Ans skew

**Sol. :** $L_1 : \vec{r} = (2\lambda+1)\hat{i} - (1+3\lambda)\hat{j} + (8\lambda+10)\hat{k}$

$L_2 : \vec{r} = (4+\mu)\hat{i} - (4\mu+3)\hat{j} + (7\mu-1)\hat{k}$

The given lines are not parallel. For coplanarity, the lines must intersect.

$\therefore \quad (2\lambda+1)\hat{i} - (1+3\lambda)\hat{j} + (8\lambda+10)\hat{k} = (4+\mu)\hat{i} - (4\mu+3)\hat{j} + (7\mu-1)\hat{k}$

$2\lambda + 1 = 4 + \mu$ .....(i)

$1 + 3\lambda = 4\mu + 3$ .....(ii)

$8\lambda + 10 = 7\mu - 1$ .....(iii)

Solving (i) & (ii), $\lambda = 2, \mu = 1$

and $\lambda = 2, \mu = 1$ does not satisfy equation (iii)

So lines are skew

Q.3 $\vec{r} = (0, 0, 0) + \lambda(3, -1, 2)$; $\vec{r} = (2, 0, 1) + \mu(6, -2, 4)$

Ans. Parallel lines

Sol. ratio of direction

$$\frac{3}{6} = \frac{-1}{2} = \frac{2}{4}$$

so lines are parallel

now solving for point of interection

$3\lambda = 6\mu + 2$ .....(i)

$-\lambda = -2\mu$ .....(ii)

& $2\lambda = 1 + 4\mu$ .....(iii)

from (i) & (ii) $6\mu = 6\mu + 2$

$\Rightarrow 0 = 2$

So there isn't any point of intersection

Q.4 $\vec{r} = (0, 0, 1) + \lambda(3, 2, 1)$; $\vec{r} = (3, 2, 2) + \mu(6, 4, 2)$

Ans. Concidental lines

Sol. ratio of direction

$$\frac{3}{6} = \frac{2}{4} = \frac{1}{2}$$

so lines are parallel

now solving for point of intersection

$3\lambda = 3 + 6\mu$ .....(i)

$2\lambda = 2 + 4\mu$ .....(ii)

$1 + \lambda = 2 + 2\mu$ .....(iii)

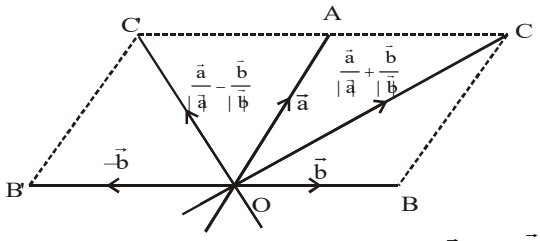
well the above equations gives some result

$\lambda = 1 + 2\mu$

so infinite point of intersection. Lines are coincidental.

## 2.3 VECTOR EQUATION OF ANGLE BISECTOR :

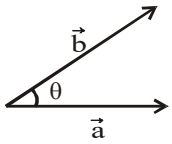
A vector in the direction of the bisector of the angle between the two vectors $\vec{a}$ & $\vec{b}$ is $\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$. Hence bisector of the angle between the two vectors $\vec{a}$ & $\vec{b}$ is $\lambda(\hat{a} + \hat{b})$, where $\lambda \in R^+$. Bisector of the exterior angle between $\vec{a}$ & $\vec{b}$ is $\lambda(\hat{a} - \hat{b})$, $\lambda \in R^+$

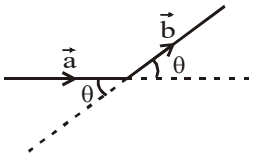



A vector in the direction of the bisector of the angle between the two vectors $\vec{a}$ and $\vec{b}$ is $\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$. Hence bisector of the angle between the two vectors $\vec{a}$ and $\vec{b}$ is $\lambda\left(\hat{a} + \hat{b}\right)$, where $\lambda \in R^+$

## 3.1 DOT PRODUCT (SCALAR PRODUCT)

**Definition :** Let $\vec{a}$ and $\vec{b}$ be two non zero vectors inclined at an angle $\theta$. Then the scalar product of $\vec{a}$ with $\vec{b}$ is denoted by $\vec{a} \cdot \vec{b}$ and is defined as $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$ ; $0 \le \theta \le \pi$.




$\vec{a}.\vec{b} = |\vec{a}||\vec{b}|\cos\theta\ (0 \le \theta \le \pi)$ Note that if $\theta$ is acute then $\vec{a}.\vec{b} > 0$ & if $\theta$ is obtuse then $\vec{a}.\vec{b} < 0$

**Properties**

1. $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
2. $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$ **(distributive)**
3. $(m\vec{a}) \cdot \vec{b} = \vec{a} \cdot (m\vec{b}) = m(\vec{a} \cdot \vec{b})$, where m is a scalar.
4. $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$; $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$
5. $\vec{a}$ is perpendicular to $\vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$
6. If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, then $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$

7. If $\vec{a}=\vec{b} \quad \Rightarrow \quad \vec{a}\,.\,\vec{c}=\vec{b}\,.\,\vec{c}$

8. $\vec{a}\,.\,\vec{a}=|\vec{a}|^2=\vec{a}^2$

$|\vec{a}+\vec{b}|^2=(\vec{a}+\vec{b}).(\vec{a}+\vec{b})=\vec{a}\,.\,\vec{a}+\vec{a}\,.\,\vec{b}+\vec{b}\,.\,\vec{a}+\vec{b}\,.\,\vec{b}$

$|\vec{a}+\vec{b}|^2=a^2+b^2+2\,\vec{a}\,.\,\vec{b}$

$(\vec{a}-\vec{b})^2=a^2+b^2-2\,\vec{a}\,.\,\vec{b}\,; \qquad (\vec{a}+\vec{b}).(\vec{a}-\vec{b})=a^2+b^2$

$(\vec{a}+\vec{b}+\vec{c})^2=a^2+b^2+c^2+2(\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a})\,; \quad (3\vec{a}-5\vec{b})^2=9\,a^2+25\,b^2-30\,\vec{a}.\vec{b}$

9. $|\vec{a}|=\sqrt{\vec{a}\,.\,\vec{a}}$

10. Any vector $\vec{a}$ can be written as $\vec{a}=(\vec{a}\,.\,\hat{i})\,\hat{i}+(\vec{a}\,.\,\hat{j})\,\hat{j}+(\vec{a}\,.\,\hat{k})\,\hat{k}$.

**Questions**

Q.1 $|\vec{a}|=4,\ |\vec{b}|=3,\ \vec{a}\wedge\vec{b}=120^\circ$. Find (i) $|\vec{a}+\vec{b}|$ (ii) $|\vec{a}-\vec{b}|$ (ii) $|3\vec{a}+2\vec{b}|$

Ans. (i) $\sqrt{13}$, (ii) $\sqrt{37}$, (ii) $\sqrt{108}$

Q.2 If $\theta=\vec{a}\wedge\vec{b}$ then prove that (i) $|\hat{a}+\hat{b}|=2\cos\dfrac{\theta}{2}$. (ii) $|\hat{a}-\hat{b}|=2\sin\dfrac{\theta}{2}$

Q.3 If $|\vec{a}|=2,\ |\vec{b}|=2,\ |\vec{c}|=1$ and $\vec{a}+\vec{b}+\vec{c}=0$ Find (i) $\vec{a}\,.\,\vec{b}+\vec{b}\,.\,\vec{c}+\vec{c}\,.\,\vec{a}$ (ii) $\vec{a}\,.\,\vec{b}$

Ans. (i) $-\dfrac{9}{2}$, (ii) $-\dfrac{7}{2}$

Q.4 If $|\vec{a}|=11,\ |\vec{b}|=23,\ |\vec{a}-\vec{b}|=30$ find $|\vec{a}+\vec{b}|$.

Ans. 20

Q.5 Prove that : $\left(|\vec{a}+\vec{b}|^2+|\vec{a}-\vec{b}|^2\right)=2\left(|\vec{a}|^2+|\vec{b}|^2\right)$

Q.6 If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular unit vectors, find $|2\vec{a}+3\vec{b}-6\vec{c}|$.

Ans. 7

## 3.2 APPLICATION OF DOT PRODUCT

### 1. Angle between two vectors

(i) If $\vec{a}$ and $\vec{b}$ be two vectors and $\theta$ be the angle between them, then

$$\cos\theta=\frac{\vec{a}.\vec{b}}{|\vec{a}|\,|\vec{b}|}=\frac{\vec{a}}{\vec{a}}\cdot\frac{\vec{b}}{\vec{b}}=\hat{a}\cdot\hat{b}$$

(ii) If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ then

$$\cos\theta = \frac{a_1b_1+a_2b_2+a_3b_3}{\sqrt{a_1^2+a_2^2+a_3^2}\sqrt{b_1^2+b_2^2+b_3^2}}$$

**Note :**

If $\vec{a}$ and $\vec{b}$ are perpendicular to each other then $a_1\ b_1 + a_2\ b_2 + a_3\ b_3 = 0$

**Ex.1** Find the angle between the vectors $4\hat{i} + \hat{j} + 3\hat{k}$ and $2\hat{i} + 2\hat{j} - \hat{k}$.

Ans. $\cos^{-1}\left(\frac{7}{3\sqrt{26}}\right)$

**Sol.** Let the required angle is $\theta$.

$$\therefore \theta = \cos^{-1}\left(\frac{4.2+1.2+3(-1)}{\sqrt{16+1+9}\sqrt{4+4+1}}\right)$$

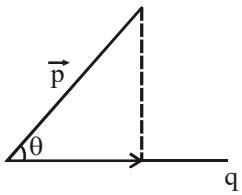
$$= \cos^{-1}\left(\frac{7}{3\sqrt{26}}\right)$$

**2. Angle between lines :**

Line $L_1$ $\quad \vec{r} = \vec{a} + \lambda\vec{p}$ ; $\quad$ Line $L_2$ $\quad \vec{r} = \vec{b} + \mu\vec{p}$

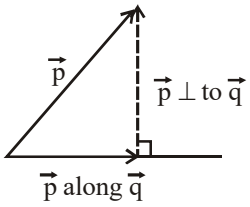
$$\cos\theta = \left|\frac{\vec{p}.\vec{q}}{|\vec{p}||\vec{q}|}\right|$$

**3. Projection of one vector along another vector**




Projection of $\vec{p}$ along $\vec{q} = |\vec{p}|\cos\theta = |\vec{p}|\frac{\vec{p}.\vec{q}}{|\vec{p}||\vec{q}|} = \frac{\vec{p}.\vec{q}}{|\vec{q}|}$

**4. Vector component of one vector along or perpendicular to another vector**




$\vec{p}_{\text{along } \vec{q}} = (\text{projection of } \vec{p} \text{ along } \vec{q})\,\hat{q}$

$$\vec{p}_{\text{along } \vec{q}} = \left(\frac{\vec{p}.\vec{q}}{|\vec{q}|}\right)\left(\frac{\vec{q}}{|\vec{q}|}\right)$$

$$\vec{p}_{\text{along } \vec{q}} + \vec{p}_{\text{ perpendicular to } \vec{q}} = \vec{p}$$

$$\vec{p}_{\text{ perpendicular to } \vec{q}} = \vec{p} - \vec{p}_{\text{along } \vec{q}}$$

Q.1 $\vec{a} = (2,-3,6)$ $\vec{b} = (2,5,-14)$ find ratio of projection of $\vec{a}$ on $\vec{b}$ to projection of $\vec{b}$ on $\vec{a}$

Ans. $\frac{7}{15}$

Sol. $\frac{\text{proj}_{\vec{b}}^{\vec{a}}}{\text{proj}_{\vec{a}}^{\vec{b}}} = \frac{\vec{a}.\hat{b}}{\vec{b}.\hat{a}} = \frac{|\vec{a}|}{|\vec{b}|} = \frac{7}{15}$

Q.2 $\vec{p} = 5\hat{i} - 2\hat{j} + 5\hat{k}$ $\vec{q} = 3\hat{i} + \hat{k}$ Find vector component of $\vec{p}$ along $\vec{q}$ and perpendicular to $\vec{q}$.

Ans. Along $2\vec{q}$ and perpendicular $(-1, -2, 3)$

Sol. vector component of $\vec{p}$ along $\vec{q}$

$= (\vec{p}.\vec{q})\vec{q}$

$= \frac{(\vec{p}.\vec{q})\vec{q}}{|\vec{q}|^2}$

$= \frac{20\vec{q}}{10} = 2\vec{q}$

vector component perpendicular to $\vec{q}$

$= \vec{p} - (\text{vector component of } \vec{p} \text{ along } \vec{q})$

$= \vec{p} - 2\vec{q}$

$= (5, -2, 5) - (6, 0, 2)$

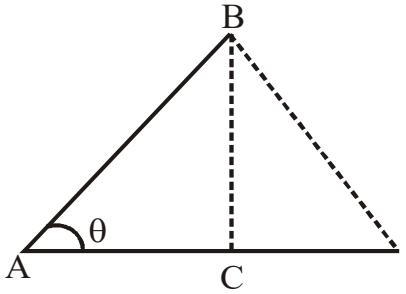
$= (-1, -2, 3)$

Q.3 In a triangle ABC. If $|\overrightarrow{BC}| = 8, |\overrightarrow{CA}| = 7, |\overrightarrow{AB}| = 10$, then the projection of the vector $\overrightarrow{AB}$ on $\overrightarrow{AC}$ is equal to:

एक त्रिभुज ABC में यदि $|\overrightarrow{BC}| = 8, |\overrightarrow{CA}| = 7, |\overrightarrow{AB}| = 10$ है, तो सदिश $\overrightarrow{AB}$ का सदिश $\overrightarrow{AC}$ पर प्रक्षेप बराबर है :

**Ans.** $\frac{85}{14}$

**Sol.**



$\left|\overrightarrow{BC}\right| = 8 \quad \left|\overrightarrow{CA}\right| = 7 \quad \left|\overrightarrow{AB}\right| = 10$

Projection $= \left|\overrightarrow{AB}\right| \cos\theta$

Projection $= 10\cos\theta$

$$\cos = \frac{\left|\overrightarrow{AB}\right|^2 + \left|\overrightarrow{CA}\right|^2 - \left|\overrightarrow{BC}\right|^2}{2 \times \left|\overrightarrow{AB}\right|\left|\overrightarrow{CA}\right|}$$

$$= \frac{100 + 49 - 64}{2 \times 10 \times 7} = \frac{85}{10 \times 2 \times 7}$$

$$\text{Projection} = 10 \times \frac{85}{10 \times 2 \times 7} = \frac{85}{14}$$

## 3.3 IMAGE AND FOOT OF PERDICULAR OF A POINT ON A LINE

**Q.1** Find the foot of perpendicular and length of the perpendicular from P (2, – 3, 1) to the line $\vec{r} = (-1,3,-2) + \lambda(2,3,-1)$.

Ans. $Q \equiv \left(\frac{-22}{7}, \frac{-3}{14}, \frac{-13}{14}\right) \quad \therefore \quad PQ = \sqrt{\frac{531}{14}}$ units.

**Sol. :** Given line is $\vec{r} = (-1,3,-2) + \lambda(2,3,-1)$ ..... (1)

$P \equiv (2, -3, 1)$

Co-ordinates of any point on line (1) may be taken as $Q \equiv (2\lambda - 1, 3\lambda + 3, -\lambda - 2)$

Direction ratios of PQ are $2\lambda - 3, 3\lambda + 6, -\lambda - 3$

Direction ratios of AB are 2, 3, – 1

Since PQ $\perp$ AB

$\therefore \quad 2(2\lambda - 3) + 3(3\lambda + 6) - 1(-\lambda - 3) = 0$

or, $\quad 14\lambda + 15 = 0 \quad \therefore \quad \lambda = \frac{-15}{14}$

$\therefore \quad Q \equiv \left(\frac{-22}{7}, \frac{-3}{14}, \frac{-13}{14}\right) \quad \therefore \quad PQ = \sqrt{\frac{531}{14}}$ units.

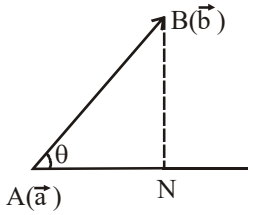
Q.2 Find its image also

Ans $\left(-\frac{58}{7}, \frac{18}{7}, -\frac{20}{7}\right)$

P Q P'

Q is the mid point of P and P'

## 3.4 DISTANCE OF A POINT FROM A LINE




Method - I

$\vec{r} = \vec{a} + \lambda\vec{p}$

Get F.O.P. N

Req. distance = BN

Method - II

AN = Projection of $\overrightarrow{AB}$ along $\vec{P}$

Required distance = BN

$$= \sqrt{(AB)^2 - (AN)^2}$$

Method - III

$\overrightarrow{NB}$ = vector component of $\overrightarrow{AB}$ perpendicular to $\vec{P}$

Required distance = NB

Method - IV

Required distance = $|\overrightarrow{AB}|\sin\theta$

$$\cos\theta = \frac{\overrightarrow{AB}.\vec{p}}{|\overrightarrow{AB}||\vec{p}|}$$

Method - V

$|\overrightarrow{AB}\times\vec{p}| = |\overrightarrow{AB}||\vec{p}|\sin\theta$

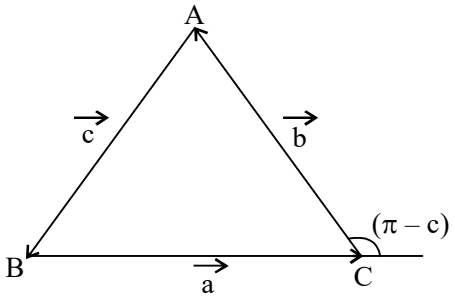
Required distance = $|\overrightarrow{AB}|\sin\theta$

$$\text{Required distance} = \frac{|\vec{AB}\times\vec{p}|}{|\vec{p}|}$$

## 3.5 PRACTICE QUESTIONS ON DOT PRODUCT

1. Prove Cosine formula for triangles.

Sol.



$\vec{a}+\vec{b}+\vec{c}=0$

$\Rightarrow \vec{a}+\vec{b}=-\vec{c}$

squaring

$a^2+b^2+2\vec{a}.\vec{b}=c^2$

$\Rightarrow a^2+b^2+2ab\cos(\pi-c)=c^2$

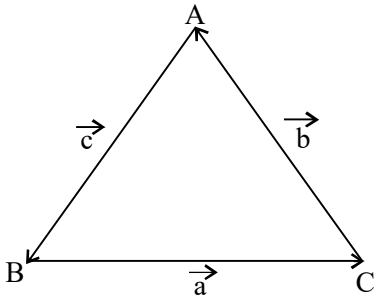
$\Rightarrow a^2+b^2-2ab\cos c=c^2$

$\Rightarrow \cos c=\dfrac{a^2+b^2-c^2}{2ab}$

similar approach can be used for cosA and cosB.

2. Prove Projection formula for triangles.

Sol.




$\vec{a}+\vec{b}+\vec{c}=0$

$\Rightarrow \vec{a}+\vec{b}=-\vec{c}$

dot with $\vec{c}$

$\Rightarrow \vec{a}.\vec{c}+\vec{b}.\vec{c}=-\vec{c}^2$

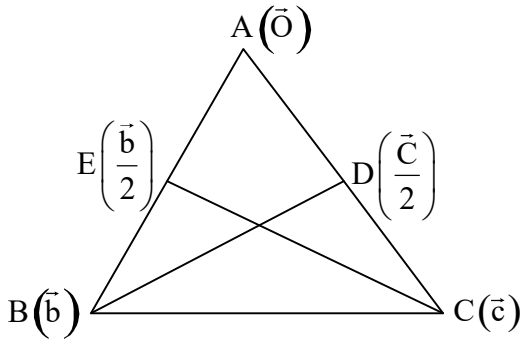
$\Rightarrow ac\cos(\pi-B)+bc\cos(\pi-A)=-c^2$

$\Rightarrow -ac\cos B-bc\cos A=-c^2$

$\Rightarrow a\cos B+b\cos A=c$

3. Prove that medians to the equal sides of an isosceles triangle are equal and converse.

Sol.




$|\vec{b}|=|\vec{c}|$

D, E are mid points of AC and AB respectively.

$$\left|\overrightarrow{BD}\right| = \left|\frac{\vec{c}}{2} - \vec{b}\right| = \sqrt{\frac{c^2}{4} + b^2 - 2\frac{\vec{c}}{2}.\vec{b}}$$

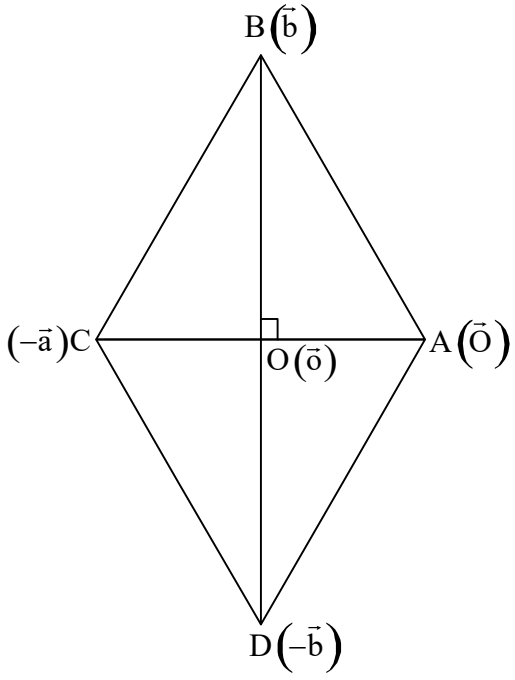
$$\left|\overrightarrow{CE}\right| = \left|\frac{\vec{b}}{2} - \vec{c}\right| = \sqrt{\frac{b^2}{4} + c^2 - 2\frac{\vec{b}}{2}.\vec{c}}$$

as b = c

so $\left|\overrightarrow{BD}\right| = \left|\overrightarrow{CE}\right|$

4. Use vectors to prove that a quadrilateral whose diagonals bisect each other at right angle is a rhombus.

Sol.

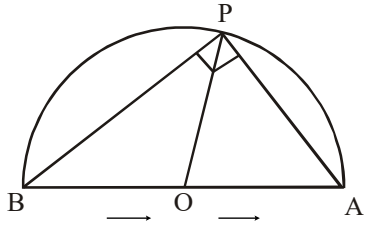



$$\left|\overrightarrow{AB}\right| = \left|\overrightarrow{BC}\right| = \left|\overrightarrow{CD}\right| = \left|\overrightarrow{DA}\right| = \sqrt{a^2 + b^2}$$

5. Use vectors to prove that angle in a semicircle is right angle.

Ans.

Sol Let O be the centre and AB the bounding diameter of the semi-circle. Let P be any point on the circumference. With O as origin.




Let $\overrightarrow{OA} = a$, $\overrightarrow{OB} = -a$ and $\overrightarrow{OP} = r$

Obviously OA = OB = OP, each being equal to radius of the semi-circle.

$\overrightarrow{AP} = r - a$ and $\overrightarrow{BP} = r - (-a) = r + a$

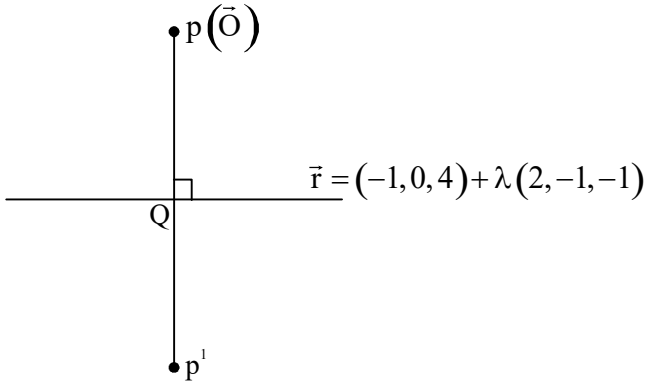
$\therefore \quad \overrightarrow{AP}.\overrightarrow{BP} = (r-a).(r+a) = r^2 - a^2 = OP^2 - OA^2 = 0$

$\Rightarrow$ AP and BP are perpendicular to each other, i.e., $\angle APB = 90^0$.

6. Find the foot of the perpendicular from the origin on the line $\vec{r} = -\hat{i} + 4\hat{k} + \lambda\left(2\hat{i} - \hat{j} - \hat{k}\right)$. Also find the position vector of its image in the line.

Ans. fop(1, –1, 3 ), image (2, –2, 6)

Sol.




Let $Q \equiv (-1+2\lambda, -\lambda, 4-\lambda)$

So $\overrightarrow{PQ} \equiv (-1+2\lambda, -\lambda, 4-\lambda)$

Now $\overrightarrow{PQ}.(2,-1,-1) = 0$

$\Rightarrow 2(-1+2\lambda) - 1(-\lambda) - 1(4-\lambda) = 0$

$\Rightarrow -2 + 4\lambda + \lambda - 4 + \lambda = 0$

$\Rightarrow \lambda = 1$

So co-ordinates of Q (1, –1, 3)

let co-ordinates of $p^1$ (x, y, z)

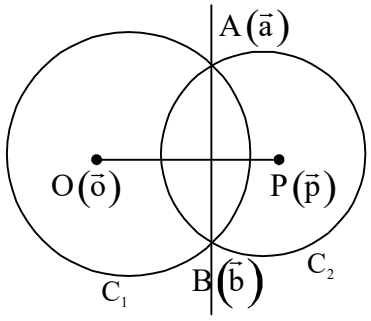
Q is the mid point of p and $p^1$

$\Rightarrow \frac{x}{2} = 1, \frac{y}{2} = -1, \frac{z}{2} = 3$

$\Rightarrow (x, y, z) = (2, -2, 6)$

7. If two circles intersect then the line joining their centres is perpendicular to the common chord.

Sol.



To prove $\left(\vec{a}-\vec{b}\right).\vec{p}=0 \Rightarrow \vec{a}.\vec{p}=\vec{b}.\vec{p}$

$|\vec{a}|=|\vec{b}|=r_1$

$|\vec{p}-\vec{a}|=|\vec{p}-\vec{b}|=r_2$

$\Rightarrow |\vec{p}-\vec{a}|^2=|\vec{p}-\vec{b}|^2$

$\Rightarrow p^2+a^2-2\vec{p}.\vec{a}=p^2+b^2-2\vec{p}.\vec{b}$

$\Rightarrow \vec{p}+\vec{a}=\vec{p}.\vec{b}$

8. If $\hat{i}+\hat{j}$ bisects the angle between $\vec{c}$ and $\hat{j}+\hat{k}$, then $\vec{c}.\hat{j}$ is equal to :

(A) 0 (B) $\frac{1}{\sqrt{2}}$ (C) $-\frac{1}{\sqrt{2}}$ (D) 1

Ans. A

Sol. $\hat{c}+\frac{\hat{j}+\hat{k}}{\sqrt{2}}=\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

$\Rightarrow \hat{c}=\frac{\hat{i}-\hat{k}}{\sqrt{2}}$

$\vec{c}.\hat{j}=\frac{\hat{c}.\hat{j}}{|\vec{c}|}=\frac{1}{|\vec{c}|}\frac{1}{\sqrt{2}}\left(\hat{i}-\hat{k}\right).\hat{j}=0$

## 4. LINEAR COMBINATION THEOREM IN A PLANE

$\vec{a}$ and $\vec{b}$ are non-colinear vectors another vector $\vec{r}$ which is coplanar with $\vec{a}$ and $\vec{b}$ can be written as linear combination of $\vec{a}$ and $\vec{b}$. $\mathbf{x\vec{a}+y\vec{b}=\vec{r}}$.

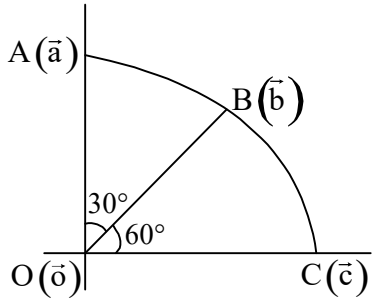
**Note**

Vector in plane of $\vec{a}$ and $\vec{b}$ will be taken as $x\vec{a}+y\vec{b}$

Q.1 Arc AC of the quadrant of a circle with centre as origin and radius unity subtends a right angle at the origin. Point B divides the arc AC in the ratio 1 : 2. Express the vector $\vec{c}$ in terms of $\vec{a}$ and $\vec{b}$.

Ans. $-\sqrt{3}\vec{a}+2\vec{b}$

Sol.



$|\vec{a}| = |\vec{b}| = |\vec{c}| = r$

Let $\vec{c} = x\vec{a} + y\vec{b}$ .....(1)

dot with $\vec{a}$

$\vec{a}.\vec{c} = xa^2 + y(\vec{a}.\vec{b})$

$\Rightarrow O = x.r^2 + y.r^2.\frac{\sqrt{3}}{2}$

$\Rightarrow x + \frac{\sqrt{3}}{2}y = 0$ .....(ii)

dot with $\vec{c}$ in (i)

$c^2 = x(\vec{a}.\vec{c}) + y(\vec{b}.\vec{c})$

$\Rightarrow r^2 = 0 + y.r^2.\frac{1}{2}$

$\Rightarrow y = 2$

From equation (ii) $x = -\sqrt{3}$

So $\vec{c} = -\sqrt{3}\vec{a} + 2\vec{b}$

Q.2 Given that $\vec{a} = \hat{i} - \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j}$ are two vectors. Find a unit vector coplanar with $\vec{a}$ and $\vec{b}$ and perpendicular to $\vec{a}$.

Ans. $\pm\left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

Sol. Let the vector is

$\vec{c} = x\vec{a} + y\vec{b}$ .....(i)

dot with $\vec{a}$

$\vec{a}.\vec{c} = xa^2 + y(\vec{a}.\vec{b})$

$\Rightarrow O = x(2) + y(-1)$

$\Rightarrow 2x - y = 0$ .....(ii)

from (1)

$c^2 = x^2a^2 + y^2b^2 + 2xy(\vec{a}.\vec{b})$

$\Rightarrow 1 = 2x^2 + 5y^2 - 2xy$

putting $y = 2x$

$2x^2 + 20x^2 - 4x^2 = 1$

$\Rightarrow 18x^2 = 1 \Rightarrow x = \pm\frac{1}{3\sqrt{2}}$

so $y = \pm\frac{2}{3\sqrt{2}}$

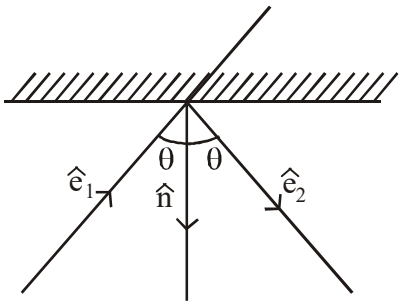
Now $\vec{c} = x\vec{a} + y\vec{b}$

$= \pm\frac{1}{3\sqrt{2}}\left(\vec{a} + 2\vec{b}\right)$

$= \pm\frac{1}{3\sqrt{2}}\left((1,-1) + (2,4)\right)$

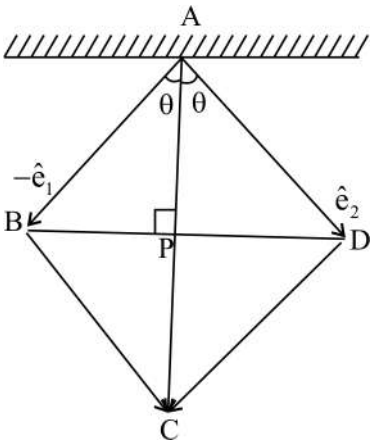
$= \pm\frac{1}{\sqrt{2}}\left(\hat{i} + \hat{j}\right)$

Q.3 Write $\hat{e}_2$ in terms of $\hat{e}_1$ and $\hat{n}$




Ans. $\vec{e}_2 = -\vec{e}_1 + 2\cos\theta\,\hat{n}$

Sol.




ABCD is a rhombers and P is mid point of diagonals

So $\overrightarrow{AP} = \frac{\hat{e}_2 - \hat{e}_1}{2} = |\cos\theta|.\hat{n}$

$\Rightarrow \hat{e}_2 - \hat{e}_1 = (2\cos\theta)\hat{n}$

$\Rightarrow \hat{e}_2 = \hat{e}_1 + (2\cos\theta).\hat{n}$

Q.4 If $\vec{a},\vec{b},\vec{c}$ are coplanar vectors, prove that $\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \end{vmatrix} = 0.$

Ans. Hint put $\vec{c} = x\vec{a} + y\vec{b}$

Sol. $\begin{vmatrix} \vec{a} & \vec{b} & x\vec{a}+y\vec{b} \\ a^2 & \vec{a}.\vec{b} & \vec{a}.\left(x\vec{a}+y\vec{b}\right) \\ \vec{a}.\vec{b} & b^2 & \vec{b}\left(x\vec{a}+y\vec{b}\right) \end{vmatrix}$

Trans forming $c_3 \to c_3 - \left(xc_1 + yc_2\right)$

$= \begin{vmatrix} \vec{a} & \vec{b} & 0 \\ a^2 & \vec{a}.\vec{b} & 0 \\ \vec{a}.\vec{b} & b^2 & 0 \end{vmatrix}$

## 5.1 CROSS PRODUCT

If $\vec{a}$ & $\vec{b}$ are two vectors & $\theta$ is the angle between them, then $\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\hat{n}$, where $\hat{n}$ is the unit vector perpendicular to both $\vec{a}$ & $\vec{b}$ such that $\vec{a}$, $\vec{b}$ & $\vec{n}$ forms a right handed screw system.

**Properties**

(i) $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

(ii) $(m\vec{a})\times\vec{b} = \vec{a} \times (m\vec{b}) = m(\vec{a}\times\vec{b})$, where m is a scalar.

(iii) $\vec{a} \times (\vec{b}+\vec{c}) = (\vec{a}\times\vec{b}) + (\vec{a}\times\vec{c})$ **(distributive)**

(iv) $\vec{a}\times\vec{b} = \vec{0} \Leftrightarrow \vec{a}$ and $\vec{b}$ are parallel (collinear) $(\vec{a} \neq 0 \ , \ \vec{b} \neq 0)$ i.e. $\vec{a} = K\vec{b}$, where K is a scalar.

(v) $\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = \vec{0}$ ; $\hat{i}\times\hat{j} = \hat{k}, \hat{j}\times\hat{k} = \hat{i}, \hat{k}\times\hat{i} = \hat{j}$

(vi) If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, then $\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$

(vii) Vector perpendicular to $\vec{a}$ and $\vec{b}$ will be taken as $\lambda(\vec{a} \times \vec{b})$

(viii) If $\vec{a} = \vec{b}$ then $\vec{a}\times\vec{c} = \vec{b}\times\vec{c}$

(ix) If $\vec{a}\times\vec{c} = \vec{b}\times\vec{c} \Rightarrow \quad \vec{a}\times\vec{c} = \vec{b}\times\vec{c} \Rightarrow \quad (\vec{a}-\vec{b})\times\vec{c} = 0 \Rightarrow \quad \vec{a}-\vec{b} = \lambda c$

Q.1 If $\left(\hat{i}+2\hat{j}-3\hat{k}\right)\times\left(2\hat{i}+\lambda\hat{j}+\mu\hat{k}\right) = 0$, find $\lambda$ and $\mu$

Ans. $\lambda = 4, \mu = -6$

Sol. $\Rightarrow (2\mu+3\lambda)\hat{i}-(\mu+6)\hat{j}+(\lambda-4)\hat{k}=0$

$\Rightarrow \lambda=4, \mu=-6$

Q.2 If $\vec{a}=2\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=6\hat{i}-3\hat{j}+2\hat{k}$ then $\vec{a}\times\vec{b}$ equals-

Ans. $\hat{i}-10\hat{j}-18\hat{k}$

**Sol. :** $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 2 & -1\\ 6 & -3 & 2\end{vmatrix}$

$=\hat{i}\ (4-3)-\hat{j}(4+6)+\hat{k}(-6-12)$

$=\hat{i}-10\hat{j}-18\hat{k}$

Q.3 $\vec{a}=(1,0,1)$, $\vec{b}=(-1,1,0)$, $\vec{c}=(0,1,-1)$ Find $\vec{d}$ such that $\vec{d}.\vec{a}=0$, $\vec{d}.\vec{b}=0$, $\vec{d}.\vec{c}=6$.

Ans. $(3,3,-3)$

Sol. Let $\vec{d}=(p,q,r)$

$\vec{d}.\vec{a}=0\Rightarrow p+r=0$ .....(i)

$\vec{d}.\vec{b}=0\Rightarrow -p+q=0$ .....(ii)

$\vec{d}.\vec{c}=0\Rightarrow q-r=6$ .....(iii)

Solving (i), (ii) and (iii)

(p, q, r) = (3, 3, –3)

Q.4 If $\overrightarrow{OA}\times\overrightarrow{OB}=(1,0,1)$, $\overrightarrow{OB}\times\overrightarrow{OC}=(-1,1,0)$, $\overrightarrow{OC}\times\overrightarrow{OA}=(0,1,2)$. Find $\overrightarrow{CB}\times\overrightarrow{CA}$

Ans. $(0,-2,-3)$

Sol. $\vec{a}\times\vec{b}=(1,0,1)$

$\vec{b}\times\vec{c}=(-1,1,0)$

$\vec{c}\times\vec{a}=(0,1,2)$

$\overrightarrow{CB}\times\overrightarrow{CA}=(\vec{b}-\vec{c})\times(\vec{a}-\vec{c})$

$=\vec{b}\times\vec{a}-\vec{b}\times\vec{c}-\vec{c}\times\vec{a}$

$=-(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a})$

$=-(0,2,3)=(0,-2,-3)$

Q.5 $\vec{a}=(1,0,2)$, $\vec{b}=(0,1,-2)$, $\vec{c}=(1,1,1)$. Find $\vec{d}$ such that $\vec{d}.\vec{a}=0$, $\vec{d}\times\vec{b}=\vec{c}\times\vec{b}$

Ans. $\left(1,\frac{7}{4},-\frac{1}{2}\right)$

Sol. $\vec{d}\times\vec{b}=\vec{c}\times\vec{b}$

$\Rightarrow \vec{d}\times\vec{b}-\vec{c}\times\vec{b}=0$

$\Rightarrow (\vec{d}\times\vec{c})\times\vec{b}=0$

$\Rightarrow \vec{d}-\vec{c}=\lambda\vec{b}$

$\Rightarrow \vec{d}=\lambda\vec{b}+\vec{c}$

$\Rightarrow \vec{d}=(0,\lambda,-2\lambda)+(1,1,1)$

$\Rightarrow \vec{d}=(1,\lambda+1,-2\lambda+1)$

Given $\vec{d}.\vec{a}=0$

$\Rightarrow (1,\lambda+1,-2\lambda+1).(1,0,2)=0$

$\Rightarrow 1+2(-2\lambda+1)=0$

$\Rightarrow \lambda=\frac{3}{4}$

So $\vec{d}=\left(1,\frac{7}{4},\frac{-1}{2}\right)$

Q.6 Find equation of line passing through point of intersection of $L_1$ and $L_2$ and perpendicular to $L_1$ and $L_2$

(i) $L_1\ \vec{r}=\vec{a}+\lambda\vec{b}$ , $L_2\ \vec{r}=\vec{a}+\mu\vec{q}$ (ii) $L_1\ \vec{r}=\vec{a}+\lambda\vec{b}$ ; $L_2\ \vec{r}=\vec{b}+\mu\vec{a}$

Ans. (i) $\vec{r}=\vec{a}+\lambda(\vec{b}\times\vec{q})$ (ii) $\vec{r}=(\vec{a}+\vec{b})+\lambda(\vec{a}\times\vec{b})$

Q.7 Let $\vec{c}$ be a vector perpendicular to the vectors $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$ if $\vec{c}.(\hat{i}+\hat{j}+3\hat{k})=8$ then the value of $\vec{c}.(\vec{a}\times\vec{b})$ is equal to _____.

माना सदिशों $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$ के लम्बवत् एक सदिश $\vec{c}$ है। यदि $\vec{c}.(\hat{i}+\hat{j}+3\hat{k})=8$ है, तो $\vec{c}.(\vec{a}\times\vec{b})$ का मान बराबर है _____।

**Ans.** 28

**Sol.** $\vec{c}=\lambda(\vec{a}\times\vec{b})$

$\vec{a}=i+j-k$

$\vec{b}=i+2j+k$

$$\vec{a}\times\vec{b}=\begin{vmatrix} i & j & k \\ 1 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix}=3i-2j+k$$

$\vec{c}=\lambda(3i-2j+k)$

$\vec{c}\cdot(i+j+3k)=8$

$\lambda(3-2+3)=8 \Rightarrow \lambda=2$

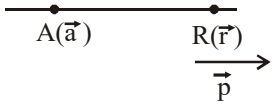
$\vec{c}=2(\vec{a}\times\vec{b})$

$$\vec{c}\cdot\left(\vec{a}\times\vec{b}\right)=2\,|\,\vec{a}\times\vec{b}\,|^2$$

$$=28$$

## 5.2 APPLICATION OF CROSS PRODUCT

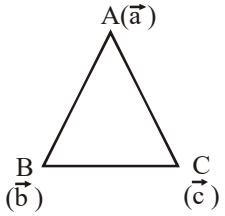
### 1. Equation of line in cross product form




$\overrightarrow{AR}\parallel\vec{p}\Rightarrow \quad A\vec{R}\times\vec{p}=0\Rightarrow(\vec{r}-\vec{a})\times\vec{p}=0$

$\vec{r}\times\vec{p}=\vec{a}\times\vec{p}$

### 2. Area of a triangle




(i) Area of triangle ABC = $\frac{1}{2}|\ \overrightarrow{AB}\ \times\ \overrightarrow{AC}\,|$

$\Delta=\frac{1}{2}$ (cross product of any two side vector)

(ii) If $\vec{a},\vec{b},\vec{c}$ are position vectors of vertices of a $\Delta$ABC then its Area = $\frac{1}{2}\,|(\vec{a}\times\vec{b})+(\vec{b}\times\vec{c})+(\vec{c}\times\vec{a})|$

**Note :**

Three points with position vectors $\vec{a}$, $\vec{b}$ , $\vec{c}$ are collinear if$(\vec{a}\times\vec{b})+(\vec{b}\times\vec{c})+(\vec{c}\times\vec{a})=0$

### 3. Area of a parallelogram

The vector product of the vectors $\vec{a}$ and $\vec{b}$ represents a vector whose modulus is equal to the area of the parallelogram whose two adjacent sides are represented by $\vec{a}$ and $\vec{b}$.

Therefore $|\vec{a}\times b|$ = area of a parallelogram whose adjacent sides are $\vec{a}$ and $\vec{b}$.

$\Delta$ = (cross product of any two adjacent side vector)

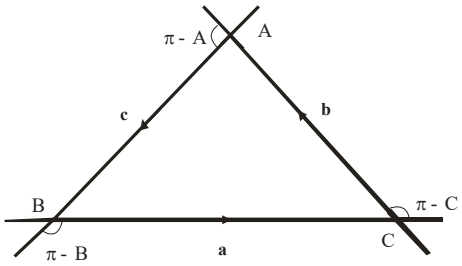
### 4. Area of a Quadrilateral

Area of a quadrilateral ABCD = $\frac{1}{2}|\,\overrightarrow{AC}\ \times\ \overrightarrow{BD}\,|$

$\Delta=\frac{1}{2}$(cross product of diagonal vectors)

Q.1 Use vectors to prove, sine rule (i.e. $\frac{\sin A}{a}=\frac{\sin B}{b}=\frac{\sin C}{c}$)

Sol Let $\overrightarrow{BC}$, $\overrightarrow{CA}$, $\overrightarrow{AB}$ represent the vectors a, b, c respectively.




Then, we have

$a + b + c = 0,$

$\Rightarrow \quad c = -(a + b)$

$\Rightarrow \quad b \times c = b \times (-a - b)$

$= -b \times a = a \times b$

Similarly, $c \times a = a \times b$

Hence, $b \times c = c \times a = a \times b$

$\Rightarrow \quad bc\sin(\pi - A) = ca\sin(\pi - B) = ab\sin(\pi - C) \quad \Rightarrow \quad bc\sin A = ca\sin B = ab\sin C$

$\Rightarrow \quad \dfrac{\sin A}{a} = \dfrac{\sin B}{b} = \dfrac{\sin C}{c}$

Q.2 Prove that 3 points with position vectors $\vec{a}+\vec{b}$, $\vec{a}-\vec{b}$ and $\vec{a}+\lambda\vec{b}$ are collinear for $\forall\ \lambda \in R$.

Sol. Let $\vec{p} = \vec{a}+\vec{b}, \vec{q} = \vec{a}-\vec{b}, \vec{r} = \vec{a}+\lambda\vec{b}$

Solving $\vec{p}\times\vec{q} + \vec{q}\times\vec{r} + \vec{r}\times\vec{p}$

$= \left(\vec{a}+\vec{b}\right)\times\left(\vec{a}-\vec{b}\right) + \left(\vec{a}-\vec{b}\right)\times\left(\vec{a}+\lambda\vec{b}\right) + \left(\vec{a}+\lambda\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)$

$= -\vec{a}\times\vec{b} + \vec{b}\times\vec{a} + \lambda\left(\vec{a}\times\vec{b}\right) - \vec{b}\times\vec{a} + \vec{a}\times\vec{b} + \lambda\left(\vec{b}\times\vec{a}\right)$

$= 0$

So $\vec{p}, \vec{q}$ and $\vec{r}$ are collivear

Q.3 For a non zero vector $\vec{a}$, if $\vec{a}.\vec{b} = \vec{a}.\vec{c}$ and $\vec{a}\times\vec{b} = \vec{a}\times\vec{c}$. Prove that $\vec{b} = \vec{c}$.

Sol. $\vec{a}.\vec{b} = \vec{a}.\vec{c}$

$\Rightarrow \vec{a}.\vec{b} - \vec{a}.\vec{c} = 0$

$\Rightarrow \vec{a}\left(.\vec{b} - \vec{c}\right) = 0 \quad \Rightarrow \vec{a} \perp \left(\vec{b} - \vec{c}\right)$

and $\vec{a}\times\vec{b} = \vec{a}\times\vec{c}$

$\Rightarrow \vec{a}\times\vec{b} - \vec{a}\times\vec{c} = 0$

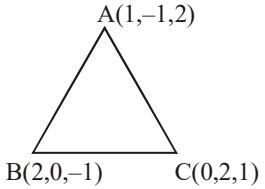
$\Rightarrow \vec{a}\times\left(\vec{b} - \vec{c}\right) = 0 \quad \Rightarrow \vec{a}11\left(\vec{b} - \vec{c}\right)$

Only possible is either

$\vec{a} = 0$ or $\vec{b} = \vec{c}$

is $\vec{a} \neq 0$, So $\vec{b} = \vec{c}$

Q.4 For the given figure find




(a) A vector of magnitude $\sqrt{6}$ perpendicular to the plane ABC

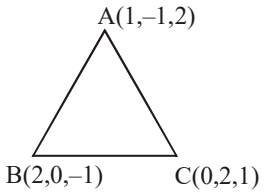
(b) Area of triangle ABC

(c) Length of the altitude from A

(d) Equation of the plane ABC

Ans. (a) $\pm(2,1,1)$ , (b) $2\sqrt{6}$ , (c) $2\sqrt{2}$ (d) 3

Sol.




$\overrightarrow{AB} = (1,1,-3)$

$\overrightarrow{AC} = (-1,3,-1)$

$\overrightarrow{AB} \times \overrightarrow{AC} = (8,4,4)$

(a) unit normal vector $= \pm\left(\dfrac{2,1,1}{\sqrt{6}}\right)$

(b) Area of $\Delta ABC = \dfrac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right|$

$= \dfrac{1}{2}.4\sqrt{6} = 2\sqrt{6}$

(c) $\overrightarrow{BC} = (-2,2,2)$

$\left|\overrightarrow{BC}\right| = 2\sqrt{3}$

Area fo triangle $= \dfrac{1}{2}.\text{base} \times \text{height} = 2\sqrt{6}$

$\Rightarrow \dfrac{1}{2}.2\sqrt{3} \times h = 2\sqrt{6}$

$\Rightarrow h = 2\sqrt{2}$

(d) Equation of the plane ABC

$2(x-1) + (y+1) + (z-2) = 0$

$\Rightarrow 2x + y + z = 3$

Q.5 If $\vec{d}_1 = 2\hat{i} + 3\hat{j} - 6\hat{k}$ and $\vec{d}_2 = 3\hat{i} - 4\hat{j} - \hat{k}$, then find area of parallelogram.

Ans. $\dfrac{1}{2}\sqrt{1274}$

Sol. $\bar{d}_1 = (2,3,-6), \bar{d}_2 (3,-4,-1)$

$\bar{d}_1 \times \bar{d}_2 = (-27,-16,-17)$

Area of parallelogram $= \dfrac{1}{2}\left|\bar{d}_1 \times \bar{d}_2\right|$

$= \dfrac{1}{2}\sqrt{729+256+289}$

$$= \frac{1}{2}\sqrt{1274}$$

Q.6 If A,B,C,D are any four points, then $|\overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD}|$ equals-

(A) Area of $\Delta$ ABC (B) 2(Area of $\Delta$ABC) (C) 3(Area of $\Delta$ABC) (D) 4 (Area of $\Delta$ABC)

Ans. D

**Sol.** Let a, b, c and d be position vectors of points A,B,C and D respectively, then

$\overrightarrow{AB} \times \overrightarrow{CD} = (\vec{b} - \vec{a}) \times (\vec{d} - \vec{c})$

$= \vec{b} \times \vec{d} - \vec{b} \times \vec{c} - \vec{a} \times \vec{d} + \vec{a} \times \vec{c}$

Similarly

$\overrightarrow{BC} \times \overrightarrow{AD}$

$= \vec{c} \times \vec{d} - \vec{c} \times \vec{a} - \vec{b} \times \vec{d} + \vec{b} \times \vec{a}$

$\overrightarrow{CA} \times \overrightarrow{BD}$

$= \vec{a} \times \vec{d} - \vec{a} \times \vec{b} - \vec{c} \times \vec{d} + \vec{c} \times \vec{b}$

Therefore given expression

$= |2\ (\vec{b} \times \vec{a} - \vec{b} \times \vec{c} + \vec{a} \times \vec{c})|$

$= 2\ |(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})|$

= 4 (Area of $\Delta$ ABC)

Q.7 Prove the identity $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = 2(\vec{a} \times \vec{b})$ and give its geometrical interpretation.

Sol. $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = \vec{a} \times \vec{a} + \vec{a} \times \vec{b} - \vec{b} \times \vec{a} - \vec{b} \times \vec{b}$

$= 2(\vec{a} \times \vec{b})$

Geometrical interpretation :-

If $\vec{a}, \vec{b}$ are adjacent sides of a parallelogeam then $\vec{a} + \vec{b}, \vec{a} - \vec{b}$ represents diagonals of it.

$\left|(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})\right| = 2.$ (area of parallelogram)

= RHS

Q.8 Let $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = 10\vec{a} + 2\vec{b}$ and $\overrightarrow{OC} = \vec{b}$, where O, A and C are non collinear points. Let 'p' denote the area of the quadrilateral OABC, and let 'q' denote the area of the parallelogram with OA and OC as adjacent sides. If p = kq. Find k.

**Ans.** 6

**Sol** p = Area of the quadrilateral OABC $\Rightarrow$ $p = \frac{1}{2}\ |\overrightarrow{OB} \times \overrightarrow{AC}| = \frac{1}{2}\ |\overrightarrow{OB} \times (\overrightarrow{OC} - \overrightarrow{OA})|$

$\Rightarrow p = \frac{1}{2}\ |(10\vec{a} + 2\vec{b}) \times (\vec{b} - \vec{a})| = \frac{1}{2}\ |10(\vec{a} \times \vec{b}) - 10(\vec{a} \times \vec{a}) + 2(\vec{b} \times \vec{b}) - 2(\vec{b} \times \vec{a})|$

$\Rightarrow p = \frac{1}{2}\ |10(\vec{a} \times \vec{b}) - 0 + 0 + 2(\vec{a} \times \vec{b})| = 6\ |\vec{a} \times \vec{b}|$ .....(i)

and q = Area of the parallelogram with OA and OC as adjacent sides

$\Rightarrow \quad q = |\overrightarrow{OA} \times \overrightarrow{OC}| = |\vec{a} \times \vec{b}|$ .....(ii)

From (i) and (ii), we get $p = 6q$

Q.9 Let $\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $8\sqrt{3}$ square units, then $\vec{a} \cdot \vec{b}$ is equal to :

माना $\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$ तथा $\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$ हैं। यदि समान्तर चर्तुभुज, जिसकी संलग्न भुजायें $\vec{a}$ तथा $\vec{b}$ हैं, का क्षेत्रफल $8\sqrt{3}$ वर्ग इकाई है, तो $\vec{a} \cdot \vec{b}$ बराबर है _____।

**Ans.** 2

**Sol.** $\begin{Vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \alpha & 3 \\ 3 & -\alpha & 1 \end{Vmatrix} = 8\sqrt{3}$

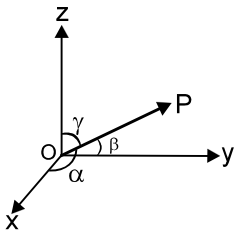
$\left|4\alpha\hat{i} + 8\hat{j} - 4\alpha\hat{k}\right| = 8\sqrt{3}$

$\alpha^2 = 4$

$\vec{a} \cdot \vec{b} = 3 - \alpha^2 + 3 = 2$

## 6.1 DIRECTION COSINES AND DIRECTION RATIOS

**(i) Direction cosines :** Let $\alpha, \beta, \gamma$ be the angles which a directed line makes with the positive directions of the axes of x, y and z respectively, then $\cos\alpha, \cos\beta, \cos\gamma$ are called the direction cosines of the line. The direction cosines are usually denoted by $\ell$, m, n.




Thus $\ell = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$.

**1.** Direction cosines are components of unit vector along that perpendicular direction

Proof: $l = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$

$\vec{p} = p_1\hat{i} + p_2\hat{j} + p_3\hat{k}$ ; $\quad |\vec{p}| = 1$ ; $\quad \vec{p}\,\hat{}\,\hat{i} = \alpha, \vec{p}\,\hat{}\,\hat{j} = \beta, \vec{p}\,\hat{}\,\hat{k} = \gamma$

$$\cos\alpha = \frac{\vec{p}.\hat{i}}{|\vec{p}||\hat{i}|} = \frac{p_1}{(1)(1)} = p_1 ; \cos\beta = \frac{\vec{p}.\hat{j}}{|\vec{p}||\hat{j}|} = \frac{p_2}{(1)(1)} = p_2$$

$$\cos\gamma = \frac{\vec{p}.\hat{k}}{|\vec{p}||\hat{k}|} = \frac{p_3}{(1)(1)} = p_3$$

**Note:**

**1.** If $\ell$, m, n be the direction cosines of a line, then $\ell^2 + m^2 + n^2 = 1$

2. Direction cosines of x–axis are (1, 0, 0)

Direction cosines of y–axis are (0, 1, 0)

Direction cosines of z–axis are (0, 0, 1)

**(ii) Direction ratios :**

Direction ratio are scalar multiple of direction cosines.

Direction ratio are components of vector of arbitary magnitude along that particular direction.

If a, b, c, are the direction ratios of any line L, then $a\hat{i}+b\hat{j}+c\hat{k}$ will be a vector parallel to the line L.

If $\ell$, m, n are direction cosines of line L, then $\ell\hat{i}+m\hat{j}+n\hat{k}$ is a unit vector parallel to the line L.

Q.1 Line $\vec{r}=(0,3,4)+\lambda(2,1,-1)$ Find D.C. and D.R.

Ans. (i) $\frac{1}{\sqrt{6}}(2, 1, -1)$ (ii) (2, 1, –1)

**Q.2** Find the direction cosines $\ell$, m, n of a line which are connected by the relations $\ell$+m +n = 0, $2mn + 2m\ell - n\ell = 0$

Ans. $\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}$ or $\frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{-1}{\sqrt{6}}$ and $\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}$ or $-\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$

**Sol. :** Given, $\ell + m + n = 0$ ..... (1)

$2mn + 2m\ell - n\ell = 0$ ..... (2)

From (1), $n = -(\ell + m)$.

Putting $n = -(\ell + m)$ in equation (2), we get,

$-2m(\ell + m) + 2m\ell + (\ell + m)\ell = 0$

or, $-2m\ell - 2m^2 + 2m\ell + \ell^2 + m\ell = 0$

or, $\ell^2 + m\ell - 2m^2 = 0$

or, $\left(\frac{\ell}{m}\right)^2 + \left(\frac{\ell}{m}\right) - 2 = 0$ [dividing by $m^2$]

or $\frac{\ell}{m} = \frac{-1\pm\sqrt{1+8}}{2} = \frac{-1\pm3}{2} = 1, -2$

**Case I.** when $\frac{\ell}{m} = 1$ : In this case $m = \ell$

From (1), $2\ell + n = 0 \Rightarrow n = -2\ell$

$\therefore$ $\ell : m : n = 1 : 1 : -2$

$\therefore$ Direction ratios of the line are 1, 1, – 2

$\therefore$ Direction cosines are

$$\pm\frac{1}{\sqrt{1^2+1^2+(-2)^2}}, \pm\frac{1}{\sqrt{1^2+1^2+(-2)^2}}, \pm\frac{-2}{\sqrt{1^2+1^2+(-2)^2}}$$

$\Rightarrow \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}$ or $-\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$

**Case II.** When $\frac{\ell}{m} = -2$ : In this case $\ell = -2m$

From (1), $-2m + m + n = 0 \quad \Rightarrow \quad n = m$

$\therefore$ $\ell : m : n = -2m : m : m$

$= -2 : 1 : 1$

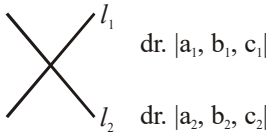
$\therefore$ Direction ratios of the line are – 2, 1, 1.

$\therefore$ Direction cosines are

$$\pm\frac{-2}{\sqrt{(-2)^2+1^2+1^2}},\ \pm\frac{1}{\sqrt{(-2)^2+1^2+1^2}},\ \pm\frac{1}{\sqrt{(-2)^2+1^2+1^2}}$$

$$\frac{-2}{\sqrt{6}},\ \frac{1}{\sqrt{6}},\ \frac{1}{\sqrt{6}} \qquad \text{or} \qquad \frac{2}{\sqrt{6}},\ \frac{-1}{\sqrt{6}},\ \frac{-1}{\sqrt{6}}$$

**Angle Between Lines**




$\vec{p}$ = direction vector of $l_1 = a_1\hat{i}+b_1\hat{j}+c_1\hat{k}$

$\vec{q}$ = direction vector of $l_2 = a_2\hat{i}+b_2\hat{j}+c_2\hat{k}$

$$\cos\theta=\left|\frac{\vec{p}.\vec{q}}{|\vec{p}||\vec{q}|}\right|$$

Q.3 Find angle between lines whose direction ratio are (2, 3, 6) & (1, 2, 2)

Ans. $\theta = \cos^{-1}\left(\frac{20}{21}\right)$

Sol. Let $\vec{a}=(2,3,6)$ and $\vec{b}=(1,2,2)$

$$\cos\theta=\frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|}=\frac{2+6+12}{7.3}=\frac{20}{21}$$

Q.4 Find D.C of line perpendicular $l_1$ and $l_2$

$l_1$ $\vec{r} = (0, 0, 1) + \lambda(1, -2, 1)$; $l_2$ $\vec{r} = (2, 1, 5) + \mu(1, 2, 3)$

Ans. $\pm\frac{1}{\sqrt{21}}(4,1,-2)$

Sol. direction ratio of line perpendicular to both

$= (1, -2, 1) \times (1, 2, 3)$

$= (-8, -2, 4)$

So D.C. $=\pm\frac{(4,1,-2)}{\sqrt{21}}$

Q.5 Find the direction cosines $\ell$, m, n of a line which are connected by the relations

$\ell - 2m + n = 0$, $\ell + 2m + 3n = 0$

Ans. $\pm\frac{1}{\sqrt{21}}(4,1,-2)$

Sol. $(\ell, m, n).(1,-2,1) = 0$

$(\ell, m, n).(1,2,3) = 0$

So direction cosive $(\ell, m, n)$ is perpendicular to both directions.

Therefore same calculation as in previous question.

Q.6 Find the measure of the angle between the lines whose direction ratios are 1, –2, 7 and 3, –2, –1.

Ans. 90°

Sol. $\sqrt{1^2 + (-2)^2 + 7^2} = \sqrt{54}$

$\sqrt{3^2 + (-2)^2 + (-1)^2} = \sqrt{14}$

$\therefore$ The actual direction cosines of the lines are

$\frac{1}{\sqrt{54}}, \frac{-2}{\sqrt{54}}, \frac{7}{\sqrt{54}}$ and $\frac{3}{\sqrt{14}}, \frac{-2}{\sqrt{14}}, \frac{-1}{\sqrt{14}}$

If $\theta$ is the angle between the lines, then

$$\cos\theta = \left(\frac{1}{\sqrt{54}}\right)\left(\frac{3}{\sqrt{14}}\right) + \left(\frac{-2}{\sqrt{54}}\right)\left(\frac{-2}{\sqrt{14}}\right) + \left(\frac{7}{\sqrt{54}}\right)\left(\frac{-1}{\sqrt{14}}\right)$$

$= \frac{3+4-7}{\sqrt{54}.\sqrt{14}} = 0 \Rightarrow \quad \theta = 90^\circ$. **Ans.**

Q.7 The angle between two lines whose direction cosines are given by $\ell + m + n = 0$, $\ell^2 + m^2 - n^2 = 0$ is-

(A) $2\pi/3$ (B) $\pi/6$ (C) $5\pi/6$ (D) none of these

Ans. A

**Sol.** Eliminating n between the given relations. We get

$\ell^2 + m^2 - (-\ell - m)^2 = 0$

or $\ell m = 0 \Rightarrow \ell = 0$

or $m = 0$

Now $\ell = 0 \Rightarrow 1\ell + 0m + 0\,n = 0$

and $\ell + m + n = 0$

Solving these equations, we get

$$\frac{\ell}{0} = \frac{m}{-1} = \frac{n}{1} \qquad ...(1)$$

Similary $m = 0 \Rightarrow \quad 0.\ell.+ 1.m + 0.n = 0$

and $\ell + m + 0.n = 0$

Solving these equations, we get

$$\frac{\ell}{1} = \frac{m}{0} = \frac{n}{-1} \quad ...(2)$$

Thus direction ratios of given lines are 0, –1, 1 and 1, 0 –1 respectively. Let $\theta$ is angle between them, then

$$\cos\theta = \frac{0\times1+(-1)\times0+1\times(-1)}{\sqrt{0^2+(-1)^2+1^2}\sqrt{1^2+0^2+(-1)^2}}$$

or $\cos\theta = -\frac{1}{2} \Rightarrow \quad \theta = 2\pi/3$ **Ans.**

## 6.2 SYMMETRICAL FORM OF A LINE

**(a) One point form :** Let $A(x_1, y_1, z_1)$ be a given point on the straight line and $\ell$, m, n be the d.c's of the line, then its

equation is $\frac{x-x_1}{\ell} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = r$ (say)

It should be noted that $P(x_1 + \ell r, y_1 + mr, z_1 + nr)$ is a general point on this line at a distance r from the point $A(x_1, y_1, z_1)$ i.e. AP = r. One should note that for AP = r; $\ell$, m, n must be d.c.'s not d.r.'s. If a, b, c are direction ratios of the line, then equation of the line is $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = r$ but here AP $\neq$ r

Ex.1 $\vec{r} = (2, 3, 4) + \lambda(2, 1, -3)$

$$\frac{x-2}{2} = \frac{y-3}{1} = \frac{z-4}{-3} \text{ or } \frac{x-2}{2} = \frac{y-3}{1} = \frac{4-z}{3}$$

Ex.2 $\vec{r} = (2, 1, 1) + \lambda(1, \frac{2}{3}, 2)$

$$\frac{x-2}{1} = \frac{y-1}{2/3} = \frac{z-1}{2} \text{ or } x-2 = \frac{3y-3}{2} = \frac{z-1}{2} \text{ or } \frac{x-2}{3} = \frac{y-1}{2} = \frac{z-1}{6}$$

Ex.3 $\vec{r} = (1, 3, 4) + \lambda(2, 1, 0)$

$$\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{0} \text{ or line } \frac{x-1}{2} = \frac{y-3}{2} \text{ and } z = 4$$

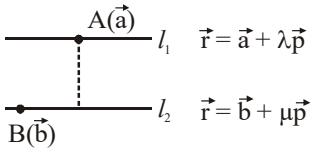
Ex.4 $\vec{r} = (2, 1, 5) + \lambda(1, 0, 0)$

$$\frac{x-2}{1} = \frac{y-1}{0} = \frac{y-5}{0} \text{ or } y = 1, z = 5$$

## 6.3 SHORTEST DISTANCE BETWEEN TWO LINES :

Case-I : If lines are intersecting then minimum distance between them 0

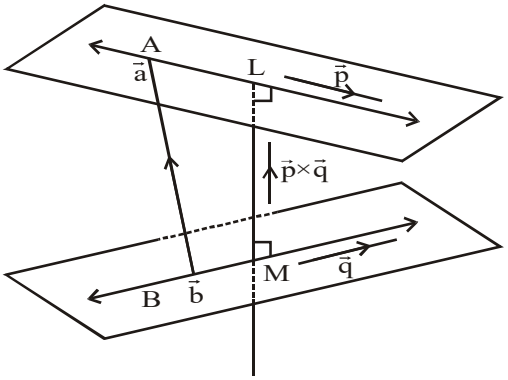
Case-II : If lines are parallel to each other



Required distance = distance of A from $l_2$

$$= \frac{|\overrightarrow{AB} \times \vec{p}|}{|\vec{p}|}$$

Case-II : If lines are skew




For Skew lines the direction of the shortest distance vector would be perpendicular to both the lines. The magnitude of the shortest distance vector would be equal to that of the projection of $\vec{AB}$ along the direction of the line of shortest distance, $\vec{LM}$ is parallel to $\vec{p} \times \vec{q}$

i.e. $\vec{LM}$ = | Projection of $\vec{AB}$ on $\vec{LM}$| = | Projection of $\vec{AB}$ on $\vec{p} \times \vec{q}$| $= \left|\frac{\vec{AB}.(\vec{p} \times \vec{q})}{\vec{p} \times \vec{q}}\right| = \left|\frac{(\vec{b} - \vec{a}).(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right|$

**The two lines directed along $\vec{p}$ & $\vec{q}$ will intersect only if shortest distance = 0 i.e.**

$(\vec{b} - \vec{a}).(\vec{p} \times \vec{q}) = 0$ i.e. $(\vec{b} - \vec{a})$ lies in the plane containing $\vec{p}$ & $\vec{q} \Rightarrow \left[(\vec{b} - \vec{a}) \quad \vec{p} \quad \vec{q}\right] = 0$

**Another Method :**

Let the equation of two non-intersecting lines be $\frac{x - x_1}{l_1} = \frac{y - y_1}{m_1} = \frac{z + z_1}{n_1} = r_1 (say)$ .....(1)

and $\frac{x - x_2}{l_2} = \frac{y - y_2}{m_2} = \frac{z + z_2}{n_2} = r_2 (say)$ .....(2)

Any point on line (1) is $P(x_1 + l_1r_1, y_1 + m_1r_1, z_1 + n_1r_1)$ and on line (2) is $Q(x_2 + l_2r_2, y_2 + m_2r_2, z_2 + n_2r_2)$ .

Let PQ be the line of shortest distance, its direction ratios will be

$[(l_1r_1 + x_1 - x_2 - l_2r_2), (m_1r_1 + y1 - y_2 - m_2r_2) , (n_1r_1 + z_1 - z_2 - n2r_2)]$

This line is perpendicular to both the given lines. By using condition of perpendicularity we obtain two equations in $r_1$ and $r_2$.

So by solving these, values of $r_1$ and $r_2$ and subsequently point P and Q can be found. The distance PQ is the shortest distance.

Q.1 $\vec{r} = \hat{i} + \hat{j} + \lambda\left(2\hat{i} - \hat{j} + \hat{k}\right);\ \vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu\left(3\hat{i} - 5\hat{j} + 2\hat{k}\right)$

Ans. $\frac{10}{\sqrt{59}}$

Sol. $\vec{r_1} = (1,1,0) + \lambda(2,-1,1) = \vec{a} + \lambda\vec{p}$

$\vec{r_2} = (2,1,-1) + \mu(3,-5,2) = \vec{b} + \mu\vec{q}$

$\vec{b} - \vec{a} = (2,1,-1) - (1,1,0) = (1,0,-1)$

$\vec{b} \times \vec{q} = (2,-1,1) \times (3,-5,2) = (3,-1,-7)$

$$\text{S.D.} = \left| \frac{(\vec{b} - \vec{c}).(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|} \right|$$

$$= \left| \frac{(1,0,-1).(3,-1,-7)}{\sqrt{59}} \right|$$

$$= \frac{10}{\sqrt{59}}$$

Q.2 Vector equation of two lines are

(a) $\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$

(b) $\vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} - (2s+1)\hat{k}$

Ans. (a) $\frac{8}{\sqrt{29}}$ (b) $\frac{3}{\sqrt{2}}$

Sol. $\vec{r_1} = (1,-2,3) + t(-1,1,-2) = \vec{a} + t\vec{p}$

$\vec{r_2} = (1,-1,-1) + s(1,2,-2) = \vec{b} + s\vec{q}$

$\vec{b} - \vec{a} = (0,1,-4)$

and $\vec{b} \times \vec{q} = (2,-4,-3)$

$$\text{S.D.} = \left| \frac{(\vec{b} - \vec{a}).(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|} \right|$$

$$= \left| \frac{(0,1,-4).(2,-4,-3)}{\sqrt{29}} \right|$$

$$= \frac{8}{\sqrt{29}}$$

Q.3 Find the shortest distance between the lines $\frac{x-6}{3} = \frac{y-7}{-1} = \frac{z-4}{1}$ and $\frac{x}{-3} = \frac{y+9}{2} = \frac{z-2}{4}$. Also find the equation of the line of the shortest distance.

Ans. $\sqrt{270}$, $\frac{x-3}{2} = \frac{y-8}{5} = \frac{z-3}{-1}$

Sol. The coordinates of any point P on the first line are $(3r+6, -r+7, r+4)$ and those of Q on the second line are $(-3R, 2R-9, 4R+2)$ where r is proportional to the distance of P from the point (6, 7, 4) and R to the distance of Q from the point (0, –9, 2). The d.c.s' of PQ are proportional to $-3R-3r-6, 2R+r-16, 4R-r-2$. The points P and Q will be nearest to each other if PQ is at right angles to both the lines.

Therefore, we have

$3(-3R-3r-6)+(-1)(2R+r-16)+1(4R-r-2)=0$

and $-3(-3R-3r-6)+2(2R+r-16)+4(4R-r-2)=0$

[Using $a_1a_2+b_1b_2+c_1c_2=0$]

i.e., $7R+11r+4=0$ and $29R+7r-22=0$

Solving these, we get $r=-1, R=1$.

P and Q are the points (3, 8, 3) and (– 3, – 7, 6) respectively.

The shortest distance PQ = $\sqrt{(-3-3)^2+(-7-8)^2+(6-3)^2}=\sqrt{270}$ units.

The equation of PQ is

$$\frac{x-3}{-3-3}=\frac{y-8}{5}=\frac{z-3}{-1} \quad \Rightarrow \quad \frac{x-3}{2}=\frac{y-8}{5}=\frac{z-3}{-1}$$

Q.4 Let $\lambda$ be an integer. If the shortest distance between the lines $x-\lambda=2y-1=-2z$ and $x=y+2\lambda=z-\lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is _____.

माना $\lambda$ एक पूर्णांक है। यदि रेखाओं $x-\lambda=2y-1=-2z$ तथा $x=y+2\lambda=z-\lambda$ के बीच की न्यूनतम दूरी $\frac{\sqrt{7}}{2\sqrt{2}}$ है, तो $|\lambda|$ बराबर है _____ ।

**Ans.** 1

**Sol.** $L_1 \quad \frac{x-\lambda}{2}=\frac{y-\frac{1}{2}}{1}=\frac{z}{-1}$

$L_2 \quad \frac{x}{1}=\frac{y+2\lambda}{1}=\frac{z-\lambda}{1}$

$\vec{p}=(2,1,-1), \vec{q}=(1,1,1)$

$A\left(\lambda, \frac{1}{2}, 0\right)$ $\qquad$ $B(0, -2\lambda, \lambda)$

$\vec{p}\times\vec{q}=2\hat{i}-3\hat{j}+\hat{k}$

$$\left|\frac{\overrightarrow{AB}\cdot(\vec{p}\times\vec{q})}{(\vec{p}\times\vec{q})}\right| = \frac{\sqrt{7}}{2\sqrt{2}}$$

$$\frac{\left|-2\lambda+3\left(2\lambda+\frac{1}{2}\right)+\lambda\right|}{\sqrt{14}} = \frac{\sqrt{7}}{2\sqrt{2}}$$

$\Rightarrow \lambda = -1$ or $\lambda = \frac{2}{5}$

$\Rightarrow |\lambda| = 1$ (as $\lambda \in I$)

## 7.1 PLANE

A plane is a surface such that if any two points are taken on it, the line segment joining them lies completely on the surface. In other words, every point on the line segment joining any two points lies on the plane.

**Normal to a plane**

A line perpendicular to a plane is called a normal to the plane. Clearly, every line lying in a plane is perpendicular to the normal to the plane.

**(i) Vector equation of plane passing through a point and normal to a given vector**

The vector equation of a plane passing through a point having position vector $\vec{n}$ is $(\vec{r}-\vec{a}).\ \vec{n} = 0$

**Reduction to cartesian form :**

If $\vec{r} = x\hat{i}+y\hat{j}+z\hat{k}$, $\vec{a} = x_1\hat{i}+y_1\hat{j}+z_1\hat{k}$ and $\vec{n} = a\hat{i}+b\hat{j}+c\hat{k}$

Then ; $(\vec{r}-\vec{a}) = (x-x_1)\hat{i}+(y-y_1)\hat{j}+(z-z_1)\hat{k}$

Then (i) can be written as

$\{(x-x_1)\hat{i}+(y-y_1)\hat{j}+(z-z_1)\hat{k}\}\cdot\{(a\hat{i}+b\hat{j}+c\hat{k})\} = 0$

$\Rightarrow\ a(x-x_1)+b(y-y_1)+c(z-z_1) = 0$

Coefficient of x, y and z in cartesian equation of plane are components of normal vector of the plane.

**For example :** The direction ratios of a vector normal to the plane $3x + 2y + 5z - 6 = 0$ are 3, 2, 5 and hence a vector normal to the plane is $3\hat{i} + 2\hat{j} + 5k$.

Every first degree equation in x, y and z represents a plane i.e. $ax+by+cz+d=0$ is the general equation of a plane.

**Equation of a plane passing through a given point**

The general equation of a plane passing through a point $(x_1, y_1, z_1)$ is

$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, where a, b and c are constants.

**Plane Parallel to the Coordinate Planes :**

(i) Equation of yz plane is $x = 0$.

(ii) Equation of zx plane is $y = 0$.

(iii) Equation of xy plane is $z = 0$.

(iv) Equation of the plane parallel to xy plane at a distance c is $z = c$ or $z = -c$.

(v) Equation of the plane parallel to yz plane at a distance c is $x = c$ or $x = -c$

(vi) Equation of the plane parallel to zx plane at a distance c is $y = c$ or $y = -c$.

## 7.2 ANGLE BETWEEN TWO PLANES

(i) **Vector form** - The angle between the two planes is defined as the angle between normals.

Let $\theta$ be the angle between planes;

$\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is given by

$$\cos\theta = \left| \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|} \right|$$

(ii) **Cartesian form** - The angle $\theta$ between the planes $a_1 x + b_1 y + c_1 z + d_1 = 0$ and

$a_2 x + b_2 y + c_2 z + d_2 = 0$ is given by $\cos\theta = \left| \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}} \right|$

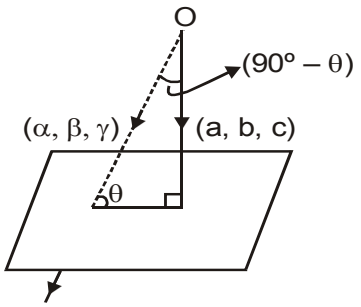
Equation of a plane parallel to the plane $ax + by + cz + d = 0$ is $ax + by + cz + d' = 0$. d' is to be found by other given condition.

## 7.3 ANGLE BETWEEN A LINE AND A PLANE

The angle between a line and a plane is the complement of the angle between the line and the normal to the plane

**Vector form :** If $\theta$ is the angle between the line $\vec{r} = \vec{a} + \lambda \vec{p}$ and plane $\vec{r} \cdot \vec{n} = d$

$$\cos(90° - \theta) = \left| \frac{\vec{p} \cdot \vec{n}}{|\vec{p}||\vec{n}|} \right|$$




$$\Rightarrow \sin\theta = \left| \frac{\vec{p} \cdot \vec{n}}{|\vec{p}||\vec{n}|} \right|$$

**Condition for a line to be parallel to a plane**

Let line $\frac{x - x_1}{\ell} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$ be parallel to plane $ax + by + cz + d = 0$ iff;

$\theta = 0$ or $\pi$ or $\sin\theta = 0$ or $a\ell + bm + cn = 0$

**Condition for a line to lie in the plane**

A line will be in a plane iff.

(i) the normal to the plane is perpendicular to the line

(ii) a point on the line lies in the plane.

Ex.1 Line $l_1$ $\vec{r} = \vec{a} + \lambda\vec{p}$ Line $l_2$ $\vec{r} = \vec{b} + \mu\vec{q}$

Plane $p_1$ $\vec{r}.\vec{n}_1 = d_1$ Plane $p_2$ $\vec{r}.\vec{n}_2 = d_2$

Find condition if

(1) $l_1 \parallel l_2$ (2) $l_1 \perp l_2$

(3) $p_1 \parallel p_2$ (4) $p_1 \perp p_2$

(5) $L_1 \perp p_1$ (6) $L_1 \parallel p_1$

(7) $l_1$ is contained in $p_1$

Find equation of plane

1. passing through point $A(\vec{a}), B(\vec{b})$ and $C(\vec{c})$

Sol. $\vec{r}.\vec{a} = \vec{n}.\vec{a}$

$= \left(\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right).\vec{a}$

$= \left(\vec{b}\times\vec{c}\right).\vec{a}$

2. passing through $A(\vec{a})$ and perpendicular to line $\vec{r} = \vec{b} + \lambda\vec{p}$.

Sol. $\vec{r}.\vec{a} = \vec{p}.\vec{a}$

3. passing through $A(\vec{a})$ and parallel to vectors $\vec{p}$ and $\vec{q}$.

Sol. $\vec{r}.\vec{a} = \left(\vec{p}\times\vec{q}\right).\vec{a}$

4. passing through $A(\vec{a})$ and parallel to palne $\vec{r}.\vec{n}_1 = d_1$

Sol. $\vec{r}.\vec{a} = \vec{n}_1.\vec{a}$

5. containg lines $\vec{r} = \vec{a} + \lambda\vec{p}, \vec{r} = a + \mu\vec{q}$

Sol. $\vec{r}.\vec{a} = \left(\vec{p}\times\vec{q}\right).\vec{a}$

6. passing through $A(\vec{a})$ and perpendicular to plane $\vec{r}.\vec{n}_1 = d_1$ and parallel to line $\vec{r} = \vec{b} + \lambda\vec{p}$.

Sol. $\vec{r}.\vec{a} = \left(\vec{n}_1\times\vec{p}\right).\vec{a}$

## 7.4 OTHER FORMS OF EQUATION OF PLANE

### 1. Parametric form of equation of plane

To find equation of plane passing point $A(\vec{a})$ and parallel to the vectors $\vec{p}$ and $\vec{q}$.

$A\vec{R}$, $\vec{p}$ and $\vec{q}$ are coplaner.

$A\vec{R} = \lambda\vec{p} + \mu\vec{q}$

$\vec{r} - \vec{a} = \lambda\vec{p} + \mu\vec{q}$

$\vec{r} = \vec{a} + \lambda\vec{p} + \mu\vec{q}$ Here normal vector $\vec{n} = \vec{p} \times \vec{q}$

### 2. Intercept form of a plane

The equation of a plane intercepting lengths a, b and c with x- axis , y-axis and z-axis respectively is

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

## 7.5 DISTANCE OF A POINT FROM A PLANE

The length of the perpendicular from a point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$

is given by $\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$

### DISTANCE BETWEEN THE PARALLEL PLANES

The distance between two parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is given by

$d = \left|\frac{(d_2 - d_1)}{\sqrt{a^2 + b^2 + c^2}}\right|$

## 7.6 POSITION OF TWO POINTS W.R.T. A PLANE :

Two points $P(x_1, y_1, z_1)$ & $Q(x_2, y_2, z_2)$ are on the same or opposite sides of a plane $ax + by + cz + d = 0$ according to $ax_1 + by_1 + cz_1 + d$ & $ax_2 + by_2 + cz_2 + d$ are of same or opposite signs. The plane divides the line joining the points P & Q externally or internally according to P and Q lying on same or opposite sides of the plane.

## 7.7 ANGLE BISECTOR PLANE BETWEEN TWO PLANES

Let the equations of the two planes be $ax + by + cz + d = 0$ and $a_1x + b_1y + c_1z + d_1 = 0$.

Then equations of bisectors of angles between them are given by $\frac{ax + by + cz + d}{\sqrt{(a^2 + b^2 + c^2)}} = \pm\left(\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{(a_1^2 + b_1^2 + c_1^2)}}\right)$

Q.1 Plane $\vec{r} = (1, 2, 1) + \lambda(2, 1, 3) + \mu(1, -1, 1)$

(i) find cartesian form of equation of plane.

Ans. $4x + y - 3z = 3$

Ans. $\vec{r} = (1, 2, 1) + \lambda(2, 1, 3) + \mu(1, -1, 1)$

$\vec{n} = (2,1,3) \times (1,-1,1) = (4,1,-3)$

So equation of plane

$4(x - 1) + (y - 2) - 3(z - 1) = 0$

$\Rightarrow 4x + y - 3z = 3$

(ii) Find vector form of equation of plane.

Ans. $\left(\vec{r}-(1,2,1)\right)\cdot(4,1,-3)=0$

Sol. $(\vec{r}-\vec{a}).\vec{n}=0$

$\Rightarrow \left(\vec{r}-(1,2,1)\right).(4,1,-3)=0$

$\Rightarrow \vec{r}.(4,1,-3)=(1,2,1).(4,1,-3)=3$

(iii) Find equation of plane passing through (1, 2, –3) and parallel to given plane.

Ans. $4x + y - 3z = 15$

Sol. Equation of parallel plane

$4x + y - 3z = \lambda$

(1, 2, –3) will satisfy it

$4.1 + 2 - 3(-3) = \lambda$

$\Rightarrow \lambda = 15$

So equation of plane

$4x + y - 3z = 15$

(iv) Find intercept of given plane on coordinate axes.

Ans. $\frac{3}{4}, 3, -1$

Sol. $4x + y - 3z = 3$

$\Rightarrow \frac{x}{(3/4)}+\frac{y}{3}-\frac{z}{1}=1$

So intersepts are $\frac{3}{4}, 3, -1$

(v) If $\vec{r} = (1, 1, 1) + \lambda(2, 1, p)$ is parallel to given plane. Find p.

Ans. 3

Sol. $\vec{p}.\vec{n} = 0$

$\Rightarrow (2, 1, p).\ (4, 1, -3) = 0$

$\Rightarrow 8 + 1 - 3p = 0$

$\Rightarrow p = 3$

(vi) If $\vec{r} = (2, 0, p) + \lambda(3, 1, q)$ is contained line given plane. Find p and q.

Ans. $p=\frac{5}{3}, q=\frac{13}{3}$

Sol. point (2, 0, p) will satisfy the plane

$4x + y - 3z = 3$

$\Rightarrow 4.2 + 0 - 3p = 3$

$\Rightarrow p = \frac{5}{3}$

If the disection of line is $\vec{p}$ then

$\vec{p}.\vec{n} = 0$

$\Rightarrow (3, 1, q).\ (4, 1, -3) = 0$

$\Rightarrow 12 + 1 - 3q = 0$

$\Rightarrow q = \frac{13}{3}$

(vii) Find equation of plane parallel to given plane which is at distance of $\sqrt{104}$ from origin.

Ans. $4x + y - 3z = \pm 52$

Sol. Equation of the parallel plane

$4x + y - 3z + \lambda = 0$

distance from origin $\left|\frac{\pi}{\sqrt{16+1+9}}\right| = \sqrt{104}$

$\Rightarrow \lambda = \pm 52$

Q.2 Find equation of plane whose normal vectors is $a\hat{i} + b\hat{j} + c\hat{k}$ and passes through $A(x_0, y_0, z_0)$

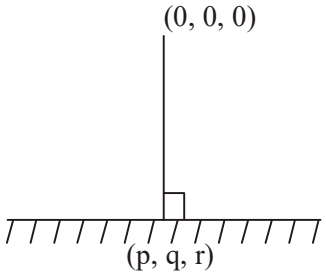
Ans. $ax + by + cz = ax_0 + by_0 + cz_0$

Sol.

Q.3 Find equation of plane such that foot of perpendicular from origin on plane is (p,q,r) equation of plane.

Ans. $px + qy + rz = p^2 + q^2 + r^2$

Sol.




$\vec{n} = (p, q, r)$

Equation of plane :

$px + qy + rz = \lambda$

(p, q, r) will satisfy it

$px + qy + rz = p^2 + q^2 + r^2$

Q.4 Find equation of plane passing through (1, 2, 3) and perpendicular to plane $x - y + 2z = 3$ and $x + y - 3z = 4$

Ans. $x + 5y + 2z = 17$

Sol. $\vec{n} = \vec{n}_1 \times \vec{n}_2$

$= (1, -1, 2) \times (1, 1, -3)$

= (1, 5, 2)

So let equation of plane is

$x + 5y + 2z = \lambda$

point (1, 2, 3) will satisfy it

$\lambda = 1 + 5.2 + 2.3$

$\lambda = 17$

So equation of plane

$x + 5y + 2z = 17$

Q.5 Find angle between plane $x + y + 2z = 9$ & $2x - y + z = 15$

Ans. $60^\circ$

Sol. $\cos\theta = \dfrac{\vec{n}_1.\vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$

$= \dfrac{(1,1,2).(2,-1,1)}{\sqrt{6}\sqrt{6}} = \dfrac{1}{2}$

$\Rightarrow \theta = 60°$

Q.6 Find equation of plane containg the lines $\vec{r} = (1, 1, 1) + \lambda(1, -2, 0)$ & $\vec{r} = (1, 1, 1) + \mu(2, 1, -1)$

Ans. $2x + y + 5z = 8$

Sol. $\vec{n} = \vec{p} \times \vec{q}$

$= (1, -2, 0) \times (2, 1, -1)$

$= (2, 1, 5)$

so equation of normal is

$2x + y + 5z = \lambda$

(1, 1, 1) will satisfy it

$\lambda = 8$

so equation of plane

$2x + y + 5z = 8$

## 8.1 FAMILY OF PLANES

Any plane passing through the line of intersection of non–parallel planes $a_1x + b_1y + c_1z + d_1 = 0$ & $a_2x + b_2y + c_2z + d_2 = 0$ is $a_1x + b_1y + c_1z + d_1 + \lambda (a_2x + b_2y + c_2z + d_2) = 0$, where $\lambda \in R$

**Q.1** Find the equation of the plane through the point (1, 1, 1) which passes through the line of intersection of the planes $x + y + z = 6$ and $2x + 3y + 4z + 5 = 0$.

Ans. $20x + 23y + 26z - 69 = 0$

**Sol. :** Given planes are $x + y + z - 6 = 0$ ..... (1)

and $2x + 3y + 4z + 5 = 0$ ..... (2)

Given point is P (1, 1, 1).

Equation of any plane through the line of intersection of planes (1) and (2) is

$x + y + z - 6 + k(2x + 3y + 4z + 5) = 0$ ..... (3)

If plane (3) passes through point P, then

$1 + 1 + 1 - 6 + k(2 + 3 + 4 + 5) = 0$ or, $k = \frac{3}{14}$

From (3) required plane is $20x + 23y + 26z - 69 = 0$

**Q.2:** The plane $x - y - z = 4$ is rotated through 90° about its line of intersection with the plane $x + y + 2z = 4$. Find its equation in the new position.

Ans. $5x + y + 4z = 20$

**Sol. :** Given planes are $x - y - z = 4$ ..... (1)

and $x + y + 2z = 4$ ..... (2)

Since the required plane passes through the line of intersection of planes (1) and (2)

$\therefore$ its equation may be taken as

$x + y + 2z - 4 + k(x - y - z - 4) = 0$

or $(1 + k)x + (1 - k)y + (2 - k)z - 4 - 4k = 0$ ..... (3)

Since planes (1) and (3) are mutually perpendicular,

$\therefore$ $(1 + k) - (1 - k) - (2 - k) = 0$

or, $1 + k - 1 + k - 2 + k = 0$ or $k = \frac{2}{3}$

Putting $k = \frac{2}{3}$ in equation (3), we get $5x + y + 4z = 20$

This is the equation of the required plane.

## 8.2 LINE OF INTERSECTION OF TWO PLANES

Line of intersection of plane $p_1$ and $p_2$ will contained on the both plane so its direction vector will be perpendicular to both there normal vectors.

To find the fixed point on line of intersection put $z = 0$ in both equation.

Q.1 Find vector form of equation of line of intersection of planes $2x + y - z = 3$ & $4x + y + z = 5$

Ans. $\vec{r} = (1, 1, 0) + \lambda(1, -3, -1)$

Sol $\vec{p}$ = direction vector of LOI

$\overrightarrow{n_1}$ = Normal vector of Plane $2x + y - z = 3$

$\overrightarrow{n_2}$ = Normal vector of Plane $4x + y + z = 5$

$\vec{p} \perp \vec{n}_1$ $\vec{p} \perp \vec{n}_2$

$$\vec{p} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 4 & 1 & 1 \end{vmatrix}$$

$= 2\hat{i} - 6\hat{j} - 2\hat{k}$

For fixed point

Put $z = 0$ in both the equations and solve the equations

$2x + y = 3$

$4x + y = 5 \Rightarrow x = 1, y = 1$

$\vec{r} = (1, 1, 0) + \lambda(1, -3, -1)$

**Method II**

Put $z = \lambda$

$2x + y - \lambda = 3 \qquad \Rightarrow 2x + y = 3 + \lambda$

$4x + y + \lambda = 5 \qquad 4x + y = 5 - \lambda$

Solving these equations $\qquad x = 1 - \lambda$ & $\quad y = 1 + 3\lambda$

$(x, y, z) = (1 - \lambda, 1 + 3\lambda, \lambda)$ or $\qquad \vec{r} = (1, 1, 0) + \lambda(-1, 3, 1)$

## 8.3 UNSYMMETRICAL FORM OF A LINE

A straight line in space is characterised by the intersection of two planes which are not parallel and therefore, the equation of a straight line is a solution of the system constituted by the equations of the two planes, $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$.

Unsymmetrical Form of a Line is given by $P_1 = 0 = P_2$. It is line of intersection of planes $P_1$ and $P_2$

**Q.1** Find the angle between the lines $x - 3y - 4 = 0, 4y - z + 5 = 0$ and $x + 3y - 11 = 0, 2y - z + 6 = 0$.

Ans. $90°$

**Sol. :** Given lines are $\left.\begin{matrix} x - 3y - 4 = 0 \\ 4y - z + 5 = 0 \end{matrix}\right\}$ ..... (1)

and $\left.\begin{matrix} x + 3y - 11 = 0 \\ 2y - z + 6 = 0 \end{matrix}\right\}$ ..... (2)

Let $\ell_1, m_1, n_1$ and $\ell_2, m_2, n_2$ be the direction cosines of lines (1) and (2) respectively

$\because$ line (1) is perpendicular to the normals of each of the planes

$x - 3y - 4 = 0$ and $4y - z + 5 = 0$

$\therefore \quad \ell_1 - 3m_1 + 0.n_1 = 0$ ..... (3)

and $\quad 0\ell_1 + 4m_1 - n_1 = 0$ ..... (4)

Solving equations (3) and (4), we get $\dfrac{\ell_1}{3-0} = \dfrac{m_1}{0-(-1)} = \dfrac{n_1}{4-0}$

or, $\dfrac{\ell_1}{3} = \dfrac{m_1}{1} = \dfrac{n_1}{4} = k$ (let).

Since line (2) is perpendicular to the normals of each of the planes

$x + 3y - 11 = 0$ and $2y - z + 6 = 0$,

$\therefore \quad \ell_2 + 3m_2 = 0$ ..... (5)

and $2m_2 - n_2 = 0$ ..... (6)

$\therefore \quad \ell_2 = -3m_2 \quad$ or, $\quad \frac{\ell_2}{-3} = m_2$ and $\quad n_2 = 2m_2 \quad$ or, $\quad \frac{n_2}{2} = m_2$.

$\therefore \quad \frac{\ell_2}{-3} = \frac{m_2}{1} = \frac{n_2}{2} = t$ (let).

If $\theta$ be the angle between lines (1) and (2), then $\cos\theta = \ell_1\ell_2 + m_1m_2 + n_1n_2$

$= (3k)(-3t) + (k)(t) + (4k)(2t) = -9kt + kt + 8kt = 0$

$\therefore \quad \theta = 90°$.

**Q.2** The distance of the point (1, – 2, 3) from the plane x – y + z = 5 measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z-1}{-6}$ is-

Ans. 1

**Sol. :** Equation of the line through (1, – 2, 3) parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z-1}{-6}$

$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6} = r$ (say)...(1)

Then any point on (1) is

(2r + 1, 3r – 2, – 6r + 3).

If this point lies on the plane x – y + z = 5, then (2r + 1) – (3r – 2) + (– 6r + 3) = 5

$\Rightarrow$ – 7r + 6 = 5, i.e. , $r = \frac{1}{7}$

Hence the point is $\left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$.

Distance between (1, – 2, 3) and $\left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$

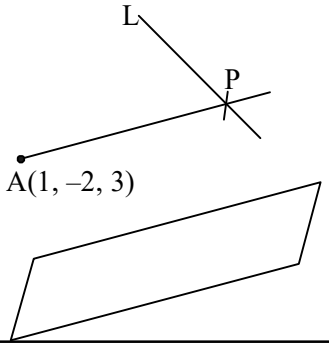
$= \sqrt{\left(\frac{4}{49} + \frac{9}{49} + \frac{36}{49}\right)} = \sqrt{\left(\frac{49}{49}\right)} = 1$

Q.3 A(1, –2, 3); Plane P x – y + z = 5 Line L $\vec{r} = (0, 1, 0) + \lambda(2, 3, -6)$

Find distance of point A from line L measured parallel to plane P.

Ans. $\sqrt{18}$

Sol.



Let $P(2\lambda, 1+3\lambda, -6\lambda)$

$\overrightarrow{AP} = (2\lambda - 1, 3\lambda + 3, -6\lambda - 3)$

$\overrightarrow{AP}$ is parallel to plane

so $\overrightarrow{AP}.\vec{n} = 0$

$\Rightarrow (2\lambda - 1, 3\lambda + 3, -6\lambda - 3).(1, -1, 1) = 0$

$\Rightarrow (2\lambda - 1) - (3\lambda + 3) - 6\lambda - 3 = 0$

$\Rightarrow -7\lambda - 7 = 0 \qquad \Rightarrow \lambda = -1$

So $\overrightarrow{AP} = (-3, 0, 3)$

$\left|\overrightarrow{AP}\right| = \sqrt{18}$

**Q.4** Find the equation of the plane through the point (1, 1, 1) which contains the line $x + y + z - 6 = 0 = 2x + 3y + 4z + 5$ :

Ans. $20x + 23y + 26z - 69 = 0$

Sol. Let the plane is

$(x + y + z - 6) + \lambda (2x + 3y + 4z + 5) = 0$ .....(1)

(1, 1, 1) will satisfy is

$-3 + \lambda (14) = 0$

$\Rightarrow \lambda = \frac{3}{14}$

from (1)

$$\left(x + y + z - 6\right) + \frac{3}{14}\left(2x + 3y + 4z + 5\right) = 0$$

$\Rightarrow 20x + 23y + 26z - 69 = 0$

## 8.4 PRACTICE QUESTIONS ON PLANE

1. The unit vector perpendicular to the plane passing through points $P\left(\hat{i} - \hat{j} + 2\hat{k}\right), Q\left(2\hat{i} - \hat{k}\right)$ and $R\left(2\hat{j} + \hat{k}\right)$ is :

(A) $2\hat{i} + \hat{j} + \hat{k}$ (B) $\sqrt{6}\left(2\hat{i} + \hat{j} + \hat{k}\right)$ (C) $\frac{1}{\sqrt{6}}\left(2\hat{i} + \hat{j} + \hat{k}\right)$ (D) $\frac{1}{6}\left(2\hat{i} + \hat{j} + \hat{k}\right)$

Ans. C

**Sol.** $\overrightarrow{PQ} = \left(2\hat{i} - \hat{k}\right) - \left(\hat{i} - \hat{j} + 2\hat{k}\right) = \hat{i} + \hat{j} - 3\hat{k}$

$\overrightarrow{PR} = \left(2\hat{i} + \hat{k}\right) - \left(\hat{i} - \hat{j} + 2\hat{k}\right) = -\hat{i} + 3\hat{j} - \hat{k}$

Now $\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -3 \\ -1 & 3 & -1 \end{vmatrix} = 8\hat{i} + 4\hat{j} + 4\hat{k}$

$\Rightarrow \left|\overrightarrow{PQ}\times\overrightarrow{PR}\right| = \sqrt{64+16+16} = 4\sqrt{6}$

$\therefore$ reqd. unit vector $= \dfrac{4(2\hat{i}+\hat{j}+\hat{k})}{4\sqrt{6}}$

$= \dfrac{1}{\sqrt{6}}\left(2\hat{i}+\hat{j}+\hat{k}\right)$

2. Show that the four points (0,–1, –1) (–4,4,4) (4,5,1) and (3,9,4) are coplanar, Find the equation of the plane containing them.

Ans. $5x - 7y + 11z + 4 = 0$

**Sol.** We shall find the equation of a passing though any three the given four points and show that the fourth points satisfies the equation.

Now, any plane passing through (0, –1, –1) is :

$a(x - 0) + b(y + 1) + c(z + 1) = 0$ ..(i)

If it passes through (–4,4,4); we have

$a(-4) + b(5) + c(5) = 0$ ..(ii)

Also, if plane passes through (4, 5,1), we have

$a(4) + b(6) + c(2) = 0$

$\Rightarrow 2a + 3b + c = 0$ ...(iii)

Solving (ii) and (iii) by cross multiplication method, we obtain

$\dfrac{a}{-5} = \dfrac{b}{7} = \dfrac{c}{-11} = k$

Putting in (i), we get

$-5kx + 7k(y + 1) - 11k(z + 1) = 0$

$\Rightarrow -5x + 7y - 11z - 4 = 0$

(required eq. of plane)

Clearly, the fourth point namely (3, 9,4) satisfies this equation. Hence the given points are coplanar and the equation of plane containing those points is

$5x - 7y + 11z + 4 = 0$

3. A plane meets the co-ordinate axis in A,B,C such that the centroid of the $\Delta ABC$ is the point (p,q,r) show that the equation of the plane is $\dfrac{x}{p}+\dfrac{y}{q}+\dfrac{z}{r}=3$

**Sol.** Let the required equation of plane be :

$\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=1$

Then the co-ordinates of A,B and C are A(a,0,0) B(0,b,0) C(0,0,c) respectively, so the centroid of the triangle ABC;

$\left(\dfrac{a}{3},\dfrac{b}{3},\dfrac{c}{3}\right)$

But the co-ordinate of the centroid are (p,q,r).

$\therefore \frac{a}{3}=p, \frac{b}{3}=q, \frac{c}{3}=r$

Putting the values of a, b and c in (i) we get The required plane as

$$\frac{x}{3p}+\frac{y}{3q}+\frac{z}{3r}=1$$

$$\frac{x}{p}+\frac{y}{q}+\frac{z}{r}=3$$

4. Find the vector equation of plane which is at a distance of 8 units from the origin and which is normal to the vector $2\hat{i}+\hat{j}+2\hat{k}$ :

Ans. $\vec{r}.\left(2\hat{i}+\hat{j}+2\hat{k}\right)=\pm 24$

**Sol.** Here, d = 8 and $\vec{n}=2\hat{i}+\hat{j}+2\hat{k}$

$$\therefore \hat{n}=\frac{\vec{n}}{|\vec{n}|}=\frac{2\hat{i}+\hat{j}+2\hat{k}}{\sqrt{2^2+1^2+2^2}}=\frac{2\hat{i}+\hat{j}+2\hat{k}}{3}$$

Hence, the required equation of plane is,

$\vec{r}.\hat{n}=d$

$$\Rightarrow \vec{r}\left(\frac{2\hat{i}+\hat{j}+2\hat{k}}{3}\right)=\pm 8$$

or $\vec{r}.\left(2\hat{i}+\hat{j}+2\hat{k}\right)=\pm 24$

5. Find the plane through the intersection of planes $\vec{r}.(2\hat{i}-3\hat{j}+4\hat{k})=1$ and $\vec{r}.(\hat{i}-\hat{j})+4=0$ and perpendicular to

$\vec{r}.(2\hat{i}-\hat{j}+\hat{k})=-8$

Ans. $\vec{r}.(-5\hat{i}+2\hat{j}+12\hat{k})=47$

**Sol.** The equation of any plane through the line of intersection of the given plane is :

$$\left\{\vec{r}.(2\hat{i}-3\hat{j}+4\hat{k})-1\right\}+\lambda\left\{\vec{r}.(\hat{i}-\hat{j})+4\right\}=0$$

or $\vec{r}.(2+\lambda)\hat{i}-(3+\lambda)\hat{j}+4\hat{k}\}=1-4\lambda$ ....(i)

If it is perpendicular to;

$\vec{r}.(2\hat{i}-j+\hat{k})+8=0$ then,

$$\{(2+\lambda)\hat{i}-(3+\lambda)\hat{j}+4\hat{k}\}.(2\hat{i}-\hat{j}+\hat{k})=0$$

$\Rightarrow 2(2+\lambda)+(3+\lambda)+4=0$

$$\Rightarrow \lambda=\frac{-11}{3}$$

Putting $\lambda = \frac{-11}{3}$ in (i), we obtian the equation of the required plane as ;

$\vec{r}.(-5\hat{i}+2\hat{j}+12\hat{k}) = 47$

6. The equation of the plane passing through the lines $\frac{x-4}{1} = \frac{y-3}{1} = \frac{z-2}{2}$ and $\frac{x-3}{1} = \frac{y-2}{-4} = \frac{z}{5}$ is :

Ans. $13x - 3y - 5z = 33$

Sol. $\vec{n} = \vec{p} \times \vec{q} = (1,1,2) \times (1,-4,5)$

$= (13, -3, -5)$

Let equation of plane is

$13x - 3y - 5z = \lambda$

(4, 3, 2) will satisfy it

$\Rightarrow \lambda = 52 - 9 - 10 = 33$

so equation of plane

$13x - 3y - 5z = 33$

7. Find the length and foot of perpendicular from the point (7, 14, 5) to the plane $2x + 4y - z = 2$

Ans. $\frac{63}{\sqrt{21}}, (1,2,8)$

**Sol.** The required length

$= \frac{2(7)+4(14)-(5)-2}{\sqrt{2^2+4^2+1^2}} = \frac{14+56-5-2}{\sqrt{4+16+1}}$

$= \frac{63}{\sqrt{21}}$

Let the co-ordinates of the foot of perpendicular from the point P(7,14,5) be M($\alpha$, $\beta$, $\gamma$,).

Then the direction ratios of PN are

($\alpha$ –7, $\beta$ – 14, $\gamma$ – 5)

i.e., d. r' s of normal to the plane :

($\alpha$ – 7, $\beta$ – 14, $\gamma$ – 5)

But the d. r' s normal to the given plane

2x + 4y – z = 2 are 2, 4, –1

Hence, $\frac{\alpha-7}{2} = \frac{\beta-14}{4} = \frac{\gamma-5}{-1} = k$

$\therefore \alpha = 2k + 7, \beta = 4k + 14, \gamma = -k + 5$ ...(i)

Since $\alpha$, $\beta$, $\gamma$ lies on the plane 2x + 4y –z = 2

$\therefore 2\alpha + 4\beta - \gamma = 2$

$\Rightarrow 2(7 + 2k) + 4(14 + 4k) - (5 - k) = 2$

$\Rightarrow 14 + 4k + 56 + 16k - 5 + k = 2$

$\Rightarrow 21k + 65 = 2$

$\Rightarrow k = -3$

Now, putting k = –3 in (i), we get

$\alpha = 1,\ \beta = 2,\ \gamma = 8$

Hence the co-ordinates of the foot of the perpendicular are (1, 2, 8)

8. Find the distance between the parallel planes $x+2y-2z+1=0$ and $2x+4y-4z+5=0$

Ans. $\frac{1}{2}$

**Sol.** We know, distance between parallel planes.

$ax + by + cz + d_1 = 0$ and

$ax + by + cz + d_2 = 0$ is,

$$\frac{|d_1 - d_2|}{\sqrt{a^2+b^2+c^2}}$$

$\therefore$ distance between $x + 2y - 2z + 1 = 0$ and $x + 2y - 2z + \frac{5}{2} = 0$

is, $\frac{|5/2-1|}{\sqrt{1+4+4}} = \frac{1}{2}$

9. Find the angle between the line $\vec{r} = \left(\hat{i}+2\hat{j}-\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}+\hat{k}\right)$ and the plane $\vec{r}.(2\hat{i}-\hat{j}+\hat{k}) = 4$ :

Ans. $\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

**Sol.** We know that the angle $\theta$ between the line

$\vec{r} = \vec{a}+\lambda\vec{b}$ and the plane $\vec{r}.\vec{n} = d$ is given by

$$\sin\theta = \frac{\vec{b}.\vec{n}}{|\vec{b}||\vec{n}|}$$

Here $\vec{b} = \hat{i}-\hat{j}+\hat{k}$ and $\vec{n} = 2\hat{i}-\hat{j}+\hat{k}$

$$\therefore \sin\theta = \frac{(\hat{i}-\hat{j}+\hat{k}).(2\hat{i}-\hat{j}+\hat{k})}{\sqrt{1+1+1}\sqrt{4+1+1}}$$

$$= \frac{4}{3\sqrt{2}} = \frac{2\sqrt{2}}{3}$$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$$

10. Find the angle between the line $\frac{x+1}{3} = \frac{y-1}{2} = \frac{z-2}{4}$ and the plane $2x + y - 3z + 4 =$

Ans. $\sin^{-1}\left(\frac{4}{\sqrt{406}}\right)$

**Sol.** The given line is parallel to the vector $\vec{b} = 3\hat{i} + 2\hat{j} + 4\hat{k}$ and the given plane is normal to the vector

$\vec{n} = 2\hat{i} + \hat{j} - 3\hat{k}$

$$\sin\theta = \frac{\vec{b}.\vec{n}}{|\vec{b}||\vec{n}|}$$

$$= \frac{(3\hat{i}+2\hat{j}+4\hat{k}).(2\hat{i}+\hat{j}-3\hat{k})}{\sqrt{3^2+2^2+4^2}\sqrt{2^2+1^2+3}}$$

$$= \frac{6+2-12}{\sqrt{29}\sqrt{14}} = \frac{-4}{\sqrt{406}}$$

$$\theta = \sin^{-1}\left(\frac{-4}{\sqrt{406}}\right)$$

11. The equation of the plane passing through the line $\frac{x-1}{5} = \frac{y+2}{6} = \frac{z-3}{4}$ and the point (4,3,7) is

Ans. $4x - 8y + 7z = 41$

**Sol.** Any plane through given line is

$A(x - 1) + B(y + 2) + C(z - 3) = 0$ ..(i)

and

$5A + 6B + 4C = 0$ ..(ii)

Since. Plane (i) passes through (4,3,7), we get

$3A + 5B + 4C = 0$ ..(iii)

Solving (ii) and (iii), we get $\frac{A}{4} = \frac{B}{-8} = \frac{C}{7}$

$\therefore$ Equation of required plane is

$4x - 8y + 7z = 41$

12. The equation of the plane through the point (2, –1, –3) and parallel to the lines $\frac{x-1}{3} = \frac{y+2}{2} = \frac{z}{-4}$ and $\frac{x}{2} = \frac{y-1}{-3} = \frac{z-2}{2}$ is :

Ans. $8x + 14y + 13z + 37 = 0$

**Sol.** Equation of plane passing throught the point (2, –1, –3) is,

Also, $A(x - 2) + B(y + 1) + C(z + 3) = 0$

Also, $3A + 2B - 4C = 0$ and

$2A - 3B + 2C = 0$

$\therefore \frac{A}{-8} = \frac{B}{-14} = \frac{C}{-13} = k$, (Let)

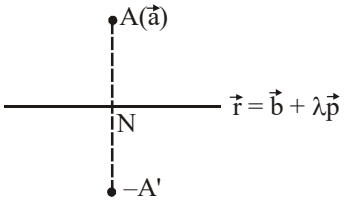
So, $A = -8k$, $B = -14k$, $C = -13k$

Equation of required plane is,

$-k[8(x-2) + 14(y+1) + 13(z+3)] = 0$

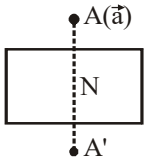
i.e., $8x + 14y + 13z + 37 = 0$

## 9.1 CONCEPT OF IMAGE IN 3D : IMAGE OF A POINT WRT A LINE




- Get foot of perpendicular N
- N is midpoint of AA'
- get A'
- A' = Image of a w.r.t. line

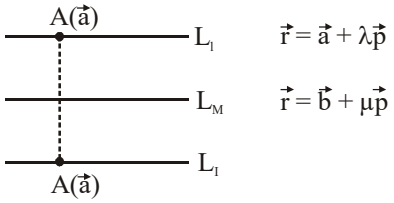
## 9.2 CONCEPT OF IMAGE IN 3D : IMAGE OF A POINT WRT A PLANE




- Get foot of perpendicular N
- N is midpoint of AA'
- get A'
- A' = Image of a w.r.t. plane

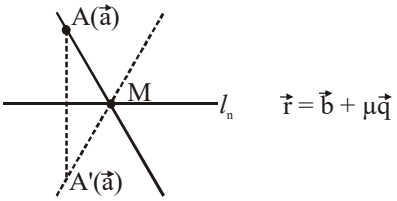
## 9.3 CONCEPT OF IMAGE IN 3D : IMAGE OF A LINE WRT A LINE

Case-I : Both lines are parallel




- Required line will be parallel to given line.
- Take image of point A wrt miror line that is A'
- Equation of required line will be $\vec{r} = \overrightarrow{a'} + t\vec{p}$

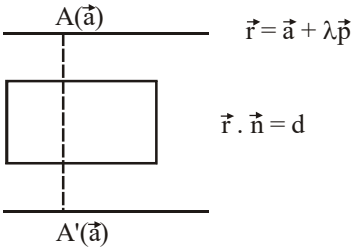
Case-II : Both lines are intersecting




- Get point of intersection M
- Get image of point A' WRT mirror line that is A'($\vec{a'}$)
- Equation of required line will be $\vec{r} = \vec{a'} + t\vec{A'M}$

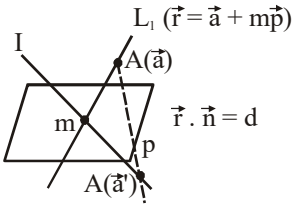
## 9.4 CONCEPT OF IMAGE IN 3D : IMAGE OF A LINE WRT A PLANE

Case-1 Line is parallel to plane




- Required line will be parallel to the given line.
- Take image of point A wrt the given plane that is A'($\vec{a'}$).
- Equation of reqiured line will be $\vec{r} = \vec{a}' + t(\vec{p})$

Case-2 Line intersects the plane




- Get point of intersection M
- Get image of point A' WRT mirror line that is A'($\vec{a'}$)
- Two point are known so equation of required line can be obtained
- Equation of required line will be $\vec{r} = \vec{a'} + t\vec{A'M}$

## 9.5 CONCEPT OF IMAGE IN 3D : IMAGE OF A PLANE WRT A PLANE

Case-1 Both planes are prallel –



$ax + by + cz = d_1$



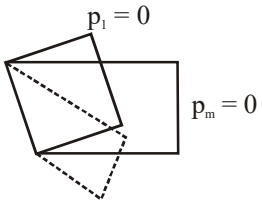
$ax + by + cz = d$



$ax + by + cz = \lambda$

where $d_1$, d and $\lambda$ are in A.P.

Case-2 Both planes are intersecting –




Equation required plane $P_1 + \lambda P_m = 0$

to get $\lambda$

M-I

- Angle between $P_1$ and $P_m$ =Angle between $P_1 + \lambda P_m$ and $P_m$

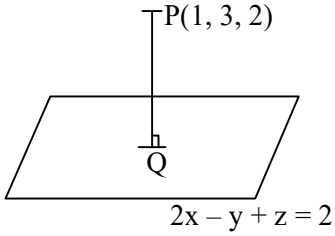
M-II

- Assume a fixed point on plane $P_1$
- Take its image wrt mirror plane
- Its image point will lie on image plane.

Q1 Find image of P(1, 3, 2) wrt plane $2x - y + z = 2$

Ans. $\left(\frac{5}{3}, \frac{8}{3}, \frac{7}{3}\right)$

Sol.




Q is foor of perpendicular

Equation of line PQ

$\vec{r} = (1,3,2) + \lambda(2,-1,1)$

let $Q(1 + 2\lambda, 3 - \lambda, 2 + \lambda)$

it will satisfy equation of plane

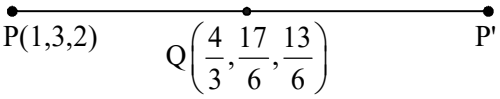
$\Rightarrow 2 + 4\lambda - 3 + \lambda + 2 + \lambda = 2$

$\Rightarrow \lambda = \frac{1}{6}$

So co-ordinetes of $Q\left(\frac{4}{3}, \frac{17}{6}, \frac{13}{6}\right)$

let P' be the image.

Now Q is the mid point of P and P'




So Co-ordinetes of $P'\left(\frac{5}{3}, \frac{8}{3}, \frac{7}{3}\right)$

Q.2 Find image of following lines wrt plane $2x - y + z = 2$

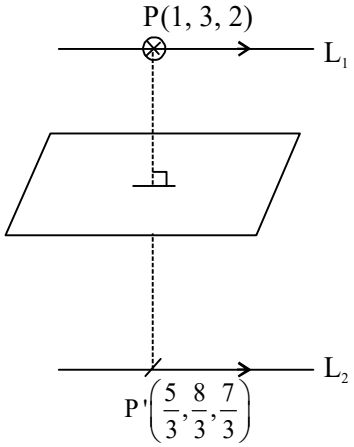
(i) $\vec{r} = (1, 3, 2) + \lambda(2, 3, -1)$

Ans. $\left(\frac{5}{3}, \frac{8}{3}, \frac{7}{3}\right) + \mu(2,3,-1)$

Sol. $\vec{p}.\vec{n} = (2,3,-1).(2,-1,1)$

$= 4 - 3 - 1 = 0$

so line is parallel to plane




so equation of line $L_2$

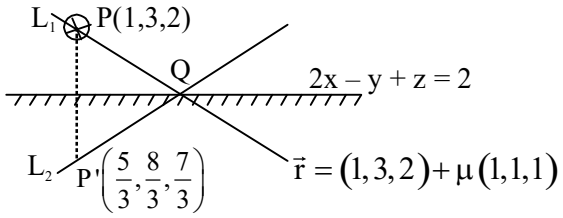
$$\vec{r} = \left(\frac{5}{3}, \frac{8}{3}, \frac{7}{3}\right) + \lambda(2,3,-1)$$

(ii) $\vec{r} = (1, 3, 2) + \mu(1, 1, 1)$

Ans. $\left(\frac{5}{3}, \frac{8}{3}, \frac{7}{3}\right) + \lambda(-1,5,1)$

Sol. $\vec{p}.\vec{n} \neq 0$

line is intrsecting the plane




Q is the pont of intersection

Let Q on line is $(1 + \mu, 3 + \mu, 2 + \mu)$

Satisfying plane

$2(1 + \mu) - (3 + \mu) + (2 + \mu) = 2$

$\Rightarrow 2 + 2\mu - 3 - \mu + 2 + \mu = 2$

$$\Rightarrow \mu = \frac{1}{2}$$

So Q is $\left(\frac{3}{2}, \frac{7}{2}, \frac{5}{2}\right)$

$$\overrightarrow{P'Q} = \left(\frac{1}{6}, \frac{-5}{6}, \frac{-1}{6}\right)$$

Equation of image $L_2$

$$\vec{r} = \left(\frac{5}{3}, \frac{8}{3}, \frac{7}{3}\right) + \lambda'(-1,5,1)$$

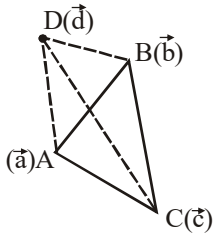
Note :

If image and foot of perpendicular of a point $P(x_1, y_1, z_1)$ wrt a plane $ax + by + cz + d = 0$ are $N(x_2, y_2, z_2)$ and $Q(x_3, y_3, z_3)$ then

$$\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{z_2 - z_1}{c} = -\left(\frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}\right)$$

$$\frac{x_3 - x_1}{a} = \frac{y_3 - y_1}{b} = \frac{z_3 - z_1}{c} = -2\left(\frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}\right)$$

## 10.1 TETRAHEDRON AND ITS PROPERTIES




Δ– ABC

1. It is a pyramid with triangular base.
2. Four vertices, 6 edges and 4 faces.

3. DA, DB, DC are lateral edges

4. Opposite edges

AB, CD

AC, BD

AD, BC

5. Coterminus edges eg : AB, AC, AD

6. Volume of tetrahedron = $\frac{1}{3}$(area of base )(Height)

7. Height of tetrahedron - projection of any lateral side along normal vector of base.

8. Centre of tetrahedron -

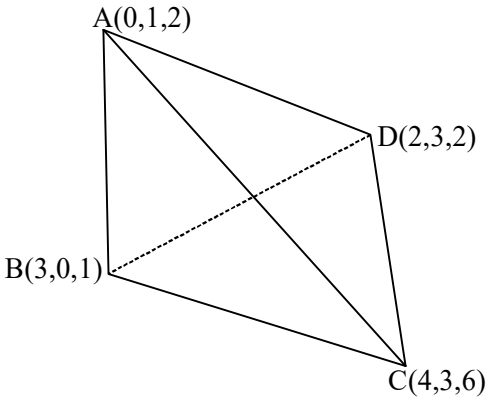
(i) its position vector is given by $\frac{\left(\vec{a}+\vec{b}+\vec{c}+\vec{d}\right)}{4}$

(ii) It is point of intersection of lines joining mid point of opposite edges.

(iii) It is point of intersection of line joining vertices to centriod of opposite face.

9. Regular tetrahedral means each face is equilateral $\Delta$.

**Q.1** For the tetrahedronABCD where A(0,1,2), B(3,0,1), C(4,3,6), D(2,3,2) where BCD is base compute the following




(i) Prove that corresponding to the point of concurrency of the join of mid points of each pair of opposite edges.

Ans. $\frac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}$

Sol. PV of centre $=\left(\frac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}\right)$

$=\left(\frac{9}{4},\frac{7}{4},\frac{11}{4}\right)$

(ii) Vector perpendicular to plane BCD.

Ans. $\mu\left(2\hat{i}+\hat{j}-\hat{k}\right)$

Sol. $\overrightarrow{BC}=(1,3,5)$

$\overrightarrow{BD}=(-1,3,1)$

$\overrightarrow{BC}\times\overrightarrow{BD}=(1,3,5)\times(-1,3,1)$

$=(-12,-6,6)$

So any vector perpendicular to plane BCD

$=\mu(2,1,-1)$

(iii) Equation of line through A and to perpendicular BCD.

Ans. $\hat{j}+2\hat{k}+\mu\left(2\hat{i}+\hat{j}-\hat{k}\right)$

Sol. $\vec{r}=(0,1,2)+\mu(2,1,-1)$

(iv) Volume of tetrahedron

Ans. V = 6

Sol. Height of tetrahedron $=\overrightarrow{AB}.\hat{n}$

$=(3,-1,-1).\dfrac{(2,1,-1)}{\sqrt{6}}$

$=\sqrt{6}$

Volume of tetrahedron = $\dfrac{1}{3}$ (Area of the base BCD) × h

$=\dfrac{1}{3}\times\left(\dfrac{1}{2}\times\left|\overrightarrow{BC}\times\overrightarrow{BD}\right|\right)\times\sqrt{6}$

$=\dfrac{1}{6}\sqrt{12^2+6^2+6^2}\times\sqrt{6}$

$=\dfrac{1}{6}.6.\sqrt{6}.\sqrt{6}$

$=6$

(v) Unit vectors normal to the plane face ABC and ADC.

Ans. $\pm\left(\dfrac{\hat{i}+8\hat{j}-5\hat{k}}{3\sqrt{10}}\right),\pm\left(\dfrac{-2\hat{i}+2\hat{j}+\hat{k}}{3}\right)$

Sol. $\overrightarrow{AB}\times\overrightarrow{AC}=(3,-1,-1)\times(4,2,4)$

$=(-2,-16,10)=-2(1,8,-5)$

unit vector normal to face ABC

$=\pm\dfrac{(1,8,-5)}{3\sqrt{10}}$

& $\overrightarrow{AD}\times\overrightarrow{AC}=(2,2,0)\times(4,2,4)$

$= (8, -8, -4) = 4(2, -2, -1)$

unit vector normal to face ADC

$$= \pm\frac{(2,-2,-1)}{3}$$

(vi) Acute angle between the planes ABC and ADC.

Ans. $\cos^{-1}\frac{1}{\sqrt{10}}$

Sol. $\vec{n}_1 = (1,8,-5)$

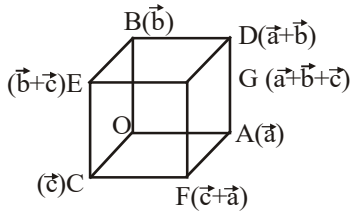
$\vec{n}_2 = (2,-2,-1)$

$$\cos\theta = \left|\frac{\vec{n}_1.\vec{n}_2}{|\vec{n}_1||\vec{n}_2|}\right|$$

$$= \left|\frac{2-16+5}{3\sqrt{10}.3}\right|$$

$$= \frac{1}{\sqrt{10}}$$

## 10.2 PARALLELOPIPED AND ITS PROPERTIES




1. It is a prism with parallelogram base.
2. 8 vertices, 12 edges, 6 faces.
3. All faces are parallelogram.
4. Sets of parallel edges–

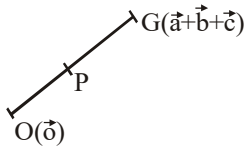
   DA, BD, EG, CF

   DB, AD, CE, FG

   DC, AF, BE, DG

5. 12 face diagonals, 4 body diagonals.
6. Volume = (area of base) × height
7. Height = projection of any lateral side vector along normal vector of base.
8. Center of parallelopiped -

(i) it is midpoint of body diagonal.




$\vec{P}\left(\frac{\vec{a}+\vec{b}+\vec{c}}{2}\right)$

$\overrightarrow{OP}=\frac{\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}}{2}$; $\overrightarrow{FP}=\frac{\overrightarrow{FC}+\overrightarrow{FG}+\overrightarrow{FA}}{2}$

## 11.1 SCALAR TRIPLE PRODUCT (STP)

The scalar triple product of three vectors $\vec{a},\vec{b}$ and $\vec{c}$ is defined as: $\vec{a}\times\vec{b}\,.\,\vec{c}=|\vec{a}||\vec{b}||\vec{c}|\sin\theta.\cos\phi$

where $\theta$ is the angle between $\vec{a},\vec{b}\left(\text{i.e.}\vec{a}\ \vec{b}=\theta\right)$

$\phi$ is the angle between $\vec{a},\vec{b}$ and $\vec{c}$ $\left(\text{i.e.}\left(\vec{a}\times\vec{b}\right)\ \vec{c}=\phi\right)$

It is (i.e. $\vec{a}\times\vec{b}.\vec{c}$ ) also written as $\left[\vec{a}\,\vec{b}\vec{c}\right]$ and spelled as box product.

**Properties**

1. If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k};\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$, then $[\vec{a}\ \vec{b}\ \vec{c}]=\begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}$

2. $[\vec{a}\ \vec{b}\ \vec{c}]=[\vec{b}\ \vec{c}\ \vec{a}]=[\vec{c}\ \vec{a}\ \vec{b}]$

3. $\vec{a}\,.\,(\vec{b}\times\vec{c})=(\vec{a}\times\vec{b})\,.\,\vec{c}$
   In a scalar triple product the position of dot and cross can be interchanged

4. $[\vec{a}\ \vec{b}\ \vec{c}]=-[\vec{a}\ \vec{c}\ \vec{b}]$

5. Scalar product of three vectors, two of which are equal or parallel is 0 $\Rightarrow$ $[\vec{a}\ \vec{b}\ \vec{c}]=0$

6. $\left[\hat{i}\hat{j}\hat{k}\right]=1$

7. $[K\vec{a}\ \vec{b}\vec{c}\ ]=K[\vec{a}\,\vec{b}\vec{c}]$

8. $[(\vec{a}+\vec{b})\ \vec{c}\ \vec{d}]=[\vec{a}\ \vec{c}\vec{d}]+[\vec{b}\vec{c}\vec{d}]$

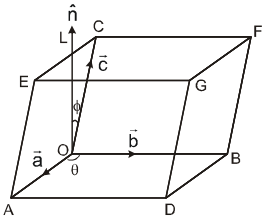
9. $\left[\vec{l}\quad\vec{m}\quad\vec{n}\right]\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\begin{vmatrix}\vec{l}.\vec{a} & \vec{l}.\vec{b} & \vec{l}.\vec{c}\\ \vec{m}.\vec{a} & \vec{m}.\vec{b} & \vec{m}.\vec{c}\\ \vec{n}.\vec{a} & \vec{n}.\vec{b} & \vec{n}.\vec{c}\end{vmatrix}$

$$\left[\vec{a}\ \vec{b}\vec{c}\right]^2 = \begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix}$$

10. If $\vec{a} = a_1\vec{\ell} + a_2\vec{m} + a_3\vec{n}; \vec{b} = b_1\vec{\ell} + b_2\vec{m} + b_3\vec{n}$ and $\vec{c} = c_1\vec{\ell} + c_2\vec{m} + c_3\vec{n}$

then $\left[\vec{a}\,\vec{b}\vec{c}\right] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \left[\vec{\ell}\,\vec{m}\,\vec{n}\right]$

## 11.2 VOLUME OF PARALLELOPIPED




Volume = (area of base) × height

Volume = (OA)(OB)sin θ (OC)cos ϕ

Volume = $\left[\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}\right]$

Volume = [STP of 3 coterminus side vectors]

## 11.3 VOLUME OF TETRAHEDRON

Volume = $\frac{1}{3}$ (area of base) × height

Volume = $\frac{1}{3} \times \frac{1}{2}$(OA)(OB)sin θ (OC)cos ϕ

Volume = $\frac{1}{6}\left[\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}\right]$

Volume = $\frac{1}{6}$[STP of 3 coterminus side vectors]

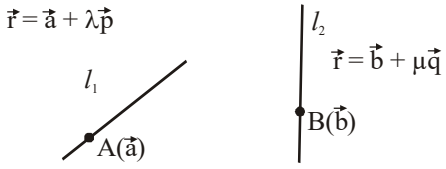
## 11.4 COPLANARITY OF THREE VECTORS

If $\vec{a}, \vec{b}, \vec{c}$ are coplanar $\Leftrightarrow \left[\vec{a}, \vec{b}, \vec{c}\right] = 0$

1. If points A, B, C, D are coplanar

$[\vec{AB} \quad \vec{AC} \quad \vec{AD} = 0]$

2. If lines $l_1$ and $l_2$ are coplanar.




If $l_1$, $l_2$ are coplanar then $[\vec{AB} \quad \vec{p} \quad \vec{q}] = 0$ or $\overrightarrow{AB}.(\vec{p} \times \vec{q}) = 0$

**Questions**

**1.** Find the value of p for which the vectors $(p+1)\hat{i} - 3\hat{j} + p\hat{k}$, $p\hat{i} + (p+1)\hat{j} - 3\hat{k}$ and $-3\hat{i} + p\hat{j} + (p+1)\hat{k}$ are linearly dependent/coplanar.

Ans. p = 1

Sol. Vectors will be coplanar if

$$\begin{vmatrix} p+1 & -3 & p \\ p & p+1 & -3 \\ -3 & p & p+1 \end{vmatrix} = 0$$

$\Rightarrow$ Either

$(p + 1) + (-3) + p = 0$

$\Rightarrow p = 1$

or $p + 1 = -3 = p$

not possible

2. Find the value of $\lambda$ for which three points with p.v.s. A (1, 0, 3); B(–1, 3, 4); C(1, 2, 1) and D($\lambda$, 2, 5) are in the same plane.

Sol. $[AB \ AC \ AD] = 0 \Rightarrow \lambda = -1$

Sol. $\overrightarrow{AB} = (-2, 3, 1)$

$\overrightarrow{AC} = (0, 2, -2)$

$\overrightarrow{AD} = (\lambda - 1, 2, 2)$

If A, B, C, D are in same plane

$\left[\overrightarrow{AB} \ \overrightarrow{AC} \ \overrightarrow{AD}\right] = 0$

$$\Rightarrow \begin{vmatrix} -2 & 3 & 1 \\ 0 & 2 & -2 \\ \lambda - 1 & 2 & 2 \end{vmatrix} = 0$$

$\Rightarrow -16 + 3(-2\lambda + 2) - 2(\lambda - 1) = 0$

$\Rightarrow -16 - 6\lambda + 6 - 2\lambda + 2 = 0$

$\Rightarrow \lambda = -1$

3. (a) Prove that $\left[\vec{a}+\vec{b},\vec{b}+\vec{c},\vec{c}+\vec{a}\right]=2\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

**Sol. :** We have $[\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}]$

$=\left\{(\vec{a}+\vec{b})\times(\vec{b}+\vec{c})\right\}.(\vec{c}+\vec{a})=\left\{\vec{a}\times\vec{b}+\vec{a}\times\vec{c}+\vec{b}\times\vec{b}+\vec{b}\times\vec{c}\right\}.(\vec{c}+\vec{a})\quad\left\{\because\vec{b}\times\vec{b}=0\right\}$

$=\left\{\vec{a}\times\vec{b}+\vec{a}\times\vec{c}+\vec{b}\times\vec{c}\right\}.(\vec{c}+\vec{a})=(\vec{a}\times\vec{b}).\vec{c}+(\vec{a}\times\vec{c}).\vec{c}+(\vec{b}\times\vec{c}).\vec{c}+(\vec{a}\times\vec{b}).\vec{a}+(\vec{a}\times\vec{c}).\vec{a}+(\vec{b}\times\vec{c}).\vec{a}$

$=\left[\vec{a}\,\vec{b}\,\vec{c}\right]+0+0+0+0+\left[\vec{b}\,\vec{c}\,\vec{a}\right]\ \{\because[\vec{a}\,\vec{c}\,\vec{c}]=0,[\vec{b}\,\vec{c}\,\vec{c}]=0,[\vec{a}\,\vec{b}\,\vec{a}]=0,[\vec{a}\,\vec{c}\,\vec{a}]=0\}$

$=[\vec{a}\,\vec{b}\,\vec{c}]+[\vec{a}\,\vec{b}\,\vec{c}]=2[\vec{a}\,\vec{b}\,\vec{c}]$

(b) Prove that $[\vec{a}-\vec{b},\vec{b}-\vec{c},\vec{c}-\vec{a}]$ is always zero.

Sol We have :

$[\vec{a}-\vec{b}\quad\vec{b}-\vec{c}\quad\vec{c}-\vec{a}]=\{(\vec{a}-\vec{b})\times(\vec{b}-\vec{c})\}\cdot(\vec{c}-\vec{a})$ [By definition]

$=(\vec{a}\times\vec{b}-\vec{a}\times\vec{c}-\vec{b}\times\vec{b}+\vec{b}\times\vec{c})\cdot(\vec{c}-\vec{a})$ [By distribution law]

$=(\vec{a}\times\vec{b}+\vec{c}\times\vec{a}+\vec{b}\times\vec{c})\cdot(\vec{c}-\vec{a})$ $[\because\ \vec{b}\times\vec{b}=\vec{0}]$

$=(\vec{a}\times\vec{b})\cdot\vec{c}-(\vec{a}\times\vec{b})\cdot\vec{a}+(\vec{c}\times\vec{a})\cdot\vec{c}-(\vec{c}\times\vec{a})\cdot\vec{a}+(\vec{b}\times\vec{c})\cdot\vec{c}-(\vec{b}\times\vec{c})\cdot\vec{a}$ [By distribution law]

$=[\vec{a}\ \vec{b}\ \vec{c}]-[\vec{a}\ \vec{b}\ \vec{a}]+[\vec{c}\ \vec{a}\ \vec{c}]-[\vec{c}\ \vec{a}\ \vec{a}]+[\vec{b}\ \vec{c}\ \vec{c}]-[\vec{b}\ \vec{c}\ \vec{a}]$

$=[\vec{a}\ \vec{b}\ \vec{c}]-[\vec{b}\ \vec{c}\ \vec{a}]$ [$\because$ When any two vectors are equal, scalar triple product is zero ]

$=[\vec{a}\ \vec{b}\ \vec{c}]-[\vec{a}\ \vec{b}\ \vec{c}]=0$ $[\because\ [\vec{b}\ \vec{c}\ \vec{a}]=[\vec{a}\ \vec{b}\ \vec{c}]]$

(c) Show that $\vec{a}\times\vec{b}=[\hat{i}\,\vec{a}\,\vec{b}]\hat{i}+[\hat{j}\,\vec{a}\,\vec{b}]\hat{j}+[\hat{k}\,\vec{a}\,\vec{b}]\hat{k}$

**Sol. :** Let $\vec{a}\times\vec{b}=x\hat{i}+y\hat{j}+z\hat{k}$

$(\vec{a}\times\vec{b}).\hat{i}=(x\hat{i}+y\hat{j}+z\hat{k}).\hat{i}$

$(\vec{a}\times\vec{b}).\hat{i}=x$

also $(\vec{a}\times\vec{b}).\hat{j}=y\quad\&\quad(\vec{a}\times\vec{b}).\hat{k}=z$

$\therefore\vec{a}\times\vec{b}=[\vec{a}\,\vec{b}\,\hat{i}]\hat{i}+[\vec{a}\,\vec{b}\,\hat{j}]\hat{j}+[\vec{a}\,\vec{b}\,\hat{k}]\hat{k}$

4. If three non-zero unequal vectors $\vec{l}=a\hat{i}+b\hat{j}+c\hat{k},\vec{m}=b\hat{i}+c\hat{j}+a\hat{k}$ and $\vec{n}=c\hat{i}+a\hat{j}+b\hat{k}$ are such that $\vec{r}.\vec{l}=\vec{r}.\vec{m}=\vec{r}.\vec{n}=0$ where a, b, $c\in R_0$ and $\vec{r}$ is non zero vector, then value of $\frac{a^2}{bc}+\frac{b^2}{ca}+\frac{c^2}{ab}$ :

Ans. 3

Sol. $\vec{r}.\vec{\ell}=\vec{r}.\vec{M}=\vec{r}.\vec{n}=0$

$\Rightarrow\vec{r}\perp\vec{\ell},\vec{r}\perp\vec{M},\vec{r}\perp\vec{n}$

$\Rightarrow\vec{\ell},\vec{m},\vec{n}$ are coplanar vector

$\Rightarrow\left[\vec{\ell}\ \vec{m}\ \vec{n}\right]=0$

$$\Rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

$\Rightarrow$ Either $a + b + c = 0$ OR $a = b = c$ [not possible]

$\Rightarrow a + b + c = 0$

Now, $\dfrac{a^2}{bc} + \dfrac{b^2}{ac} + \dfrac{c^2}{cb} = \dfrac{a^3 + b^3 + c^3}{abc} = 3$

$\left\{ \because a + b + c = 0 \quad \Rightarrow a^3 + b^3 + c^3 = 3abc \right\}$

5. $\vec{a}.(\vec{b} + \vec{c}) \times (\vec{a} + \vec{b} + \vec{c})$ is equal to :

(A) 0 (B) 2 [abc] (C) [abc] (D) None of these

Ans. A

**Sol.** $\vec{a}.(\vec{b} + \vec{c}) \times (\vec{a} + \vec{b} + \vec{c})$

$= \vec{a}.[(\vec{b} + \vec{c}) \times \vec{a} + (\vec{b} + \vec{c}) \times \vec{b} + (\vec{b} + \vec{c}) \times \vec{c}]$

$= \vec{a}.[(\vec{b} \times \vec{a} + \vec{c} \times \vec{a} + \vec{b} \times \vec{b} + \vec{c} \times \vec{c} \times \vec{b} + \vec{c} \times \vec{c}]$

$= \vec{a}.[\vec{b} \times \vec{a} + \vec{c} \times \vec{a}]$

$= [\vec{a}\ \vec{b}\ \vec{c}] + [\vec{a}\ \vec{c}\ \vec{a}]$

$= 0$

6. If $\vec{a} = -3\hat{i} + 7\hat{j} + 5\hat{k}$, $\vec{b} = -3\hat{i} + 7\hat{j} - 3\hat{k}$ and $\vec{c} = 7\hat{i} - 5\hat{j} - 3\hat{k}$ are the coterminus edges of a parallelopiped then its volume is-

(A) 108 (B) 210 (C) 272 (D) 308

Ans. C

**Sol.** Required volume $= [\vec{a}\ \ \vec{b}\ \ \vec{c}]$

$$= \begin{vmatrix} -3 & 7 & 5 \\ -3 & 7 & -3 \\ 7 & -5 & -3 \end{vmatrix}$$

$= \{-3(-21 - 15) - 7(9 + 21) + 5(15 - 49)]$

$= |108 - 210 - 170|$

$= 272$

7. Let P be a point not on the plane that passes through Q, R and S. Show that the distance d from P to the plane is

$d = \left| \dfrac{[\vec{a}\,\vec{b}\,\vec{c}]}{|\vec{a} \times \vec{b}|} \right|$, where $\vec{a} = \overrightarrow{QR}$ ; $\vec{b} = \overrightarrow{QS}$ and $\vec{c} = \overrightarrow{QP}$.

Ans.

Sol. distance of p from plane

$= \overrightarrow{PQ}.\hat{n}$

where $\hat{n} = \dfrac{\overrightarrow{QR} \times \overrightarrow{QS}}{\left|\overrightarrow{QR} \times \overrightarrow{QS}\right|} = \dfrac{\vec{a} \times \vec{b}}{\left|\vec{a} \times \vec{b}\right|}$

So distance $= \left|\dfrac{\vec{c}.\left(\vec{a} \times \vec{b}\right)}{\left|\vec{a} \times \vec{b}\right|}\right| = \left|\dfrac{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}{\left|\vec{a} \times \vec{b}\right|}\right|$

8. Prove that : $\left[\vec{a}\,\vec{b}\,\vec{c}\right]\left(\vec{p} \times \vec{q}\right) = \begin{vmatrix} \vec{p}.\vec{a} & \vec{q}.\vec{a} & \vec{a} \\ \vec{p}.\vec{b} & \vec{q}.\vec{b} & \vec{b} \\ \vec{p}.\vec{c} & \vec{q}.\vec{c} & \vec{c} \end{vmatrix}$

Hints. Proof : $\left[\vec{a}\,\vec{b}\,\vec{c}\right]\left(\vec{p} \times \vec{q}\right) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{vmatrix} p_1 & q_1 & \hat{i} \\ p_2 & q_2 & \hat{j} \\ p_3 & q_3 & \hat{k} \end{vmatrix} = \begin{vmatrix} \vec{a}.\vec{p} & \vec{a}.\vec{q} & \vec{a} \\ \vec{b}.\vec{p} & \vec{b}.\vec{q} & \vec{b} \\ \vec{c}.\vec{p} & \vec{c}.\vec{q} & \vec{c} \end{vmatrix}$

## 12. VECTOR TRIPLE PRODUCT (VTP)

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be any three vectors, then the expression $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector & is called a vector triple product.

Consider the expression $\vec{a} \times (\vec{b} \times \vec{c})$ which itself is a vector, since it is a cross product of two vectors $\vec{a}$ and $(\vec{b} \times \vec{c})$. Now $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector perpendicular to the plane containing $\vec{a}$ and $(\vec{b} \times \vec{c})$ but $\vec{b} \times \vec{c}$ is a vector perpendicular to the plane containing $\vec{b}$ and $\vec{c}$, therefore $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector which lies in the plane of $\vec{b}$ and $\vec{c}$ and perpendicular to $\vec{a}$.

$(\vec{a} \text{ x } \vec{b}) \text{ x } \vec{c} = (\vec{a} \,.\, \vec{c})\vec{b} - (\vec{b} \,.\, \vec{c})\vec{a}$

1. $\left(\vec{a} \times \vec{b}\right) \times \vec{c}$ = (dot of others)middle vector – (dot of others)other vector

2. Whenever we assume a vector which is coplane with $\vec{p}$ and $\vec{q}$ and perpendicular to $\vec{r}$ we will take it as $\lambda\left(\vec{p} \times \vec{q}\right) \times \vec{r}$

**1.** If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors, then prove that $\left[\vec{a} \times \vec{b}\ \vec{b} \times \vec{c}\ \vec{c} \times \vec{a}\right] = \left[\vec{a}\,\vec{b}\,\vec{c}\right]^2 = \begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix}$

**Sol. :** We have, $\left[\vec{a} \times \vec{b}\ \vec{b} \times \vec{c}\ \vec{c} \times \vec{a}\right] = \left\{\left(\vec{a} \times \vec{b}\right) \times \left(\vec{b} \times \vec{c}\right)\right\}.\left(\vec{c} \times \vec{a}\right)$

$= \left\{\vec{d} \times \left(\vec{b} \times \vec{c}\right)\right\}.\left(\vec{c} \times \vec{a}\right)$ (where, $\vec{d} = (\vec{a} \times \vec{b})$)

$$= \left\{\left(\vec{d}.\vec{c}\right)\vec{b} - \left(\vec{d}.\vec{b}\right)\vec{c}\right\}.\left(\vec{c}\times\vec{a}\right) = \left\{\left(\left(\vec{a}\times\vec{b}\right).\vec{c}\right)\vec{b} - \left(\left(\vec{a}\times\vec{b}\right).\vec{b}\right)\vec{c}\right\}.\left(\vec{c}\times\vec{a}\right)$$

$$= \left\{\left[\vec{a}\vec{b}\vec{c}\right]\vec{b} - 0\right\}.\left(\vec{c}\times\vec{a}\right) \quad \left(\because [\vec{a}\vec{b}\vec{b}] = 0\right)$$

$$= \left[\vec{a}\vec{b}\vec{c}\right]\left\{\vec{b}.\left(\vec{c}\times\vec{a}\right)\right\} = \left[\vec{a}\vec{b}\vec{c}\right]\left[\vec{b}\vec{c}\vec{a}\right] = \left[\vec{a}\vec{b}\vec{c}\right]\left[\vec{a}\vec{b}\vec{c}\right] = \left[\vec{a}\vec{b}\vec{c}\right]^2$$

2. If $\left.\begin{array}{l}\vec{V}_1 = \vec{a}\times\left(\vec{b}\times\vec{c}\right)\\ \vec{V}_2 = \vec{b}\times\left(\vec{c}\times\vec{a}\right)\\ \vec{V}_3 = \vec{c}\times\left(\vec{a}\times\vec{b}\right)\end{array}\right]$, then which of the following is true :

(A) $\vec{V}_1, \vec{V}_2, \vec{V}_3$ are coplanar

(B) $\vec{V}_1, \vec{V}_2, \vec{V}_3$ can form the sides of a triangle

(C) $\vec{V}_1 + \vec{V}_2 + \vec{V}_3$ is a null vector

(D) $\vec{V}_1, \vec{V}_2, \vec{V}_3$ are linearly dependend

Ans. ABCD

Sol. $\vec{V}_1 = \vec{a}\times\left(\vec{b}\times\vec{c}\right)$

$$= \left(\vec{a}.\vec{c}\right)\vec{b} - \left(\vec{a}.\vec{b}\right).\vec{c}$$

$\vec{V}_2 = \vec{b}\times\left(\vec{c}\times\vec{a}\right)$

$$= \left(\vec{b}.\vec{a}\right)\vec{c} - \left(\vec{b}.\vec{c}\right).\vec{a}$$

$\vec{V}_3 = \vec{c}\times\left(\vec{a}\times\vec{b}\right)$

$$= \left(\vec{c}.\vec{b}\right)\vec{a} - \left(\vec{c}.\vec{a}\right)\vec{b}$$

$\Rightarrow \vec{V}_1 + \vec{V}_2 + \vec{V}_3 = 0$

3. (i) If $\hat{a}\times\left(\hat{b}\times\hat{c}\right) = \frac{1}{2}\hat{b}$, where $\hat{b}$ and $\hat{c}$ are non collinear then find the angle between $\hat{a}$ and $\hat{b}$ and between $\hat{a}$ and $\hat{c}$.

Ans.

Sol. $\hat{a}\times\left(\hat{b}\times\hat{c}\right) = \frac{1}{2}\hat{b}$

$\Rightarrow \left(\hat{a}.\vec{c}\right)\vec{b} - \left(\hat{a}.\hat{b}\right)\hat{c} = \frac{1}{2}\hat{b}$

as $\hat{b}.\hat{c}$ are non-collinear so they are linearly independent

$\Rightarrow \left(\hat{a}.\vec{c}\right) = \frac{1}{2}$ & $\hat{a}.\hat{b} = 0$

$\Rightarrow \cos\theta = \frac{1}{2}$ $\Rightarrow \hat{a} \perp \hat{b}$

$\Rightarrow \theta = 60°$

(ii) Let three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}, \vec{a}.\vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$, where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2\left|\vec{a} + \vec{b} + \vec{c}\right|^2$ is _____.

माना तीन सदिश $\vec{a}, \vec{b}$ तथा $\vec{c}$ इस प्रकार हैं कि $\vec{c}$, सदिशों $\vec{a}$ तथा $\vec{b}$ के समतल में है, $\vec{a}.\vec{c} = 7$ है तथा $\vec{b}$, सदिश $\vec{c}$ के लम्बवत् है, जबकि $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ तथा $\vec{b} = 2\hat{i} + \hat{k}$ हैं, तो $2\left|\vec{a} + \vec{b} + \vec{c}\right|^2$ बराबर है _____।

Ans. (i) $\frac{\pi}{2}$ ; $\frac{\pi}{3}$ (ii) 75

**Sol.** (ii) $\vec{c} = \lambda\left(\left(\vec{a} \times \vec{b}\right) \times \vec{b}\right)$

$\vec{c} = \lambda\left(\left(\vec{a} \cdot \vec{b}\right)\vec{b}\right) - \left(\vec{b} \cdot \vec{b}\right)\vec{a}\right)$

$\vec{c} = \lambda\left(-\vec{b} - 5\vec{a}\right)$

$\vec{a} \cdot \vec{c} = \lambda\left(-\vec{a} \cdot \vec{b} - 5\vec{a} \cdot \vec{a}\right) = 7$

$\lambda = \frac{-1}{2}$

$\vec{c} = \frac{5\vec{a} + \vec{b}}{2} = \frac{-3i + 5i + 6k}{2}$

$\vec{a} + \vec{b} + \vec{c} = \frac{-i + 7j + 10k}{2}$

$\left|\vec{a} + \vec{b} + \vec{c}\right| = \frac{1}{2} \times \sqrt{150}$

$2\left|\vec{a} + \vec{b} + \vec{c}\right|^2 = 75$

4. Show that : $\hat{i} \times \left(\vec{a} \times \hat{i}\right) + \hat{j} \times \left(\hat{a} \times \hat{j}\right) + \hat{k} \times \left(\vec{a} \times \hat{k}\right) = 2\vec{a}$

**Sol. :** Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$.

Then $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\hat{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k})$

$= \{(\hat{i} \,.\, \hat{i})\vec{a} - (\hat{i} \,.\, \vec{a})\hat{i}\} + \{(\hat{j} \,.\, \hat{j})\vec{a} - (\hat{j} \,.\, \vec{a})\hat{j}\} + \{(\hat{k} \,.\, \hat{k})\vec{a} - (\hat{k} \,.\, \vec{a})\hat{k}\}$

$= \{(\vec{a} - (\hat{i} \,.\, \vec{a})\,\hat{i}\} + \{\vec{a} - (\hat{j} \,.\, \vec{a})\hat{j}\} + \{\vec{a} - (\hat{k} \,.\, \vec{a})\hat{k}\}$

$= 3\vec{a} - \{(\hat{i}\,.\,\vec{a})\hat{i} + (\hat{j}\,.\,\vec{a})\hat{j} + (\hat{k}\,.\,\vec{a})\hat{k} = 3\vec{a} - (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) = 3\vec{a} - \vec{a} = 2\vec{a}$

5. If $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{c}$ is a vector such that $\left(\vec{a}\times\vec{b}\right)\times\vec{c} = \vec{a}\times\left(\vec{b}\times\vec{c}\right)$ & $|\vec{c}| = 6$. Then the volume of parallelopiped whose co-terminouse edges are $\vec{a}$, $\vec{b}$, $\vec{b}\times\vec{c}$ will be :

(A) 180 (B) 214 (C) 232 (D) 244

Ans D

Sol. $\left(\vec{a}\times\vec{b}\right)\times\vec{c} = \vec{a}\times\left(\vec{b}\times\vec{c}\right)$

$\Rightarrow (\vec{a}\cdot\vec{c})\vec{b} - (\vec{b}\cdot\vec{c})\vec{a} = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$

$\Rightarrow (\vec{b}.\vec{c})\vec{a} = (\vec{a}.\vec{b})\vec{c}$

$\Rightarrow \vec{a}$ and $\vec{c}$ are wllirve wehers

Ler $\vec{c} = \lambda\vec{a}$

$|\vec{c}| = \lambda|\vec{a}|$

$\Rightarrow 6 = 3\lambda$

$\Rightarrow \lambda = 2$

$\Rightarrow \vec{c} = 2\vec{a}$

Volume of parallelopiped

$= \left|\left[\vec{a}\ \vec{b}\quad \vec{b}\times\vec{c}\right]\right|$

$= \left|\left[\vec{a}\ \vec{b}\quad \vec{b}\times(2\vec{a})\right]\right|$

$v = 2\left|(\vec{a}\times\vec{b}).(\vec{a}\times\vec{b})\right|$ ..... (1)

$\vec{a}\times\vec{b} = (1,2,2)\times(2,-3,1) = (8,3,-7)$

So $\left|\vec{a}\times\vec{b}\right|^2 = 64 + 9 + 49 = 122$

from equation (1)

v = 2 (122) = 244

6. Prove that : $\left\{\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}\times\vec{c}\right)\right\}.\vec{d} = \left(\vec{a}.\vec{d}\right)\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

Ans. $\left[\vec{a}\vec{b}\vec{c}\right]\left(\vec{a}.\vec{d}\right)$

Sol. $\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}\times\vec{c}\right) = \vec{p}\times\left(\vec{a}\times\vec{c}\right)$

(ler $\vec{a}\times\vec{b} = \vec{p}$)

$= (\vec{p}.\vec{c})\vec{a} - (\vec{p}.\vec{a})\vec{c}$

$= \left((\vec{a}\times\vec{b}).\vec{c}\right)\vec{a} - \left((\vec{a}\times\vec{b}).\vec{a}\right)\vec{c}$

$= \left[\vec{a}\,\vec{b}\,\vec{c}\right]\vec{a}$

so $\left\{\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}\times\vec{c}\right)\right\}.\vec{d} = \left[\vec{a}\vec{b}\vec{c}\right]\left(\vec{a}.\vec{d}\right)$

7. Prove that : $\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{c}\times\vec{d}\right) = \begin{vmatrix}\vec{a}.\vec{c} & \vec{a}.\vec{d}\\ \vec{b}.\vec{c} & \vec{b}.\vec{d}\end{vmatrix}$

Sol. LHS : $\left(\vec{a}\times\vec{b}\right).\left(\vec{c}\times\vec{d}\right)$

$= \vec{a}.\left(\vec{b}\times\left(\vec{c}\times\vec{d}\right)\right)$

$= \vec{a}.\left\{\left(\vec{b}.\vec{a}\right)\vec{c} - \left(\vec{b}.\vec{c}\right)\vec{d}\right\}$

$= \left(\vec{a}.\vec{c}\right)\left(\vec{b}.\vec{d}\right) - \left(\vec{b}.\vec{c}\right)\left(\vec{a}.\vec{d}\right)$

= RHS

8. Prove that : $\vec{d}.\left\{\vec{a}\times\left(\vec{b}\times\left(\vec{c}\times\vec{d}\right)\right)\right\} = \left(\vec{b}.\vec{d}\right)\left[\vec{a}\,\vec{c}\,\vec{d}\right]$

Ans. $\left(\vec{b}.\vec{d}\right)\left[\vec{a}\,\vec{c}\,\vec{d}\right]$

Sol. $\vec{d}.\left(\vec{a}\times\left(\vec{b}\times\left(\vec{c}\times\vec{d}\right)\right)\right)$

$= \vec{d}.\left(\vec{a}\times\left(\left(\vec{b}.\vec{d}\right)\vec{c} - \left(\vec{b}.\vec{c}\right)\vec{d}\right)\right)$

$= \vec{d}.\left(\left(\vec{b}.\vec{d}\right)\left(\vec{a}\times\vec{c}\right) - \left(\vec{b}.\vec{c}\right)\left(\vec{a}\times\vec{d}\right)\right)$

$= \left(\vec{b}.\vec{d}\right)\left[\vec{d}\ \vec{a}\ \vec{c}\right] - \left(\vec{b}.\vec{c}\right)\left[\vec{d}\ \vec{a}\ \vec{d}\right]$

$= \left(\vec{b}.\vec{d}\right)\left[\vec{a}\,\vec{c}\,\vec{d}\right]$

9. Prove that : $\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{c}\times\vec{d}\right) + \left(\vec{b}\times\vec{c}\right)\cdot\left(\vec{a}\times\vec{d}\right) + \left(\vec{c}\times\vec{a}\right).\left(\vec{b}\times\vec{d}\right) = 0$

Ans. 0

Sol. $\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{c}\times\vec{d}\right) + \left(\vec{b}\times\vec{c}\right)\cdot\left(\vec{a}\times\vec{d}\right) + \left(\vec{c}\times\vec{a}\right).\left(\vec{b}\times\vec{d}\right)$

$= \begin{vmatrix}\vec{a}.\vec{c} & \vec{a}.\vec{d}\\ \vec{b}.\vec{c} & \vec{b}.\vec{d}\end{vmatrix} + \begin{vmatrix}\vec{b}.\vec{a} & \vec{b}.\vec{d}\\ \vec{c}.\vec{a} & \vec{c}.\vec{d}\end{vmatrix} + \begin{vmatrix}\vec{c}.\vec{b} & \vec{c}.\vec{d}\\ \vec{a}.\vec{b} & \vec{a}.\vec{d}\end{vmatrix}$

$= \left(\vec{a}.\vec{c}\right)\left(\vec{b}.\vec{d}\right) - \left(\vec{b}.\vec{c}\right)\left(\vec{a}.\vec{d}\right) + \left(\vec{b}.\vec{a}\right)\left(\vec{c}.\vec{d}\right) - \left(\vec{c}.\vec{a}\right)\left(\vec{b}.\vec{d}\right) + \left(\vec{c}.\vec{b}\right)\left(\vec{a}.\vec{d}\right) - \left(\vec{a}.\vec{b}\right)\left(\vec{c}.\vec{d}\right)$

= 0

10. Prove that : $\left(\vec{a}\times\vec{b}\right)\times\left(\vec{c}\times\vec{d}\right) + \left(\vec{a}\times\vec{c}\right)\times\left(\vec{d}\times\vec{b}\right) + \left(\vec{a}\times\vec{d}\right)\times\left(\vec{b}\times\vec{c}\right) = -2\left[\vec{b}\,\vec{c}\,\vec{d}\right]\vec{a}$

Ans. $\left[\vec{a}\ \vec{b}\ \vec{c}\right]\vec{d}-\left[\vec{a}\,\vec{c}\,\vec{d}\right]\vec{b}$

Sol. $\left(\vec{a}\times\vec{b}\right)\times\left(\vec{c}\times\vec{d}\right)=\left[\vec{a}\vec{c}\vec{d}\right]\vec{b}-\left[\vec{b}\,\vec{c}\,\vec{d}\right]\vec{a}$

$\left(\vec{a}\times\vec{c}\right)\times\left(\vec{d}\times\vec{b}\right)=\left[\vec{a}\,\vec{c}\,\vec{b}\right]\vec{d}-\left[\vec{a}\,\vec{c}\,\vec{d}\right]\vec{b}$

$\left(\vec{a}\times\vec{d}\right)\times\left(\vec{b}\times\vec{c}\right)=\left[\vec{a}\ \vec{b}\ \vec{c}\right]\vec{d}-\left[\vec{a}\,\vec{c}\,\vec{d}\right]\vec{b}$

## 13. LINEAR COMBINATION THEOREM IN SPACE

Let $\vec{a},\ \vec{b},\ \vec{c}$ be non-zero, non-coplanar vectors in space. Then any vector $\vec{r}$, can be uniquely expressed as a linear combination of $\vec{a},\ \vec{b},\ \vec{c}$

$\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$ .

**1.** Express the non coplanar vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ in terms of $\vec{b}\times\vec{c}$, $\vec{c}\times\vec{a}$, $\vec{a}\times\vec{b}$.

Ans. $\vec{a}=\dfrac{\vec{a}.\vec{a}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}\left(\vec{b}\times\vec{c}\right)+\dfrac{\vec{a}.\vec{b}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}\left(\vec{c}\times\vec{a}\right)+\dfrac{\vec{a}.\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}\left(\vec{a}\times\vec{b}\right)$

Sol. Let $\vec{a}=p\left(\vec{b}\times\vec{c}\right)+q\left(\vec{c}\times\vec{a}\right)+r\left(\vec{a}\times\vec{b}\right)$ .....(1)

dot with $\vec{a}$

$$p=\frac{\vec{a}.\vec{a}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$

dot with $\vec{b}$

$$q=\frac{\vec{a}.\vec{b}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$

dot with $\vec{c}$

$$r=\frac{\vec{a}.\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$

$$\Rightarrow\vec{a}=\frac{\vec{a}.\vec{a}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}\left(\vec{b}\times\vec{c}\right)+\frac{\vec{a}.\vec{b}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}\left(\vec{c}\times\vec{a}\right)+\frac{\vec{a}.\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}\left(\vec{a}\times\vec{b}\right)$$

similar for $\vec{b}$ and $\vec{c}$.

2. Express $\vec{b}\times\vec{c}$, $\vec{c}\times\vec{a}$, $\vec{a}\times\vec{b}$ in terms of 3 non coplanar vectors $\vec{a}$, $\vec{b}$, $\vec{c}$

Ans. $\vec{a}\times\vec{b} = x\vec{a}+y\vec{b}+z\vec{c}$

Sol. Let $\vec{a}\times\vec{b}=p\vec{a}+q\vec{b}+r\vec{c}$ .....(i)

dot with $\left(\vec{a}\times\vec{b}\right)$

$$\left(\vec{a}\times\vec{b}\right).\left(\vec{a}\times\vec{b}\right)=r\left[\vec{c}\,\vec{a}\,\vec{b}\right]$$

$$\Rightarrow r=\frac{\left|\vec{a}\times b\right|^2}{\left[\vec{a}\vec{b}\vec{c}\right]} \qquad \text{..... (ii)}$$

dot with $\left(\vec{b}\times\vec{c}\right)$ in (i)

$$\left(\vec{a}\times b\right).\left(\vec{b}\times\vec{c}\right)=p\left[\vec{a}\,\vec{b}\,\vec{c}\right]$$

$$\Rightarrow p\left[\vec{a}\,\vec{b}\,\vec{c}\right]=\begin{vmatrix}\vec{a}.\vec{b} & \vec{a}.\vec{c}\\ \vec{b}.\vec{b} & \vec{b}.\vec{c}\end{vmatrix}$$

$$\Rightarrow p=\frac{\left(\vec{a}.\vec{b}\right)\left(\vec{b}.\vec{c}\right)-b^2\left(\vec{a}.\vec{c}\right)}{\left[\vec{a}\vec{b}\vec{c}\right]} \qquad \text{..... (iii)}$$

dot with $\left(\vec{c}\times\vec{a}\right)$ in (i)

$$\left(\vec{c}\times\vec{b}\right).\left(\vec{c}\times\vec{a}\right)=q\vec{b}.\left(\vec{c}\times\vec{a}\right)$$

$$\Rightarrow q\left[\vec{a}\vec{b}\vec{c}\right]=\begin{vmatrix}\vec{a}.\vec{c} & \vec{a}.\vec{a}\\ \vec{b}.\vec{c} & \vec{b}.\vec{a}\end{vmatrix}$$

$$=\left(\vec{a}.\vec{c}\right)\left(\vec{b}.\vec{a}\right)-\left(\vec{b}.\vec{c}\right)a^2$$

$$\Rightarrow q=\frac{\left(\vec{a}.\vec{c}\right)\left(\vec{a}.\vec{b}\right)-a^2\left(\vec{b}.\vec{c}\right)}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]} \qquad \text{..... (iv)}$$

from (i), (ii), (iii) and (iv)

$$\vec{a}\times\vec{b}=\frac{\left(\left(\vec{a}.\vec{b}\right)\left(\vec{b}.\vec{c}\right)-b^2\left(\vec{a}.\vec{c}\right)\right)\vec{a}+\left(\left(\vec{a}.\vec{b}\right)\left(\vec{a}.\vec{c}\right)-a^2\left(\vec{b}.\vec{c}\right)\right)\vec{b}+\left(a^2b^2-\left(\vec{a}.\vec{b}\right)^2\right)\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$

3. Given the vector $\vec{a}$ and $\vec{b}$ orthogonal to each other. Find the vector $\vec{V}$ in terms of $\vec{a}$ and $\vec{b}$ satisfying $\vec{V}$.
$\vec{a}=0$; $\vec{V}.\vec{b}=1$ and $\left[\vec{V}\,\vec{a}\,\vec{b}\right]=1$.

Ans. $\vec{V}=\frac{1}{\left|\vec{b}\right|^2}\vec{b}+\frac{1}{\left|\vec{a}\times\vec{b}\right|^2}\left(\vec{a}\times\vec{b}\right)$

Sol. Let $\vec{v}=p\vec{a}+q\vec{b}+r\left(\vec{a}\times\vec{b}\right)$ ..... (i)

dot with $\vec{a}$

$O=pa^2\Rightarrow p=0$ .....(ii)

dot with $\vec{b}$

$$\vec{v}.\vec{b} = 0 + qb^2$$

$$\Rightarrow q = \frac{1}{b^2} \qquad \text{.....(iii)}$$

dot with $\left(\vec{a}\times\vec{b}\right)$

$$\vec{v}.\left(\vec{a}\times\vec{b}\right) = r\left|\vec{a}\times\vec{b}\right|^2$$

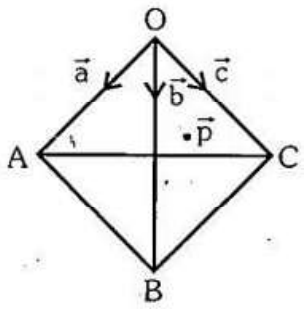
$$\Rightarrow r = \frac{\left[\vec{v}\,\vec{a}\,\vec{b}\right]}{\left|\vec{a}\times\vec{b}\right|^2} = \frac{1}{\left|\vec{a}\times\vec{b}\right|^2} \qquad \text{.....(iv)}$$

from (i), (ii), (iii) and (iv)

$$\vec{v} = \frac{1}{b^2}\vec{b} + \frac{1}{\left|\vec{a}\times\vec{b}\right|^2}\left(\vec{a}\times\vec{b}\right)$$

4. Show that the p.v. of curcumcentre of a tetrahedron OABC is (where 'O' is the origin)

$$\frac{\vec{a}^2\left(\vec{b}\times\vec{c}\right)+\vec{b}^2\left(\vec{c}\times\vec{a}\right)+\vec{c}^2\left(\vec{a}\times\vec{b}\right)}{2\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$




Hints : The position vector of circumcenter of tetrahedron in terms of $\vec{a},\vec{b},\vec{c}$ as shown with O as origin is

$$\vec{p} = x(\vec{b}\times\vec{c}) + y(\vec{c}\times\vec{a}) + z(\vec{a}\times\vec{b})$$

dot with $\vec{a}$

$$\vec{p}.\vec{a} = x\left[\vec{a}\quad\vec{b}\quad\vec{c}\right] \Rightarrow x = \frac{\vec{p}.\vec{a}}{\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]} \text{ and so on}$$

$$\left|\vec{p}\right|^2 = \left|\vec{p}-\vec{a}\right|^2 \Rightarrow \vec{p}.\vec{a} = \frac{\vec{a}^2}{2} \text{ and so on}$$

$$\vec{p} = \frac{\vec{a}^2(\vec{b}\times\vec{c})+\vec{b}^2(\vec{c}\times\vec{a})+\vec{c}^2(\vec{a}\times\vec{b})}{2\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]}$$

5. Let $\vec{a} = 3\hat{i} - 5\hat{k}$, $\vec{b} = 2\hat{i} + 7\hat{j}$ and $\vec{c} = \hat{i} + \hat{j} + \hat{k}$. Consider $\vec{r}$ such that $\vec{r}.\vec{a} = -1$, $\vec{r}.\vec{b} = 6$ and $\vec{r}.\vec{c} = 5$. Then the vector component of $2\hat{i} + 3\hat{j} + 4\hat{k}$ along $\vec{r}$ is $n\left(\frac{l\hat{i} + m\hat{k}}{l^2 + m^2}\right)$, where $l$ and m are coprimes, then $l^2 + m^2 + n^2$ is

Ans. 209

Sol. $\vec{a}=(3,0,-5),\vec{b}=(2,7,0),\vec{c}=(1,1,1)$

Let $\vec{r}=(p,q,r)$

$\vec{r}.\vec{a}=-1$

$\Rightarrow 3p-5r=-1$ .....(i)

$\vec{r}.\vec{b}=6$

$\Rightarrow 2p+7q=6$ .....(ii)

$\vec{r}.\vec{c}=5$

$\Rightarrow p+q+r=5$ .....(iii)

Solving (i), (ii) and (iii)

$p=3,\ q=0,\ r=2$

so $\vec{r}=3\hat{i}+2\hat{k}$

so vector component of $2\hat{i}+3\hat{j}+4\hat{k}$ along $\vec{r}$

$$=\left(\frac{\left(2\hat{i}+3\hat{j}+4\hat{k}\right).\left(3\hat{i}+2\hat{k}\right)}{\sqrt{13}}\right)\left(\frac{3\hat{i}+2\hat{k}}{\sqrt{13}}\right)$$

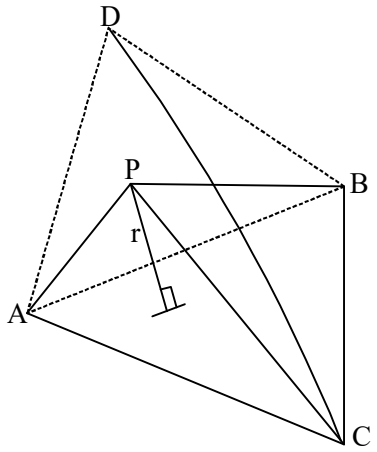
$$=\frac{14\left(3\hat{i}+2\hat{k}\right)}{\sqrt{3^2+2^2}}$$

so $\ell=3,m=2$ and $n=14$

$\Rightarrow \ell^2+m^2+n^2=9+4+196=209$

6. If $S_1$, $S_2$, $S_3$ and $S_4$ are the areas of the four triangular faces of the tetrahedron with volume V. If r is the radius of the sphere touching the four faces then prove that $V=\frac{1}{3}(S_1+S_2+S_3+S_4)r$

Sol (RHS denotes the volume of four smaller tetrahedrons each with height r and area of the base on $S_1$, $S_2$, $S_3$ and $S_4$)



Volume of tetrahedrons
(P–ABC) + (P–BCD) + (P–ACD) + (P – ABD)
= volume of tetrahedrons ABCD

$$\Rightarrow \frac{1}{3}s_1.r+\frac{1}{3}s_2r+\frac{1}{3}s_3r+\frac{1}{3}s_4r=v$$

$$\Rightarrow v=\frac{1}{3}(s_1+s_2+s_3+s_4)r$$

**Paragraph for Q7 to Q10: Reciprocal System of Vectors**

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be a system of three non-coplanar vectors. Then the system of vectors $\vec{a}'$, $\vec{b}'$ and $\vec{c}'$ which satisfies $\vec{a}.\vec{a}'=\vec{b}.\vec{b}'=\vec{c}.\vec{c}'=1$ and $\vec{a}.\vec{b}'=\vec{a}.\vec{c}'=\vec{b}.\vec{a}'=\vec{b}.\vec{c}'=\vec{c}.\vec{a}'=\vec{c}.\vec{b}'=0$, is called the reciprocal system to the vectors $\vec{a},\vec{b},\vec{c}$ .

7. find $\vec{a}'$, $\vec{b}'$ and $\vec{c}'$ in terms of $\vec{a}$, $\vec{b}$ and $\vec{c}$

Ans. $\vec{a}'=\frac{\vec{b}\times\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$, $\vec{b}'=\frac{\vec{c}\times\vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}$, $\vec{c}'=\frac{\vec{a}\times\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$.

Sol. Applying formula

$$\vec{r}=\frac{(\vec{r}.\vec{a})(\vec{b}\times\vec{c})+(\vec{r}.\vec{b})(\vec{c}\times\vec{a})+(\vec{r}.\vec{c})(\vec{a}\times\vec{b})}{[\vec{a}\,\vec{b}\,\vec{c}]}$$

seplacing $\vec{r}$ by $\vec{a}'$

$$\vec{a}'=\frac{\vec{b}\times\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}$$

Similarly $\vec{b}'=\frac{\vec{c}\times\vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}$ and $\vec{c}'=\frac{\vec{a}\times\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$.

8. Prove that $\vec{a}\times\vec{a}'+\vec{b}\times\vec{b}'+\vec{c}\times\vec{c}'=0$

Ans.

Sol. $\vec{a}\times\vec{a}'+\vec{b}\times\vec{b}'+\vec{c}\times\vec{c}'$

$$=\frac{\vec{a}\times(\vec{b}\times\vec{c})+\vec{b}\times(\vec{c}\times\vec{a})+\vec{c}\times(\vec{a}\times\vec{b})}{[\vec{a}\,\vec{b}\,\vec{c}]}$$

$$=\frac{(\vec{a}.\vec{c})\vec{b}-(\vec{a}.\vec{b})\vec{c}+(\vec{b}.\vec{a})\vec{c}-(\vec{b}.\vec{c})\vec{c}+(\vec{c}.\vec{b})\vec{a}-(\vec{c}.\vec{a})\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$$

$= 0$

9. Prove that $(\vec{a}+\vec{b}+\vec{c}).(\vec{a}'+\vec{b}'+\vec{c}')=3$

Ans.

Sol. $\left(\vec{a}+\vec{b}+\vec{c}\right).\left(\vec{a}'+\vec{b}'+\vec{c}'\right)$

$= (1 + 0 + 0) + (0 + 1 + 0) + (0 + 0 + 1)$

$= 3$

10. Prove that $\left[\vec{a},\vec{b},\vec{c}\right]=\dfrac{1}{\left[\vec{a}',\vec{b}',\vec{c}'\right]}$

Sol. $\left[\vec{a}'\ \vec{b}'\ \vec{c}'\right]=\left[\dfrac{\vec{b}\times\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}\ \dfrac{\vec{c}\times\vec{a}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}\ \dfrac{\vec{a}\times\vec{b}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}\right]$

$=\dfrac{1}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]^3}\left[\vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}\ \ \vec{a}\times\vec{b}\right]$

$=\dfrac{\left[\vec{a}\,\vec{b}\,\vec{c}\right]^2}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]^3}$

$=\dfrac{1}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$

## 13.2 VECTOR EQUATION

Solving a vector equation means determining an unknown vector (or a number of vectors satisfying the given conditions)

Do not introduce any new vector.

Try to isolate the unknown vector

**Q.1** Solve the vector equation : $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}, \vec{r}.\vec{c}=0$ provided that $\vec{c}$ is not perpendicular to $\vec{b}$.

Ans. $\vec{r}=\vec{a}-\left(\dfrac{\vec{a}.\vec{c}}{\vec{b}.\vec{c}}\right)\vec{b}$

**Sol. :** We are given;

$\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$

$\Rightarrow (\vec{r}-\vec{a})\times\vec{b}=0$

Hence $(\vec{r}-\vec{a})$ and $\vec{b}$ are parallel

$\Rightarrow \vec{r}-\vec{a}=t\,\vec{b}$ ... (i)

and we know $\vec{r}.\vec{c}=0$,

$\therefore$ taking dot product of (i) by $\vec{c}$ we get

$\vec{r}.\vec{c}-\vec{a}.\vec{c}=t(\vec{b}.\vec{c})$

$0-\vec{a}.\vec{c}=t(\vec{b}.\vec{c})$

or $t=-\left(\dfrac{\vec{a}.\vec{c}}{\vec{b}.\vec{c}}\right)$ ... (ii)

$\therefore$ from (i) and (ii) solution of $\vec{r}$ is ;

$$\vec{r} = \vec{a} - \left(\frac{\vec{a}.\vec{c}}{\vec{b}.\vec{c}}\right)\vec{b}.$$

Q.2 Find vector $\vec{c}$ such that $\vec{a}\,.\,\vec{c} = 1$ & $\vec{c}\times\vec{a} = \vec{b}$ is

Ans. $\vec{c} = \frac{1}{|\vec{a}|^2}\left(\vec{a} - \vec{b}\times\vec{a}\right)$ (Hint : cross product with $\vec{a}$ in $\vec{c}\times\vec{a} = \vec{b}$)

Sol. $\vec{a}\,.\,\vec{c} = 1$

$\vec{c}\times\vec{a} = \vec{b}$

cross with $\vec{a}$

$\vec{a}\times\left(\vec{c}\times\vec{a}\right) = \left(\vec{a}\times\vec{b}\right)$

$\Rightarrow a^2\vec{c} - \left(\vec{a}.\vec{c}\right)\vec{a} = \vec{a}\times\vec{b}$

$\Rightarrow a^2\vec{c} - \vec{a} = \vec{a}\times\vec{b}$

$\Rightarrow \vec{c} = \frac{\vec{a} + \vec{a}\times\vec{b}}{a^2}$

**Q.3** If $\vec{x}\times\vec{a} + k\vec{x} = \vec{b}$, where k is a scalar and $\vec{a}, \vec{b}$ are any two vectors, then determine $\vec{x}$ in terms of $\vec{a}$, $\vec{b}$ and k.

Ans. $\vec{x} = \frac{1}{a^2 + k^2}\left[k\vec{b} + (\vec{a}\times\vec{b}) + \frac{(\vec{a}\,.\,\vec{b})}{k}\vec{a}\right]$

**Sol. :** $\vec{x}\times\vec{a} + k\vec{x} = \vec{b}$ .....(i)

Premultiply the given equation vectorially by $\vec{a}$

$\vec{a}\times(\vec{x}\times\vec{a}) + k(\vec{a}\times\vec{x}) = \vec{a}\times\vec{b}$

$\Rightarrow (\vec{a}\,.\,\vec{a})\,\vec{x} - (\vec{a}\,.\,\vec{x})\,\vec{a} + k(\vec{a}\times\vec{x}) = \vec{a}\times\vec{b}$ .....(ii)

Premultiply (i) scalarly by $\vec{a}$

$[\vec{a}\ \vec{x}\ \vec{a}] + k\,(\vec{a}\,.\,\vec{x}) = \vec{a}\,.\,\vec{b}$

$k(\vec{a}\,.\,\vec{x}) = \vec{a}\,.\,\vec{b}$ .....(iii)

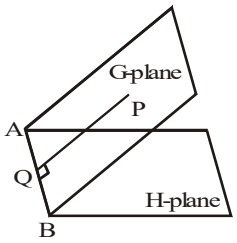
Substituting $\vec{x}\times\vec{a}$ from (i) and $\vec{a}\,.\,\vec{x}$ from (iii) in (ii) we get

$$\vec{x} = \frac{1}{a^2 + k^2}\left[k\vec{b} + (\vec{a}\times\vec{b}) + \frac{(\vec{a}\,.\,\vec{b})}{k}\vec{a}\right]$$

## 14. LINE OF GREATEST SLOPE

Consider two planes G-plane and H-plane. H-plane is treated as a horizontal plane or reference plane. G-plane is a given plane. Let AB be the line of intersection of G-plane & H-plane. Line of greatest slope is a line which is contained by G-plane & perpendicular to line of intersection of G-plane & H-plane. Obviously, infinitely many such lines of greatest slopes are contained by G-plane. Generally an additional information is given in problem so

that a unique line of greatest slope can be found out.




Direction vector of line of greatest slope = $\left(\overrightarrow{n_H} \times \overrightarrow{n_G}\right) \times \overrightarrow{n_G}$

**Q.1** Assuming the plane $4x-3y+7z=0$ to be horizontal, find the equation of the line of greatest slope through the point $(2,1,1)$ in the plane $2x+y-5z=0$.

Ans. $\dfrac{x-2}{3}=\dfrac{y-1}{-1}=\dfrac{z-1}{1}$

**Sol.** direction of line $=\left(\overline{n_H} \times \overline{n_G}\right) \times \overline{n_G}$

$=\left(\overline{n_H} \cdot \overline{n_G}\right)\overline{n_G}-\left(\overline{n_G} \cdot \overline{n_G}\right)\overline{n_H}$

$=((4,-3,7)\cdot(2,1,-5))\overline{n_G}-((2,1,-5)\cdot(2,1,-5))\overline{n_H}$

$=-30\overline{n_G}-30\overline{n_H}$

$=-30\left(\overline{n_G}+\overline{n_H}\right)$

$=-30\ ((4,-3,7)+(2,1,-5))$

$=-30(6,-2,2)$

so direction ratio of line $=(3,1,-4)$

so line passing through $(2,1,1)$ and having dirs $(3,1,-4)$

$\dfrac{x-2}{3}=\dfrac{y-1}{-1}=\dfrac{3-1}{1}$